

UNIVERSITY OF CAMBRIDGE

STEP III Past Papers 1998-2019

Section A: Pure Mathematics

1 Let

$$\mathrm{f}(x) = \sin^2 x + 2\cos x + 1$$

for $0 \leqslant x \leqslant 2\pi$. Sketch the curve $y = \mathrm{f}(x)$, giving the coordinates of the stationary points. Now let

$$\mathrm{g}(x) = \frac{a\mathrm{f}(x) + b}{c\mathrm{f}(x) + d} \qquad\qquad ad \neq bc\,,\ d \neq -3c\,,\ d \neq c\,.$$

Show that the stationary points of $y = \mathrm{g}(x)$ occur at the same values of x as those of $y = \mathrm{f}(x)$, and find the corresponding values of $\mathrm{g}(x)$.

Explain why, if $d/c < -3$ or $d/c > 1$, $|\mathrm{g}(x)|$ cannot be arbitrarily large.

2 Let

$$\mathrm{I}(a,b) = \int_0^1 t^a (1-t)^b \,\mathrm{d}t \qquad (a \geqslant 0,\ b \geqslant 0).$$

(i) Show that $\mathrm{I}(a,b) = \mathrm{I}(b,a)$,

(ii) Show that $\mathrm{I}(a,b) = \mathrm{I}(a+1,b) + \mathrm{I}(a,b+1)$.

(iii) Show that $(a+1)\mathrm{I}(a,b) = b\mathrm{I}(a+1,b-1)$ when a and b are positive and hence calculate $\mathrm{I}(a,b)$ when a and b are positive integers.

3 The value V_N of a bond after N days is determined by the equation

$$V_{N+1} = (1+c)V_N - d \qquad (c > 0,\ d > 0),$$

where c and d are given constants. By looking for solutions of the form $V_T = Ak^T + B$ for some constants A, B and k, or otherwise, find V_N in terms of V_0.

What is the solution for $c = 0$? Show that this is the limit (for fixed N) as $c \to 0$ of your solution for $c > 0$.

4 Show that the equation (in plane polar coordinates) $r = \cos\theta$, for $-\frac{\pi}{2} \leqslant \theta \leqslant \frac{\pi}{2}$, represents a circle.

Sketch the curve $r = \cos 2\theta$ for $0 \leqslant \theta \leqslant 2\pi$, and describe the curves $r = \cos 2n\theta$, where n is an integer. Show that the area enclosed by such a curve is independent of n.

Sketch also the curve $r = \cos 3\theta$ for $0 \leqslant \theta \leqslant 2\pi$.

5 The exponential of a square matrix $\mathbf{A}$ is defined to be

$$\exp(\mathbf{A}) = \sum_{r=0}^{\infty} \frac{1}{r!}\mathbf{A}^r \,,$$

where $\mathbf{A}^0 = \mathbf{I}$ and $\mathbf{I}$ is the identity matrix.

Let

$$\mathbf{M} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} .$$

Show that $\mathbf{M}^2 = -\mathbf{I}$ and hence express $\exp(\theta\mathbf{M})$ as a single 2×2 matrix, where θ is a real number. Explain the geometrical significance of $\exp(\theta\mathbf{M})$.

Let

$$\mathbf{N} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} .$$

Express similarly $\exp(s\mathbf{N})$, where s is a real number, and explain the geometrical significance of $\exp(s\mathbf{N})$.

For which values of θ does

$$\exp(s\mathbf{N})\ \exp(\theta\mathbf{M}) \ = \ \exp(\theta\mathbf{M})\ \exp(s\mathbf{N})$$

for all s? Interpret this fact geometrically.

6 (i) Show that four vertices of a cube, no two of which are adjacent, form the vertices of a regular tetrahedron. Hence, or otherwise, find the volume of a regular tetrahedron whose edges are of unit length.

(ii) Find the volume of a regular octahedron whose edges are of unit length.

(iii) Show that the centres of the faces of a cube form the vertices of a regular octahedron. Show that its volume is half that of the tetrahedron whose vertices are the vertices of the cube.

[*A regular tetrahedron (octahedron) has four (eight) faces, all equilateral triangles.*]

7 Sketch the graph of $\mathrm{f}(s) = e^s(s-3)+3$ for $0 \leqslant s < \infty$. Taking $e \approx 2.7$, find the smallest positive integer, m, such that $\mathrm{f}(m) > 0$.

Now let

$$\mathrm{b}(x) = \frac{x^3}{e^{x/T}-1}$$

where T is a positive constant. Show that $\mathrm{b}(x)$ has a single turning point in $0 < x < \infty$. By considering the behaviour for small x and for large x, sketch $\mathrm{b}(x)$ for $0 \leqslant x < \infty$.

Let

$$\int_0^\infty \mathrm{b}(x)\,\mathrm{d}x = B,$$

which may be assumed to be finite. Show that $B = KT^n$ where K is a constant, and n is an integer which you should determine.

Given that $B \approx 2\int_0^{Tm} \mathrm{b}(x)\mathrm{d}x$, use your graph of $\mathrm{b}(x)$ to find a rough estimate for K.

8 (i) Show that the line $\mathbf{r} = \mathbf{b} + \lambda\mathbf{m}$, where $\mathbf{m}$ is a unit vector, intersects the sphere $\mathbf{r}.\mathbf{r} = a^2$ at two points if

$$a^2 > \mathbf{b}.\mathbf{b} - (\mathbf{b}.\mathbf{m})^2\,.$$

Write down the corresponding condition for there to be precisely one point of intersection. If this point has position vector $\mathbf{p}$, show that $\mathbf{m}.\mathbf{p} = 0$.

(ii) Now consider a second sphere of radius a and a plane perpendicular to a unit vector $\mathbf{n}$. The centre of the sphere has position vector $\mathbf{d}$ and the minimum distance from the origin to the plane is l. What is the condition for the plane to be tangential to this second sphere?

(iii) Show that the first and second spheres intersect at right angles (*i.e.* the two radii to each point of intersection are perpendicular) if

$$\mathbf{d}.\mathbf{d} = 2a^2\,.$$

Section B: Mechanics

9 A uniform right circular cone of mass m has base of radius a and perpendicular height h from base to apex. Show that its moment of inertia about its axis is $\frac{3}{10}ma^2$, and calculate its moment of inertia about an axis through its apex parallel to its base.
[*Any theorems used should be stated clearly.*]

The cone is now suspended from its apex and allowed to perform small oscillations. Show that their period is

$$2\pi\sqrt{\frac{4h^2+a^2}{5gh}}\,.$$

[*You may assume that the centre of mass of the cone is a distance* $\frac{3}{4}h$ *from its apex.*]

10 Two identical spherical balls, moving on a horizontal, smooth table, collide in such a way that both momentum and kinetic energy are conserved. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be the velocities of the balls before the collision and let $\mathbf{v}_1'$ and $\mathbf{v}_2'$ be the velocities of the balls after the collision, where $\mathbf{v}_1$, $\mathbf{v}_2$, $\mathbf{v}_1'$ and $\mathbf{v}_2'$ are two-dimensional vectors. Write down the equations for conservation of momentum and kinetic energy in terms of these vectors. Hence show that their relative speed is also conserved.

Show that, if one ball is initially at rest but after the collision both balls are moving, their final velocities are perpendicular.

Now suppose that one ball is initially at rest, and the second is moving with speed V. After a collision in which they lose a proportion k of their original kinetic energy ($0 \leqslant k \leqslant 1$), the direction of motion of the second ball has changed by an angle θ. Find a quadratic equation satisfied by the final speed of the second ball, with coefficients depending on k, V and θ. Hence show that $k \leqslant \frac{1}{2}$.

11 Consider a simple pendulum of length l and angular displacement θ, which is **not** assumed to be small. Show that

$$\frac{1}{2}l\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = g(\cos\theta - \cos\gamma)\,,$$

where γ is the maximum value of θ. Show also that the period P is given by

$$P = 2\sqrt{\frac{l}{g}}\int_0^{\gamma}\left(\sin^2(\gamma/2) - \sin^2(\theta/2)\right)^{-\frac{1}{2}}\,\mathrm{d}\theta\,.$$

By using the substitution $\sin(\theta/2) = \sin(\gamma/2)\sin\phi$, and then finding an approximate expression for the integrand using the binomial expansion, show that for small values of γ the period is approximately

$$2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16}\right)\,.$$

Section C: Probability and statistics

12 The mountain villages A, B, C and D lie at the vertices of a tetrahedron, and each pair of villages is joined by a road. After a snowfall the probability that any road is blocked is p, and is independent of the conditions of any other road. The probability that, after a snowfall, it is possible to travel from any village to any other village by some route is P. Show that

$$P = 1 - p^2(6p^3 - 12p^2 + 3p + 4).$$

13 Write down the probability of obtaining k heads in n tosses of a fair coin. Now suppose that k is known but n is unknown. A *maximum likelihood estimator* (MLE) of n is defined to be a value (which must be an integer) of n which maximizes the probability of k heads.

A friend has thrown a fair coin a number of times. She tells you that she has observed one head. Show that in this case there are *two* MLEs of the number of tosses she has made.

She now tells you that in a repeat of the exercise she has observed k heads. Find the two MLEs of the number of tosses she has made.

She next uses a coin biased with probability p (known) of showing a head, and again tells you that she has observed k heads. Find the MLEs of the number of tosses made. What is the condition for the MLE to be unique?

14 A hostile naval power possesses a large, unknown number N of submarines. Interception of radio signals yields a small number n of their identification numbers X_i $(i = 1, 2, ..., n)$, which are taken to be independent and uniformly distributed over the continuous range from 0 to N. Show that Z_1 and Z_2, defined by

$$Z_1 = \frac{n+1}{n}\text{Max}\{X_1, X_2, ..., X_n\} \qquad \text{and} \qquad Z_2 = \frac{2}{n}\sum_{i=1}^{n} X_i \ ,$$

both have means equal to N.

Calculate the variance of Z_1 and of Z_2. Which estimator do you prefer, and why?

STEP III

- Recent STEP papers, together with Hints and Solutions, can be obtained from OCR Publications, Mill Wharf, Mill Street, Birmingham, B6 4BU .
- There is a useful booklet entitled *Advanced Problems in Mathematics* which is also available from OCR, at a modest price.
- You can get individual help from the Mathematics Millenium Project on-line maths site http://www.nrich.maths.org.uk/.

Section A: Pure Mathematics

1 Consider the cubic equation

$$x^3 - px^2 + qx - r = 0 \,,$$

where $p \neq 0$ and $r \neq 0$.

(i) If the three roots can be written in the form ak^{-1}, a and ak for some constants a and k, show that one root is q/p and that $q^3 - rp^3 = 0$.

(ii) If $r = q^3/p^3$, show that q/p is a root and that the product of the other two roots is $(q/p)^2$. Deduce that the roots are in geometric progression.

(iii) Find a necessary and sufficient condition involving p, q and r for the roots to be in arithmetic progression.

2 (i) Let $\mathrm{f}(x) = (1 + x^2)e^x$. Show that $\mathrm{f}'(x) \geqslant 0$ and sketch the graph of $\mathrm{f}(x)$. Hence, or otherwise, show that the equation

$$(1 + x^2)\mathrm{e}^x = k,$$

where k is a constant, has exactly one real root if $k > 0$ and no real roots if $k \leqslant 0$.

(ii) Determine the number of real roots of the equation

$$(\mathrm{e}^x - 1) - k \tan^{-1} x = 0$$

in the cases (a) $0 < k \leqslant 2/\pi$ and (b) $2/\pi < k < 1$.

3 Justify, by means of a sketch, the formula

$$\lim_{n\to\infty} \left\{ \frac{1}{n} \sum_{m=1}^{n} \mathrm{f}(1 + m/n) \right\} = \int_1^2 \mathrm{f}(x)\,\mathrm{d}x \,.$$

Show that

$$\lim_{n\to\infty} \left\{ \frac{1}{n+1} + \frac{1}{n+2} + \cdots + \frac{1}{n+n} \right\} = \ln 2 \,.$$

Evaluate

$$\lim_{n\to\infty} \left\{ \frac{n}{n^2+1} + \frac{n}{n^2+4} + \cdots + \frac{n}{n^2+n^2} \right\} .$$

4 A polyhedron is a solid bounded by F plane faces, which meet in E edges and V vertices. You may assume *Euler's formula*, that $V - E + F = 2$.

In a regular polyhedron the faces are equal regular m-sided polygons, n of which meet at each vertex. Show that

$$F = \frac{4n}{h}\,,$$

where $h = 4 - (n-2)(m-2)$.

By considering the possible values of h, or otherwise, prove that there are only five regular polyhedra, and find V, E and F for each.

5 The sequence u_0, u_1, u_2, ... is defined by

$$u_0 = 1, \quad u_1 = 1, \qquad u_{n+1} = u_n + u_{n-1} \quad \text{for } n \geqslant 1\,.$$

Prove that

$$u_{n+2}^2 + u_{n-1}^2 = 2(u_{n+1}^2 + u_n^2)\,.$$

Using induction, or otherwise, prove the following result:

$$u_{2n} = u_n^2 + u_{n-1}^2 \quad \text{and} \quad u_{2n+1} = u_{n+1}^2 - u_{n-1}^2$$

for any positive integer n.

6 A closed curve is given by the equation

$$x^{2/n} + y^{2/n} = a^{2/n} \qquad (*)$$

where n is an odd integer and a is a positive constant. Find a parametrization $x = x(t)$, $y = y(t)$ which describes the curve anticlockwise as t ranges from 0 to 2π.

Sketch the curve in the case $n = 3$, justifying the main features of your sketch.

The area A enclosed by such a curve is given by the formula

$$A = \frac{1}{2}\int_0^{2\pi} \left[x(t)\frac{\mathrm{d}y(t)}{\mathrm{d}t} - y(t)\frac{\mathrm{d}x(t)}{\mathrm{d}t}\right] \mathrm{d}t\,.$$

Use this result to find the area enclosed by $(*)$ for $n = 3$.

7 Let a be a non-zero real number and define a binary operation on the set of real numbers by

$$x * y = x + y + axy\,.$$

Show that the operation $*$ is associative.

Show that $(G, *)$ is a group, where G is the set of all real numbers except for one number which you should identify.

Find a subgroup of $(G, *)$ which has exactly 2 elements.

8 The function $y(x)$ is defined for $x \geqslant 0$ and satisfies the conditions

$$y = 0 \quad \text{and} \quad \frac{\mathrm{d}y}{\mathrm{d}x} = 1 \quad \text{at } x = 0.$$

When x is in the range $2(n-1)\pi < x < 2n\pi$, where n is a positive integer, $y(t)$ satisfies the differential equation

$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + n^2y = 0.$$

Both y and $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ are continuous at $x = 2n\pi$ for $n = 0,\ 1,\ 2,\ \ldots$.

(i) Find $y(x)$ for $0 \leqslant x \leqslant 2\pi$.

(ii) Show that $y(x) = \frac{1}{2}\sin 2x$ for $2\pi \leqslant x \leqslant 4\pi$, and find $y(x)$ for all $x \geqslant 0$.

(iii) Show that

$$\int_0^\infty y^2 \mathrm{d}x = \pi \sum_{n=1}^{\infty} \frac{1}{n^2}\,.$$

Section B: Mechanics

9 The gravitational force between two point particles of masses m and m' is mutually attractive and has magnitude

$$\frac{Gmm'}{r^2},$$

where G is a constant and r is the distance between them.

A particle of unit mass lies on the axis of a thin uniform circular ring of radius r and mass m, at a distance x from its centre. Explain why the net force on the particle is directed towards the centre of the ring and show that its magnitude is

$$\frac{Gmx}{(x^2+r^2)^{3/2}}.$$

The particle now lies inside a thin hollow spherical shell of uniform density, mass M and radius a, at a distance b from its centre. Show that the particle experiences no gravitational force due to the shell.

10 A chain of mass m and length l is composed of n small smooth links. It is suspended vertically over a horizontal table with its end just touching the table, and released so that it collapses inelastically onto the table. Calculate the change in momentum of the $(k+1)$th link from the bottom of the chain as it falls onto the table.

Write down an expression for the total impulse sustained by the table in this way from the whole chain. By approximating the sum by an integral, show that this total impulse is approximately

$$\tfrac{2}{3}m\sqrt{(2gl)}$$

when n is large.

11 Calculate the moment of inertia of a uniform thin circular hoop of mass m and radius a about an axis perpendicular to the plane of the hoop through a point on its circumference.

The hoop, which is rough, rolls with speed v on a rough horizontal table straight towards the edge and rolls over the edge without initially losing contact with the edge. Show that the hoop will lose contact with the edge when it has rotated about the edge of the table through an angle θ, where

$$\cos\theta = \frac{1}{2} + \frac{v^2}{2ag}.$$

Section C: Probability and Statistics

12 In the game of endless cricket the scores X and Y of the two sides are such that

$$\mathrm{P}(X = j,\ Y = k) = e^{-1}\frac{(j+k)\lambda^{j+k}}{j!k!},$$

for some positive constant λ, where $j, k = 0,\ 1,\ 2,\ \ldots$.

(i) Find $\mathrm{P}(X + Y = n)$ for each $n > 0$.

(ii) Show that $2\lambda e^{2\lambda-1} = 1$.

(iii) Show that $2xe^{2x-1}$ is an increasing function of x for $x > 0$ and deduce that the equation in (ii) has at most one solution and hence determine λ.

(iv) Calculate the expectation $\mathrm{E}(2^{X+Y})$.

13 The cakes in our canteen each contain exactly four currants, each currant being randomly placed in the cake. I take a proportion X of a cake where X is a random variable with density function

$$\mathrm{f}(x) = Ax$$

for $0 \leqslant x \leqslant 1$ where A is a constant.

(i) What is the expected number of currants in my portion?

(ii) If I find all four currants in my portion, what is the probability that I took more than half the cake?

14 In the basic version of Horizons (H1) the player has a maximum of n turns, where $n \geqslant 1$. At each turn, she has a probability p of success, where $0 < p < 1$. If her first success is at the rth turn, where $1 \leqslant r \leqslant n$, she collects r pounds and then withdraws from the game. Otherwise, her winnings are nil. Show that in H1, her expected winnings are

$$p^{-1}\left[1 + nq^{n+1} - (n+1)q^n\right] \quad \text{pounds},$$

where $q = 1 - p$.

The rules of H2 are the same as those of H1, except that n is randomly selected from a Poisson distribution with parameter λ. If $n = 0$ her winnings are nil. Otherwise she plays H1 with the selected n. Show that in H2, her expected winnings are

$$\frac{1}{p}\left(1 - e^{-\lambda p}\right) - \lambda q e^{-\lambda p} \quad \text{pounds}.$$

Section A: Pure Mathematics

1 Sketch on the same axes the two curves C_1 and C_2, given by

$$\begin{aligned} C_1: \quad xy &= 1\,, \\ C_2: \quad x^2 - y^2 &= 2\,. \end{aligned}$$

The curves intersect at P and Q. Given that the coordinates of P are (a, b) (which you need not evaluate), write down the coordinates of Q in terms of a and b.

The tangent to C_1 through P meets the tangent to C_2 through Q at the point M, and the tangent to C_2 through P meets the tangent to C_1 through Q at N. Show that the coordinates of M are $(-b, a)$ and write down the coordinates of N.

Show that $PMQN$ is a square.

2 Use the substitution $x = 2 - \cos\theta$ to evaluate the integral

$$\int_{3/2}^{2} \left(\frac{x-1}{3-x}\right)^{\frac{1}{2}} \mathrm{d}x.$$

Show that, for $a < b$,

$$\int_{p}^{q} \left(\frac{x-a}{b-x}\right)^{\frac{1}{2}} \mathrm{d}x = \frac{(b-a)(\pi + 3\sqrt{3} - 6)}{12},$$

where $p = (3a + b)/4$ and $q = (a + b)/2$.

3 Given that $\alpha = \mathrm{e}^{i\frac{\pi}{3}}$, prove that $1 + \alpha^2 = \alpha$.

A triangle in the Argand plane has vertices A, B, and C represented by the complex numbers p, $q\alpha^2$ and $-r\alpha$ respectively, where p, q and r are positive real numbers. Sketch the triangle ABC.

Three equilateral triangles ABL, BCM and CAN (each lettered clockwise) are erected on sides AB, BC and CA respectively. Show that the complex number representing N is $(1 - \alpha)p - \alpha^2 r$ and find similar expressions for the complex numbers representing L and M.

Show that lines LC, MA and NB all meet at the origin, and that these three line segments have the common length $p + q + r$.

4 The function $\mathrm{f}(x)$ is defined by

$$\mathrm{f}(x) = \frac{x(x-2)(x-a)}{x^2-1}.$$

Prove algebraically that the line $y = x + c$ intersects the curve $y = \mathrm{f}(x)$ if $|a| \geqslant 1$, but there are values of c for which there are no points of intersection if $|a| < 1$.

Find the equation of the oblique asymptote of the curve $y = \mathrm{f}(x)$. Sketch the graph in the two cases (i) $a < -1$; and (ii) $-1 < a < -\frac{1}{2}$. (You need not calculate the turning points.)

5 Given two non-zero vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ we define $\Delta(\mathbf{a}, \mathbf{b})$ by $\Delta(\mathbf{a}, \mathbf{b}) = a_1 b_2 - a_2 b_1$.

Let A, B and C be points with position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, respectively, no two of which are parallel. Let P, Q and R be points with position vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$, respectively, none of which are parallel.

(i) Show that there exists a 2×2 matrix $\mathbf{M}$ such that P and Q are the images of A and B under the transformation represented by $\mathbf{M}$.

(ii) Show that $\Delta(\mathbf{a}, \mathbf{b})\,\mathbf{c} + \Delta(\mathbf{c}, \mathbf{a})\,\mathbf{b} + \Delta(\mathbf{b}, \mathbf{c})\,\mathbf{a} = \mathbf{0}$.

Hence, or otherwise, prove that a necessary and sufficient condition for the points P, Q, and R to be the images of points A, B and C under the transformation represented by some 2×2 matrix $\mathbf{M}$ is that

$$\Delta(\mathbf{a}, \mathbf{b}) : \Delta(\mathbf{b}, \mathbf{c}) : \Delta(\mathbf{c}, \mathbf{a}) = \Delta(\mathbf{p}, \mathbf{q}) : \Delta(\mathbf{q}, \mathbf{r}) : \Delta(\mathbf{r}, \mathbf{p})\,.$$

6 Given that

$$x^4 + px^2 + qx + r = (x^2 - ax + b)(x^2 + ax + c),$$

express p, q and r in terms of a, b and c.

Show also that a^2 is a root of the cubic equation

$$u^3 + 2pu^2 + (p^2 - 4r)u - q^2 = 0.$$

Explain why this equation always has a non-negative root, and verify that $u = 9$ is a root in the case $p = -1$, $q = -6$, $r = 15$.

Hence, or otherwise, express

$$y^4 - 8y^3 + 23y^2 - 34y + 39$$

as a product of two quadratic factors.

7 Given that

$$\mathrm{e} = 1 + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{r!} + \cdots ,$$

use the binomial theorem to show that

$$\left(1 + \frac{1}{n}\right)^n < \mathrm{e}$$

for any positive integer n.

The product $\mathrm{P}(n)$ is defined, for any positive integer n, by

$$\mathrm{P}(n) = \frac{3}{2} \cdot \frac{5}{4} \cdot \frac{9}{8} \cdot \ldots \cdot \frac{2^n + 1}{2^n}.$$

Use the arithmetic-geometric mean inequality,

$$\frac{a_1 + a_2 + \cdots + a_n}{n} \geqslant (a_1 \cdot a_2 \cdot \ldots \cdot a_n)^{\frac{1}{n}} ,$$

to show that $\mathrm{P}(n) < \mathrm{e}$ for all n .

Explain briefly why $\mathrm{P}(n)$ tends to a limit as $n \to \infty$. Show that this limit, L, satisfies $2 < L \leqslant \mathrm{e}$.

8 The sequence a_n is defined by $a_0 = 1$, $a_1 = 1$, and

$$a_n = \frac{1 + a_{n-1}^2}{a_{n-2}} \qquad (n \geqslant 2).$$

Prove by induction that

$$a_n = 3a_{n-1} - a_{n-2} \qquad (n \geqslant 2).$$

Hence show that

$$a_n = \frac{\alpha^{2n-1} + \alpha^{-(2n-1)}}{\sqrt{5}} \qquad (n \geqslant 1),$$

where $\alpha = \dfrac{1 + \sqrt{5}}{2}$.

Section B: Mechanics

9 Two small discs of masses m and μm lie on a smooth horizontal surface. The disc of mass μm is at rest, and the disc of mass m is projected towards it with velocity $\mathbf{u}$. After the collision, the disc of mass μm moves in the direction given by unit vector $\mathbf{n}$. The collision is perfectly elastic.

(i) Show that the speed of the disc of mass μm after the collision is $\dfrac{2\mathbf{u}.\mathbf{n}}{1+\mu}$.

(ii) Given that the two discs have equal kinetic energy after the collision, find an expression for the cosine of the angle between $\mathbf{n}$ and $\mathbf{u}$ and show that $3-\sqrt{8} \leqslant \mu \leqslant 3+\sqrt{8}$.

10 A sphere of radius a and weight W rests on horizontal ground. A thin uniform beam of weight $3\sqrt{3}\,W$ and length $2a$ is freely hinged to the ground at X, which is a distance $\sqrt{3}\,a$ from the point of contact of the sphere with the ground. The beam rests on the sphere, lying in the same vertical plane as the centre of the sphere. The coefficients of friction between the beam and the sphere and between the sphere and the ground are μ_1 and μ_2 respectively.

Given that the sphere is on the point of slipping at its contacts with both the ground and the beam, find the values of μ_1 and μ_2.

11 A thin beam is fixed at a height $2a$ above a horizontal plane. A uniform straight rod ACB of length $9a$ and mass m is supported by the beam at C. Initially, the rod is held so that it is horizontal and perpendicular to the beam. The distance AC is $3a$, and the coefficient of friction between the beam and the rod is μ.

The rod is now released. Find the minimum value of μ for which B strikes the horizontal plane before slipping takes place at C.

Section C: Statistics

12 In a lottery, any one of N numbers, where N is large, is chosen at random and independently for each player by machine. Each week there are $2N$ players and one winning number is drawn. Write down an exact expression for the probability that there are three or fewer winners in a week, given that you hold a winning ticket that week. Using the fact that

$$\left(1-\frac{a}{n}\right)^n \approx \mathrm{e}^{-a}$$

for n much larger than a, or otherwise, show that this probability is approximately $\frac{2}{3}$.

Discuss briefly whether this probability would increase or decrease if the numbers were chosen by the players.

Show that the expected number of winners in a week, given that you hold a winning ticket that week, is $3 - N^{-1}$.

13 A set of n dice is rolled repeatedly. For each die the probability of showing a six is p. Show that the probability that the first of the dice to show a six does so on the rth roll is

$$q^{nr}(q^{-n}-1)$$

where $q = 1 - p$.

Determine, and simplify, an expression for the probability generating function for this distribution, in terms of q and n. The first of the dice to show a six does so on the Rth roll. Find the expected value of R and show that, in the case $n = 2$, $p = 1/6$, this value is $36/11$.

Show that the probability that the last of the dice to show a six does so on the rth roll is

$$(1-q^r)^n - (1-q^{r-1})^n.$$

Find, for the case $n = 2$, the probability generating function. The last of the dice to show a six does so on the Sth roll. Find the expected value of S and evaluate this when $p = 1/6$.

14 The random variable X takes only the values x_1 and x_2 (where $x_1 \neq x_2$), and the random variable Y takes only the values y_1 and y_2 (where $y_1 \neq y_2$). Their joint distribution is given by

$$\mathrm{P}(X = x_1, Y = y_1) = a \ ; \quad \mathrm{P}(X = x_1, Y = y_2) = q - a \ ; \quad \mathrm{P}(X = x_2, Y = y_1) = p - a \ .$$

Show that if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$ then

$$(a - pq)(x_1 - x_2)(y_1 - y_2) = 0.$$

Hence show that two random variables each taking only two distinct values are independent if $\mathrm{E}(XY) = \mathrm{E}(X)\mathrm{E}(Y)$.

Give a joint distribution for two random variables A and B, each taking the three values -1, 0 and 1 with probability $\frac{1}{3}$, which have $\mathrm{E}(AB) = \mathrm{E}(A)\mathrm{E}(B)$, but which are not independent.

Section A: Pure Mathematics

1 Given that $y = \ln\left(x + \sqrt{(x^2+1)}\right)$, show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{1}{\sqrt{(x^2+1)}}$.

Prove by induction that, for $n \geqslant 0$,

$$\left(x^2+1\right) y^{(n+2)} + (2n+1)\, xy^{(n+1)} + n^2 y^{(n)} = 0 \,,$$

where $y^{(n)} = \dfrac{\mathrm{d}^n y}{\mathrm{d}x^n}$ and $y^{(0)} = y$.

Using this result in the case $x = 0$, or otherwise, show that the Maclaurin series for y begins

$$x - \frac{x^3}{6} + \frac{3x^5}{40}$$

and find the next non-zero term.

2 Show that $\cosh^{-1} x = \ln\left(x + \sqrt{(x^2-1)}\right)$.

Show that the area of the region defined by the inequalities $y^2 \geqslant x^2 - 8$ and $x^2 \geqslant 25y^2 - 16$ is $(72/5)\ln 2$.

3 Consider the equation

$$x^2 - bx + c = 0 \,,$$

where b and c are real numbers.

(i) Show that the roots of the equation are real and positive if and only if $b > 0$ and $b^2 \geqslant 4c > 0$, and sketch the region of the b-c plane in which these conditions hold.

(ii) Sketch the region of the b-c plane in which the roots of the equation are real and less than 1 in magnitude.

4 In this question, the function $\sin^{-1}$ is defined to have domain $-1 \leqslant x \leqslant 1$ and range $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$ and the function $\tan^{-1}$ is defined to have the real numbers as its domain and range $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

(i) Let

$$\mathrm{g}(x) = \frac{2x}{1+x^2}\,, \qquad -\infty < x < \infty\,.$$

Sketch the graph of $\mathrm{g}(x)$ and state the range of g.

(ii) Let

$$\mathrm{f}\,(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)\,, \qquad -\infty < x < \infty\,.$$

Show that $\mathrm{f}(x) = 2\tan^{-1} x$ for $-1 \leqslant x \leqslant 1$ and $\mathrm{f}(x) = \pi - 2\tan^{-1} x$ for $x \geqslant 1$.

Sketch the graph of $\mathrm{f}(x)$.

5 Show that the equation $x^3 + px + q = 0$ has exactly one real solution if $p \geqslant 0$.

A parabola C is given parametrically by

$$x = at^2, \quad y = 2at \qquad (a > 0)\ .$$

Find an equation which must be satisfied by t at points on C at which the normal passes through the point $(h,\ k)$. Hence show that, if $h \leqslant 2a$, exactly one normal to C will pass through $(h,\ k)$.

Find, in Cartesian form, the equation of the locus of the points from which exactly two normals can be drawn to C. Sketch the locus.

6 The plane

$$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

meets the co-ordinate axes at the points A, B and C. The point M has coordinates $\left(\frac{1}{2}a, \frac{1}{2}b, \frac{1}{2}c\right)$ and O is the origin.

Show that OM meets the plane at the centroid $\left(\frac{1}{3}a, \frac{1}{3}b, \frac{1}{3}c\right)$ of triangle ABC. Show also that the perpendiculars to the plane from O and from M meet the plane at the orthocentre and at the circumcentre of triangle ABC respectively.

Hence prove that the centroid of a triangle lies on the line segment joining its orthocentre and circumcentre, and that it divides this line segment in the ratio $2 : 1$.

[The *orthocentre* of a triangle is the point at which the three altitudes intersect; the *circumcentre* of a triangle is the point equidistant from the three vertices.]

7 Sketch the graph of the function $\ln x - \frac{1}{2}x^2$.

Show that the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2xy}{x^2 - 1}$$

describes a family of parabolas each of which passes through the points $(1\,,\ 0)$ and $(-1\,,\ 0)$ and has its vertex on the y–axis.

Hence find the equation of the curve that passes through the point $(1\,,\ 1)$ and intersects each of the above parabolas orthogonally. Sketch this curve.

[Two curves intersect *orthogonally* if their tangents at the point of intersection are perpendicular.]

8 (i) Prove that the equations

$$|z - (1 + i)|^2 = 2 \qquad (*)$$

and

$$|z - (1 - i)|^2 = 2\,|z - 1|^2$$

describe the same locus in the complex z–plane. Sketch this locus.

(ii) Prove that the equation

$$\arg\left(\frac{z - 2}{z}\right) = \frac{\pi}{4} \qquad (**)$$

describes part of this same locus, and show on your sketch which part.

(iii) The complex number w is related to z by

$$w = \frac{2}{z}\,.$$

Determine the locus produced in the complex w–plane if z satisfies $(*)$. Sketch this locus and indicate the part of this locus that corresponds to $(**)$.

Section B: Mechanics

9 B_1 and B_2 are parallel, thin, horizontal fixed beams. B_1 is a vertical distance $d\sin\alpha$ above B_2, and a horizontal distance $d\cos\alpha$ from B_2, where $0 < \alpha < \pi/2$. A long heavy plank is held so that it rests on the two beams, perpendicular to each, with its centre of gravity at B_1. The coefficients of friction between the plank and B_1 and B_2 are μ_1 and μ_2, respectively, where $\mu_1 < \mu_2$ and $\mu_1 + \mu_2 = 2\tan\alpha$.

The plank is released and slips over the beams experiencing a force of resistance from each beam equal to the limiting frictional force (i.e. the product of the appropriate coefficient of friction and the normal reaction). Show that it will come to rest with its centre of gravity over B_2 in a time

$$\pi\left(\frac{d}{g(\mu_2-\mu_1)\cos\alpha}\right)^{\frac{1}{2}}.$$

10 Three ships A, B and C move with velocities $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{u}$ respectively. The velocities of A and B relative to C are equal in magnitude and perpendicular. Write down conditions that $\mathbf{u}$, $\mathbf{v}_1$ and $\mathbf{v}_2$ must satisfy and show that

$$\left|\mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1+\mathbf{v}_2)\right|^2 = \left|\tfrac{1}{2}(\mathbf{v}_1-\mathbf{v}_2)\right|^2$$

and

$$\left(\mathbf{u} - \tfrac{1}{2}(\mathbf{v}_1+\mathbf{v}_2)\right)\cdot(\mathbf{v}_1-\mathbf{v}_2) = 0\,.$$

Explain why these equations determine, for given $\mathbf{v}_1$ and $\mathbf{v}_2$, two possible velocities for C, provided $\mathbf{v}_1 \neq \mathbf{v}_2$.

If $\mathbf{v}_1$ and $\mathbf{v}_2$ are equal in magnitude and perpendicular, show that if $\mathbf{u} \neq \mathbf{0}$ then $\mathbf{u} = \mathbf{v}_1 + \mathbf{v}_2$.

11 A uniform cylinder of radius a rotates freely about its axis, which is fixed and horizontal. The moment of inertia of the cylinder about its axis is I. A light string is wrapped around the cylinder and supports a mass m which hangs freely. A particle of mass M is fixed to the surface of the cylinder. The system is held at rest with the particle vertically below the axis of the cylinder, and then released. Find, in terms of I, a, M, m, g and θ, the angular velocity of the cylinder when it has rotated through angle θ.

Show that the cylinder will rotate without coming to a halt if $m/M > \sin\alpha$, where α satisifes $\alpha = \tan\frac{1}{2}\alpha$ and $0 < \alpha < \pi$.

Section C: Statistics

12 A bag contains b black balls and w white balls. Balls are drawn at random from the bag and when a white ball is drawn it is put aside.

(i) If the black balls drawn are also put aside, find an expression for the expected number of black balls that have been drawn when the last white ball is removed.

(ii) If instead the black balls drawn are put back into the bag, prove that the expected number of times a black ball has been drawn when the first white ball is removed is b/w. Hence write down, in the form of a sum, an expression for the expected number of times a black ball has been drawn when the last white ball is removed.

13 In a game for two players, a fair coin is tossed repeatedly. Each player is assigned a sequence of heads and tails and the player whose sequence appears first wins. Four players, A, B, C and D take turns to play the game. Each time they play, A is assigned the sequence TTH (i.e. Tail then Tail then Head), B is assigned THH, C is assigned HHT and D is assigned HTT.

(i) A and B play the game. Let p_{HH}, p_{HT}, p_{TH} and p_{TT} be the probabilities of A winning the game given that the first two tosses of the coin show HH, HT, TH and TT, respectively. Explain why $p_{\mathrm{TT}} = 1$, and why $p_{\mathrm{HT}} = \frac{1}{2}\,p_{\mathrm{TH}} + \frac{1}{2}\,p_{\mathrm{TT}}$. Show that $p_{\mathrm{HH}} = p_{\mathrm{HT}} = \frac{2}{3}$ and that $p_{\mathrm{TH}} = \frac{1}{3}$. Deduce that the probability that A wins the game is $\frac{2}{3}$.

(ii) B and C play the game. Find the probability that B wins.

(iii) Show that if C plays D, then C is more likely to win than D, but that if D plays A, then D is more likely to win than A.

14 A random variable X is distributed uniformly on $[\,0\,,\,a\,]$. Show that the variance of X is $\frac{1}{12}a^2$.

A sample, X_1 and X_2, of two independent values of the random variable is drawn, and the variance V of the sample is determined. Show that $V = \frac{1}{4}\,(X_1 - X_2)^2$, and hence prove that $2V$ is an unbiased estimator of the variance of X.

Find an exact expression for the probability that the value of V is less than $\frac{1}{12}a^2$ and estimate the value of this probability correct to one significant figure.

Section A: Pure Mathematics

1 Find the area of the region between the curve $y = \dfrac{\ln x}{x}$ and the x-axis, for $1 \leqslant x \leqslant a$. What happens to this area as a tends to infinity?

Find the volume of the solid obtained when the region between the curve $y = \dfrac{\ln x}{x}$ and the x-axis, for $1 \leqslant x \leqslant a$, is rotated through 2π radians about the x-axis. What happens to this volume as a tends to infinity?

2 Prove that $\arctan a + \arctan b = \arctan\left(\dfrac{a+b}{1-ab}\right)$ when $0 < a < 1$ and $0 < b < 1$.

Prove by induction that, for $n \geqslant 1$,

$$\sum_{r=1}^{n} \arctan\left(\frac{1}{r^2+r+1}\right) = \arctan\left(\frac{n}{n+2}\right)$$

and hence find

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2+r+1}\right).$$

Hence prove that

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{r^2-r+1}\right) = \frac{\pi}{2}.$$

3 Let

$$\mathrm{f}(x) = a\sqrt{x} - \sqrt{x-b},$$

where $x \geqslant b > 0$ and $a > 1$. Sketch the graph of $\mathrm{f}(x)$. Hence show that the equation $\mathrm{f}(x) = c$, where $c > 0$, has no solution when $c^2 < b\left(a^2-1\right)$. Find conditions on c^2 in terms of a and b for the equation to have exactly one or exactly two solutions.

Solve the equations (i) $3\sqrt{x} - \sqrt{x-2} = 4$ and (ii) $3\sqrt{x} - \sqrt{x-3} = 5$.

4 Show that if x and y are positive and $x^3 + x^2 = y^3 - y^2$ then $x < y$.

Show further that if $0 < x \leqslant y - 1$, then $x^3 + x^2 < y^3 - y^2$.

Prove that there does not exist a pair of *positive* integers such that the difference of their cubes is equal to the sum of their squares.

Find all the pairs of integers such that the difference of their cubes is equal to the sum of their squares.

5 Give a condition that must be satisfied by p, q and r for it to be possible to write the quadratic polynomial $px^2 + qx + r$ in the form $p\,(x + h)^2$, for some h.

Obtain an equation, which you need not simplify, that must be satisfied by t if it is possible to write

$$\left(x^2 + \tfrac{1}{2}bx + t\right)^2 - \left(x^4 + bx^3 + cx^2 + dx + e\right)$$

in the form $k\,(x + h)^2$, for some k and h.

Hence, or otherwise, write $x^4 + 6x^3 + 9x^2 - 2x - 7$ as a product of two quadratic factors.

6 Find all the solution curves of the differential equation

$$y^4 \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^4 = \left(y^2 - 1\right)^2$$

that pass through either of the points

(i) $\left(0,\ \frac{\sqrt{3}}{2}\right)$,

(ii) $\left(0,\ \frac{\sqrt{5}}{2}\right)$.

Show also that $y = 1$ and $y = -1$ are solutions of the differential equation. Sketch all these solution curves on a single set of axes.

7 Given that α and β are acute angles, show that $\alpha+\beta=\pi/2$ if and only if $\cos^2\alpha+\cos^2\beta=1$.

In the x–y plane, the point A has coordinates $(0,s)$ and the point C has coordinates $(s,0)$, where $s>0$. The point B lies in the first quadrant ($x>0$, $y>0$). The lengths of AB, OB and CB are respectively a, b and c.

Show that

$$(s^2+b^2-a^2)^2+(s^2+b^2-c^2)^2=4s^2b^2$$

and hence that

$$(2s^2-a^2-c^2)^2+(2b^2-a^2-c^2)^2=4a^2c^2\ .$$

Deduce that

$$(a-c)^2\leqslant 2b^2\leqslant(a+c)^2\ .$$

8 Four complex numbers u_1, u_2, u_3 and u_4 have unit modulus, and arguments θ_1, θ_2, θ_3 and θ_4, respectively, with $-\pi<\theta_1<\theta_2<\theta_3<\theta_4<\pi$.

Show that

$$\arg(u_1-u_2)=\frac{1}{2}(\theta_1+\theta_2-\pi)+2n\pi$$

where $n=0$ or 1. Deduce that

$$\arg((u_1-u_2)(u_4-u_3))=\arg((u_1-u_4)(u_3-u_2))+2n\pi$$

for some integer n.

Prove that

$$|(u_1-u_2)(u_4-u_3)|+|(u_1-u_4)(u_3-u_2)|=|(u_1-u_3)(u_4-u_2)|\ .$$

Section B: Mechanics

9 A tall container made of light material of negligible thickness has the form of a prism, with a square base of area a^2. It contains a volume ka^3 of fluid of uniform density. The container is held so that it stands on a rough plane, which is inclined at angle θ to the horizontal, with two of the edges of the base of the container horizontal. In the case $k > \frac{1}{2}\tan\theta$, show that the centre of mass of the fluid is at a distance x from the lower side of the container and at a distance y from the base of the container, where

$$\frac{x}{a} = \frac{1}{2} - \frac{\tan\theta}{12k}\ , \qquad \frac{y}{a} = \frac{k}{2} + \frac{\tan^2\theta}{24k}\ .$$

Determine the corresponding coordinates in the case $k < \frac{1}{2}\tan\theta$.

The container is now released. Given that $k < \frac{1}{2}$, show that the container will topple if $\theta > 45°$.

10 A light hollow cylinder of radius a can rotate freely about its axis of symmetry, which is fixed and horizontal. A particle of mass m is fixed to the cylinder, and a second particle, also of mass m, moves on the rough inside surface of the cylinder. Initially, the cylinder is at rest, with the fixed particle on the same horizontal level as its axis and the second particle at rest vertically below this axis. The system is then released. Show that, if θ is the angle through which the cylinder has rotated, then

$$\ddot{\theta} = \frac{g}{2a}\left(\cos\theta - \sin\theta\right)\ ,$$

provided that the second particle does not slip.

Given that the coefficient of friction is $\dfrac{3+\sqrt{3}}{6}$, show that the second particle starts to slip when the cylinder has rotated through $60°$.

11 A particle moves on a smooth triangular horizontal surface AOB with angle $AOB = 30°$. The surface is bounded by two vertical walls OA and OB and the coefficient of restitution between the particle and the walls is e, where $e < 1$. The particle, which is initially at point P on the surface and moving with velocity u_1, strikes the wall OA at M_1, with angle $PM_1A = \theta$, and rebounds, with velocity v_1, to strike the wall OB at N_1, with angle $M_1N_1B = \theta$. Find e and $\dfrac{v_1}{u_1}$ in terms of θ.

The motion continues, with the particle striking side OA at M_2, M_3, ... and striking side OB at N_2, N_3, ... Show that, if $\theta < 60°$, the particle reaches O in a finite time.

Section C: Probability and Statistics

12 In a game, a player tosses a biased coin repeatedly until two successive tails occur, when the game terminates. For each head which occurs the player wins £1. If E is the expected number of tosses of the coin in the course of a game, and p is the probability of a head, explain why

$$E = p\,(1 + E) + (1 - p)\,p\,(2 + E) + 2\,(1 - p)^2\ ,$$

and hence determine E in terms of p. Find also, in terms of p, the expected winnings in the course of a game.

A second game is played, with the same rules, except that the player continues to toss the coin until r successive tails occur. Show that the expected number of tosses in the course of a game is given by the expression $\dfrac{1 - q^r}{pq^r}$, where $q = 1 - p$.

13 A continuous random variable is said to have an exponential distribution with parameter λ if its density function is $\mathrm{f}(t) = \lambda \mathrm{e}^{-\lambda t}\ \ (0 \leqslant t < \infty)$. If X_1 and X_2, which are independent random variables, have exponential distributions with parameters λ_1 and λ_2 respectively, find an expression for the probability that either X_1 or X_2 (or both) is less than x. Prove that if X is the random variable whose value is the lesser of the values of X_1 and X_2, then X also has an exponential distribution.

Route A and Route B buses run from my house to my college. The time between buses on each route has an exponential distribution and the mean time between buses is 15 minutes for Route A and 30 minutes for Route B. The timings of the buses on the two routes are independent. If I emerge from my house one day to see a Route A bus and a Route B bus just leaving the stop, show that the median wait for the next bus to my college will be approximately 7 minutes.

14 Prove that, for any two discrete random variables X and Y,

$$\mathrm{Var}\,[X+Y] = \mathrm{Var}\,[X] + \mathrm{Var}\,[Y] + 2\,\mathrm{Cov}\,[X,\,Y]\,,$$

where $\mathrm{Var}\,[X]$ is the variance of X and $\mathrm{Cov}\,[X,\,Y]$ is the covariance of X and Y.

When a Grandmaster plays a sequence of m games of chess, she is, independently, equally likely to win, lose or draw each game. If the values of the random variables W, L and D are the numbers of her wins, losses and draws respectively, justify briefly the following claims:

(i) $W + L + D$ has variance 0;

(ii) $W + L$ has a binomial distribution.

Find the value of $\dfrac{\mathrm{Cov}\,[W,\,L]}{\sqrt{\mathrm{Var}\,[W]\,\mathrm{Var}\,[L]}}$.

Section A: Pure Mathematics

1 Given that $x + a > 0$ and $x + b > 0$, and that $b > a$, show that

$$\frac{\mathrm{d}}{\mathrm{d}x} \arcsin\left(\frac{x+a}{x+b}\right) = \frac{\sqrt{b-a}}{(x+b)\sqrt{a+b+2x}}$$

and find $\dfrac{\mathrm{d}}{\mathrm{d}x} \operatorname{arcosh}\left(\dfrac{x+b}{x+a}\right)$.

Hence, or otherwise, integrate, for $x > -1$,

(i) $\displaystyle\int \frac{1}{(x+1)\sqrt{x+3}}\,\mathrm{d}x$,

(ii) $\displaystyle\int \frac{1}{(x+3)\sqrt{x+1}}\,\mathrm{d}x$.

[You may use the results $\dfrac{\mathrm{d}}{\mathrm{d}x}\arcsin x = \dfrac{1}{\sqrt{1-x^2}}$ and $\dfrac{\mathrm{d}}{\mathrm{d}x}\operatorname{arcosh} x = \dfrac{1}{\sqrt{x^2-1}}$.]

2 Show that ${}^{2r}\mathrm{C}_r = \dfrac{1 \times 3 \times \cdots \times (2r-1)}{r!} \times 2^r$, for $r \geqslant 1$.

(i) Give the first four terms of the binomial series for $(1-p)^{-\frac{1}{2}}$.

By choosing a suitable value for p in this series, or otherwise, show that

$$\sum_{r=0}^{\infty} \frac{{}^{2r}\mathrm{C}_r}{8^r} = \sqrt{2}\,.$$

(ii) Show that

$$\sum_{r=0}^{\infty} \frac{(2r+1)\ {}^{2r}\mathrm{C}_r}{5^r} = \left(\sqrt{5}\right)^3.$$

[**Note:** ${}^{n}\mathrm{C}_r$ is an alternative notation for $\binom{n}{r}$ for $r \geqslant 1$, and ${}^{0}\mathrm{C}_0 = 1$.]

3 If m is a positive integer, show that $(1+x)^m + (1-x)^m \neq 0$ for any real x.

The function f is defined by

$$\mathrm{f}(x) = \frac{(1+x)^m - (1-x)^m}{(1+x)^m + (1-x)^m} \ .$$

Find and simplify an expression for $\mathrm{f}'(x)$.

In the case $m = 5$, sketch the curves $y = \mathrm{f}(x)$ and $y = \dfrac{1}{\mathrm{f}(x)}$.

4 A curve is defined parametrically by

$$x = t^2 \ , \quad y = t(1+t^2) \ .$$

The tangent at the point with parameter t, where $t \neq 0$, meets the curve again at the point with parameter T, where $T \neq t$. Show that

$$T = \frac{1-t^2}{2t} \quad \text{and} \quad 3t^2 \neq 1 \ .$$

Given a point P_0 on the curve, with parameter t_0, a sequence of points P_0, P_1, P_2, ... on the curve is constructed such that the tangent at P_i meets the curve again at P_{i+1}. If $t_0 = \tan\frac{7\pi}{18}$, show that $P_3 = P_0$ but $P_1 \neq P_0$. Find a second value of t_0, with $t_0 > 0$, for which $P_3 = P_0$ but $P_1 \neq P_0$.

5 Find the coordinates of the turning point on the curve $y = x^2 - 2bx + c$. Sketch the curve in the case that the equation $x^2 - 2bx + c = 0$ has two distinct real roots. Use your sketch to determine necessary and sufficient conditions on b and c for the equation $x^2 - 2bx + c = 0$ to have two distinct real roots. Determine necessary and sufficient conditions on b and c for this equation to have two distinct positive roots.

Find the coordinates of the turning points on the curve $y = x^3 - 3b^2x + c$ (with $b > 0$) and hence determine necessary and sufficient conditions on b and c for the equation $x^3 - 3b^2x + c = 0$ to have three distinct real roots. Determine necessary and sufficient conditions on a, b and c for the equation $(x-a)^3 - 3b^2(x-a) + c = 0$ to have three distinct positive roots.

Show that the equation $2x^3 - 9x^2 + 7x - 1 = 0$ has three distinct positive roots.

6 Show that

$$2\sin\tfrac{1}{2}\theta\,\cos r\theta = \sin\left(r+\tfrac{1}{2}\right)\theta - \sin\left(r-\tfrac{1}{2}\right)\theta\,.$$

Hence, or otherwise, find all solutions of the equation

$$\cos a\theta + \cos(a+1)\theta + \cdots + \cos(b-2)\theta + \cos(b-1)\theta = 0\,,$$

where a and b are positive integers with $a < b-1$.

7 In the x–y plane, the point A has coordinates $(a\,,0)$ and the point B has coordinates $(0\,,b)$, where a and b are positive. The point P, which is distinct from A and B, has coordinates (s,t). X and Y are the feet of the perpendiculars from P to the x–axis and y–axis respectively, and N is the foot of the perpendicular from P to the line AB. Show that the coordinates $(x\,,y)$ of N are given by

$$x = \frac{ab^2 - a(bt-as)}{a^2+b^2}\,, \quad y = \frac{a^2b + b(bt-as)}{a^2+b^2}\,.$$

Show that, if $\left(\dfrac{t-b}{s}\right)\left(\dfrac{t}{s-a}\right) = -1$, then N lies on the line XY.

Give a geometrical interpretation of this result.

8 (i) Show that the gradient at a point $(x\,,\,y)$ on the curve

$$(y+2x)^3\,(y-4x) = c\,,$$

where c is a constant, is given by

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{16x-y}{2y-5x}\,.$$

(ii) By considering the derivative with respect to x of $(y+ax)^n\,(y+bx)$, or otherwise, find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{10x-4y}{3x-y}\,.$$

Section B: Mechanics

9 A particle P of mass m is constrained to move on a vertical circle of smooth wire with centre O and of radius a. L is the lowest point of the circle and H the highest and $\angle LOP = \theta$. The particle is attached to H by an elastic string of natural length a and modulus of elasticity αmg, where $\alpha > 1$. Show that, if $\alpha > 2$, there is an equilibrium position with $0 < \theta < \pi$.

Given that $\alpha = 2 + \sqrt{2}$, and that $\theta = \dfrac{\pi}{2} + \phi$, show that

$$\ddot{\phi} \approx -\frac{g(\sqrt{2}+1)}{2a}\,\phi$$

when ϕ is small.

For this value of α, explain briefly what happens to the particle if it is given a small displacement when $\theta = \dfrac{\pi}{2}$.

10 A particle moves along the x-axis in such a way that its acceleration is k$x\dot{x}$ where k is a positive constant. When $t = 0$, $x = d$ (where $d > 0$) and $\dot{x} = U$.

(i) Find x as a function of t in the case $U = kd^2$ and show that x tends to infinity as t tends to $\dfrac{\pi}{2dk}$.

(ii) If $U < 0$, find x as a function of t and show that it tends to a limit, which you should state in terms of d and U, as t tends to infinity.

11 Point B is a distance d due south of point A on a horizontal plane. Particle P is at rest at B at $t = 0$, when it begins to move with constant acceleration a in a straight line with fixed bearing β. Particle Q is projected from point A at $t = 0$ and moves in a straight line with constant speed v. Show that if the direction of projection of Q can be chosen so that Q strikes P, then

$$v^2 \geqslant ad\,(1 - \cos\beta)\ .$$

Show further that if $v^2 > ad(1 - \cos\beta)$ then the direction of projection of Q can be chosen so that Q strikes P before P has moved a distance d.

Section C: Probability and Statistics

12 Brief interruptions to my work occur on average every ten minutes and the number of interruptions in any given time period has a Poisson distribution. Given that an interruption has just occurred, find the probability that I will have less than t minutes to work before the next interruption. If the random variable T is the time I have to work before the next interruption, find the probability density function of T.

I need an uninterrupted half hour to finish an important paper. Show that the expected number of interruptions before my first uninterrupted period of half an hour or more is $e^3 - 1$. Find also the expected length of time between interruptions that are less than half an hour apart. Hence write down the expected wait before my first uninterrupted period of half an hour or more.

13 In a rabbit warren, underground chambers A, B, C and D are at the vertices of a square, and burrows join A to B, B to C, C to D and D to A. Each of the chambers also has a tunnel to the surface. A rabbit finding itself in any chamber runs along one of the two burrows to a neighbouring chamber, or leaves the burrow through the tunnel to the surface. Each of these three possibilities is equally likely.

Let p_A, p_B, p_C and p_D be the probabilities of a rabbit leaving the burrow through the tunnel from chamber A, given that it is currently in chamber A, B, C or D, respectively.

(i) Explain why $p_A = \frac{1}{3} + \frac{1}{3}p_B + \frac{1}{3}p_D$.

(ii) Determine p_A.

(iii) Find the probability that a rabbit which starts in chamber A does not visit chamber C, given that it eventually leaves the burrow through the tunnel in chamber A.

14 Write down the probability generating function for the score on a standard, fair six-faced die whose faces are labelled 1, 2, 3, 4, 5, 6. Hence show that the probability generating function for the sum of the scores on two standard, fair six-faced dice, rolled independently, can be written as

$$\frac{1}{36}t^2\left(1+t\right)^2\left(1-t+t^2\right)^2\left(1+t+t^2\right)^2 \ .$$

Write down, in factorised form, the probability generating functions for the scores on two fair six-faced dice whose faces are labelled with the numbers 1, 2, 2, 3, 3, 4 and 1, 3, 4, 5, 6, 8, and hence show that when these dice are rolled independently, the probability of any given sum of the scores is the same as for the two standard fair six-faced dice.

Standard, fair four-faced dice are tetrahedra whose faces are labelled 1, 2, 3, 4, the score being taken from the face which is not visible after throwing, and each score being equally likely. Find all the ways in which two fair four-faced dice can have their faces labelled with positive integers if the probability of any given sum of the scores is to be the same as for the two standard fair four-faced dice.

OCR
RECOGNISING ACHIEVEMENT

OXFORD CAMBRIDGE AND RSA EXAMINATIONS

Sixth Term Examination Papers
administered on behalf of the Cambridge Colleges

MATHEMATICS III 9475

Wednesday **30 JUNE 2004** Afternoon 3 hours

Additional materials:
Answer paper
Graph paper
Formulae booklet

Candidates may not use electronic calculators

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

- Write your name, Centre number and candidate number in the spaces on the answer paper/ answer booklet.
- Begin each answer on a new page.

INFORMATION FOR CANDIDATES

- Each question is marked out of 20. There is no restriction of choice.
- You will be assessed on the **six** questions for which you gain the highest marks.
- You are advised to concentrate on no more than **six** questions. Little credit will be given to fragmentary answers.
- You are provided with Mathematical Formulae and Tables.
- **Electronic calculators are not permitted.**

This question paper consists of 8 printed pages.

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[Turn over

Section A: Pure Mathematics

1 Show that

$$\int_0^a \frac{\sinh x}{2\cosh^2 x - 1}\,\mathrm{d}x = \frac{1}{2\sqrt{2}} \ln\left(\frac{\sqrt{2}\cosh a - 1}{\sqrt{2}\cosh a + 1}\right) + \frac{1}{2\sqrt{2}} \ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)$$

and find

$$\int_0^a \frac{\cosh x}{1 + 2\sinh^2 x}\,\mathrm{d}x\,.$$

Hence show that

$$\int_0^\infty \frac{\cosh x - \sinh x}{1 + 2\sinh^2 x}\,\mathrm{d}x = \frac{\pi}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \ln\left(\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)\,.$$

By substituting $u = \mathrm{e}^x$ in this result, or otherwise, find

$$\int_1^\infty \frac{1}{1+u^4}\,\mathrm{d}u\,.$$

2 The equation of a curve is $y = \mathrm{f}(x)$ where

$$\mathrm{f}(x) = x - 4 - \frac{16\,(2x+1)^2}{x^2\,(x-4)}\,.$$

(i) Write down the equations of the vertical and oblique asymptotes to the curve and show that the oblique asymptote is a tangent to the curve.

(ii) Show that the equation $\mathrm{f}(x) = 0$ has a double root.

(iii) Sketch the curve.

3 Given that $\mathrm{f}''(x) > 0$ when $a \leqslant x \leqslant b$, explain with the aid of a sketch why

$$(b-a)\,\mathrm{f}\left(\frac{a+b}{2}\right) < \int_a^b \mathrm{f}(x)\,\mathrm{d}x < (b-a)\,\frac{\mathrm{f}(a)+\mathrm{f}(b)}{2}\,.$$

By choosing suitable a, b and $\mathrm{f}(x)$, show that

$$\frac{4}{(2n-1)^2} < \frac{1}{n-1} - \frac{1}{n} < \frac{1}{2}\left(\frac{1}{n^2} + \frac{1}{(n-1)^2}\right),$$

where n is an integer greater than 1.

Deduce that

$$4\left(\frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots\right) < 1 < \frac{1}{2} + \left(\frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \cdots\right).$$

Show that

$$\frac{1}{2}\left(\frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \cdots\right) < \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \cdots$$

and hence show that

$$\frac{3}{2} < \sum_{n=1}^{\infty} \frac{1}{n^2} < \frac{7}{4}\,.$$

4 The triangle OAB is isosceles, with $OA = OB$ and angle $AOB = 2\alpha$ where $0 < \alpha < \frac{\pi}{2}$. The semi-circle C_0 has its centre at the midpoint of the base AB of the triangle, and the sides OA and OB of the triangle are both tangent to the semi-circle. $\mathrm{C}_1, \mathrm{C}_2, \mathrm{C}_3, \ldots$ are circles such that C_n is tangent to C_{n-1} and to sides OA and OB of the triangle.

Let r_n be the radius of C_n. Show that

$$\frac{r_{n+1}}{r_n} = \frac{1-\sin\alpha}{1+\sin\alpha}\,.$$

Let S be the total area of the semi-circle C_0 and the circles C_1, C_2, C_3, ... Show that

$$S = \frac{1+\sin^2\alpha}{4\sin\alpha}\,\pi r_0^2\,.$$

Show that there are values of α for which S is more than four fifths of the area of triangle OAB.

[Turn over

5 Show that if $\cos(x-\alpha) = \cos\beta$ then either $\tan x = \tan(\alpha+\beta)$ or $\tan x = \tan(\alpha-\beta)$. By choosing suitable values of x, α and β, give an example to show that if $\tan x = \tan(\alpha+\beta)$, then $\cos(x-\alpha)$ need not equal $\cos\beta$.

Let ω be the acute angle such that $\tan\omega = \frac{4}{3}$.

(i) For $0 \leqslant x \leqslant 2\pi$, solve the equation

$$\cos x - 7\sin x = 5$$

giving both solutions in terms of ω.

(i) For $0 \leqslant x \leqslant 2\pi$, solve the equation

$$2\cos x + 11\sin x = 10$$

showing that one solution is twice the other and giving both in terms of ω.

6 Given a sequence $w_0, w_1, w_2, \ldots$, the sequence $F_1, F_2, \ldots$ is defined by

$$F_n = w_n^2 + w_{n-1}^2 - 4w_n w_{n-1}.$$

Show that $F_n - F_{n-1} = (w_n - w_{n-2})(w_n + w_{n-2} - 4w_{n-1})$ for $n \geqslant 2$.

(i) The sequence $u_0, u_1, u_2, \ldots$ has $u_0 = 1$, and $u_1 = 2$ and satisfies

$$u_n = 4u_{n-1} - u_{n-2} \quad (n \geqslant 2).$$

Prove that $u_n^2 + u_{n-1}^2 = 4u_n u_{n-1} - 3$ for $n \geqslant 1$.

(ii) A sequence $v_0, v_1, v_2, \ldots$ has $v_0 = 1$ and satisfies

$$v_n^2 + v_{n-1}^2 = 4v_n v_{n-1} - 3 \quad (n \geqslant 1). \qquad (*)$$

(a) Find v_1 and prove that, for each $n \geqslant 2$, either $v_n = 4v_{n-1} - v_{n-2}$ or $v_n = v_{n-2}$.

(b) Show that the sequence, with period 2, defined by

$$v_n = \begin{cases} 1 & \text{for } n \text{ even} \\ 2 & \text{for } n \text{ odd} \end{cases}$$

satisfies $(*)$.

(c) Find a sequence v_n with period 4 which has $v_0 = 1$, and satisfies $(*)$.

7 For $n = 1, 2, 3, \ldots$, let

$$I_n = \int_0^1 \frac{t^{n-1}}{(t+1)^n}\,\mathrm{d}t\,.$$

By considering the greatest value taken by $\dfrac{t}{t+1}$ for $0 \leqslant t \leqslant 1$ show that $I_{n+1} < \frac{1}{2}I_n$.

Show also that $I_{n+1} = -\dfrac{1}{n\,2^n} + I_n$.

Deduce that $I_n < \dfrac{1}{n\,2^{n-1}}$.

Prove that

$$\ln 2 = \sum_{r=1}^{n} \frac{1}{r\,2^r} + I_{n+1}$$

and hence show that $\frac{2}{3} < \ln 2 < \frac{17}{24}$.

8 Show that if

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{f}(x)y + \frac{\mathrm{g}(x)}{y}$$

then the substitution $u = y^2$ gives a linear differential equation for $u(x)$.

Hence or otherwise solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{y}{x} - \frac{1}{y}\,.$$

Determine the solution curves of this equation which pass through $(1\,,1)$, $(2\,,2)$ and $(4\,,4)$ and sketch graphs of all three curves on the same axes.

[Turn over

Section B: Mechanics

9 A circular hoop of radius a is free to rotate about a fixed horizontal axis passing through a point P on its circumference. The plane of the hoop is perpendicular to this axis. The hoop hangs in equilibrium with its centre, O, vertically below P. The point A on the hoop is vertically below O, so that POA is a diameter of the hoop.

A mouse M runs at constant speed u round the rough inner surface of the lower part of the hoop. Show that the mouse can choose its speed so that the hoop remains in equilibrium with diameter POA vertical.

Describe what happens to the hoop when the mouse passes the point at which angle $AOM = 2\arctan\mu$, where μ is the coefficient of friction between mouse and hoop.

10 A particle P of mass m is attached to points A and B, where A is a distance $9a$ vertically above B, by elastic strings, each of which has modulus of elasticity $6mg$. The string AP has natural length $6a$ and the string BP has natural length $2a$. Let x be the distance AP.

The system is released from rest with P on the vertical line AB and $x = 6a$. Show that the acceleration $\ddot{x}$ of P is $\frac{4g}{a}(7a - x)$ for $6a < x < 7a$ and $\frac{g}{a}(7a - x)$ for $7a < x < 9a$.

Find the time taken for the particle to reach B.

11 Particles P, of mass 2, and Q, of mass 1, move along a line. Their distances from a fixed point are x_1 and x_2, respectively where $x_2 > x_1$. Each particle is subject to a repulsive force from the other of magnitude $\frac{2}{z^3}$, where $z = x_2 - x_1$.

Initially, $x_1 = 0$, $x_2 = 1$, Q is at rest and P moves towards Q with speed 1. Show that z obeys the equation $\frac{\mathrm{d}^2z}{\mathrm{d}t^2} = \frac{3}{z^3}$.

By first writing $\frac{\mathrm{d}^2z}{\mathrm{d}t^2} = v\frac{\mathrm{d}v}{\mathrm{d}z}$, where $v = \frac{\mathrm{d}z}{\mathrm{d}t}$, show that $z = \sqrt{4t^2 - 2t + 1}$.

By considering the equation satisfied by $2x_1 + x_2$, find x_1 and x_2 in terms of t.

Section C: Probability and Statistics

12 A team of m players, numbered from 1 to m, puts on a set of a m shirts, similarly numbered from 1 to m. The players change in a hurry, so that the shirts are assigned to them randomly, one to each player.

Let C_i be the random variable that takes the value 1 if player i is wearing shirt i, and 0 otherwise. Show that $\mathrm{E}[C_1] = \frac{1}{m}$ and find $\mathrm{Var}[C_1]$ and $\mathrm{Cov}[C_1, C_2]$.

Let $N = C_1 + C_2 + \cdots + C_m$ be the random variable whose value is the number of players who are wearing the correct shirt. Show that $\mathrm{E}[N] = \mathrm{Var}[N] = 1$.

Explain why a Normal approximation to N is not likely to be appropriate for any m, but that a Poisson approximation might be reasonable.

In the case $m = 4$, find, by listing equally likely possibilities or otherwise, the probability that no player is wearing the correct shirt and verify that an appropriate Poisson approximation to N gives this probability with a relative error of about 2%. [Use $\mathrm{e} \approx 2\frac{72}{100}$.]

13 A men's endurance competition has an unlimited number of rounds. In each round, a competitor has, independently, a probability p of making it through the round; otherwise, he fails the round. Once a competitor fails a round, he drops out of the competition; before he drops out, he takes part in every round. The grand prize is awarded to any competitor who makes it through a round which all the other remaining competitors fail; if all the remaining competitors fail at the same round the grand prize is not awarded.

If the competition begins with three competitors, find the probability that:

(i) all three drop out in the same round;

(ii) two of them drop out in round r (with $r \geqslant 2$) and the third in an earlier round;

(iii) the grand prize is awarded.

[Turn over

14 In this question, $\Phi(z)$ is the cumulative distribution function of a standard normal random variable.

A random variable is known to have a Normal distribution with mean μ and standard deviation either σ_0 or σ_1, where $\sigma_0 < \sigma_1$. The mean, $\overline{X}$, of a random sample of n values of X is to be used to test the hypothesis $H_0 : \sigma = \sigma_0$ against the alternative $H_1 : \sigma = \sigma_1$.

Explain carefully why it is appropriate to use a two sided test of the form: accept H_0 if $\mu - c < \overline{X} < \mu + c$, otherwise accept H_1.

Given that the probability of accepting H_1 when H_0 is true is α, determine c in terms of n, σ_0 and z_α, where z_α is defined by $\Phi(z_\alpha) = 1 - \frac{\alpha}{2}$.

The probability of accepting H_0 when H_1 is true is denoted by β. Show that β is independent of n.

Given that $\Phi(1.960) \approx 0.975$ and that $\Phi(0.063) \approx 0.525$, determine, approximately, the minimum value of $\frac{\sigma_1}{\sigma_0}$ if α and β are both to be less than 0.05.

Section A: Pure Mathematics

1 Show that $\sin A = \cos B$ if and only if $A = (4n+1)\dfrac{\pi}{2} \pm B$ for some integer n.

Show also that $\left|\sin x \pm \cos x\right| \leqslant \sqrt{2}$ for all values of x and deduce that there are no solutions to the equation $\sin(\sin x) = \cos(\cos x)$.

Sketch, on the same axes, the graphs of $y = \sin(\sin x)$ and $y = \cos(\cos x)$. Sketch, not on the previous axes, the graph of $y = \sin(2\sin x)$.

2 Find the general solution of the differential equation $\dfrac{\mathrm{d}y}{\mathrm{d}x} = -\dfrac{xy}{x^2+a^2}$, where $a \neq 0$, and show that it can be written in the form $y^2(x^2+a^2) = c^2$, where c is an arbitrary constant. Sketch this curve.

Find an expression for $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^2+y^2)$ and show that

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}(x^2+y^2) = 2\left(1 - \frac{c^2}{(x^2+a^2)^2}\right) + \frac{8c^2x^2}{(x^2+a^2)^3}\,.$$

(i) Show that, if $0 < c < a^2$, the points on the curve whose distance from the origin is least are $\left(0\,,\ \pm\dfrac{c}{a}\right)$.

(ii) If $c > a^2$, determine the points on the curve whose distance from the origin is least.

3 Let $\mathrm{f}(x) = x^2 + px + q$ and $\mathrm{g}(x) = x^2 + rx + s$. Find an expression for $\mathrm{f}(\mathrm{g}(x))$ and hence find a necessary and sufficient condition on a, b and c for it to be possible to write the quartic expression $x^4 + ax^3 + bx^2 + cx + d$ in the form $\mathrm{f}(\mathrm{g}(x))$, for some choice of values of p, q, r and s.

Show further that this condition holds if and only if it is possible to write the quartic expression $x^4 + ax^3 + bx^2 + cx + d$ in the form $(x^2 + vx + w)^2 - k$, for some choice of values of v, w and k.

Find the roots of the quartic equation $x^4 - 4x^3 + 10x^2 - 12x + 4 = 0$.

4 The sequence u_n $(n = 1, 2, \ldots)$ satisfies the recurrence relation

$$u_{n+2} = \frac{u_{n+1}}{u_n}(ku_n - u_{n+1})$$

where k is a constant.

If $u_1 = a$ and $u_2 = b$, where a and b are non-zero and $b \neq ka$, prove by induction that

$$u_{2n} = \left(\frac{b}{a}\right)u_{2n-1}$$

$$u_{2n+1} = cu_{2n}$$

for $n \geqslant 1$, where c is a constant to be found in terms of k, a and b. Hence express u_{2n} and u_{2n-1} in terms of a, b, c and n.

Find conditions on a, b and k in the three cases:

(i) the sequence u_n is geometric;

(ii) the sequence u_n has period 2;

(iii) the sequence u_n has period 4.

5 Let P be the point on the curve $y = ax^2 + bx + c$ (where a is non-zero) at which the gradient is m. Show that the equation of the tangent at P is

$$y - mx = c - \frac{(m-b)^2}{4a}\ .$$

Show that the curves $y = a_1x^2 + b_1x + c_1$ and $y = a_2x^2 + b_2x + c_2$ (where a_1 and a_2 are non-zero) have a common tangent with gradient m if and only if

$$(a_2 - a_1)m^2 + 2(a_1b_2 - a_2b_1)m + 4a_1a_2(c_2 - c_1) + a_2b_1^2 - a_1b_2^2 = 0\ .$$

Show that, in the case $a_1 \neq a_2$, the two curves have exactly one common tangent if and only if they touch each other. In the case $a_1 = a_2$, find a necessary and sufficient condition for the two curves to have exactly one common tangent.

6 In this question, you may use without proof the results

$$4\cosh^3 y - 3\cosh y = \cosh(3y) \quad \text{and} \quad \operatorname{arcosh} y = \ln\left(y + \sqrt{y^2-1}\right).$$

[**Note:** arcosh$y$ is another notation for $\cosh^{-1} y$]

Show that the equation $x^3 - 3a^2x = 2a^3\cosh T$ is satisfied by $2a\cosh\left(\frac{1}{3}T\right)$ and hence that, if $c^2 \geqslant b^3 > 0$, one of the roots of the equation $x^3 - 3bx = 2c$ is $u + \dfrac{b}{u}$, where $u = \left(c + \sqrt{c^2-b^3}\right)^{\frac{1}{3}}$.

Show that the other two roots of the equation $x^3 - 3bx = 2c$ are the roots of the quadratic equation $x^2 + \left(u + \dfrac{b}{u}\right)x + u^2 + \dfrac{b^2}{u^2} - b = 0$, and find these roots in terms of u, b and ω, where $\omega = \frac{1}{2}(-1 + i\sqrt{3})$.

Solve completely the equation $x^3 - 6x = 6$.

7 Show that if $\displaystyle\int \frac{1}{u\,\mathrm{f}(u)}\,\mathrm{d}u = \mathrm{F}(u) + c$, then $\displaystyle\int \frac{m}{x\,\mathrm{f}(x^m)}\,\mathrm{d}x = \mathrm{F}(x^m) + c$, where $m \neq 0$.

Find:

(i) $\displaystyle\int \frac{1}{x^n - x}\,\mathrm{d}x$;

(ii) $\displaystyle\int \frac{1}{\sqrt{x^n + x^2}}\,\mathrm{d}x$.

8 In this question, a and c are distinct non-zero complex numbers. The complex conjugate of any complex number z is denoted by z^*.

Show that

$$|a - c|^2 = aa^* + cc^* - ac^* - ca^*$$

and hence prove that the triangle OAC in the Argand diagram, whose vertices are represented by 0, a and c respectively, is right angled at A if and only if $2aa^* = ac^* + ca^*$.

Points P and P' in the Argand diagram are represented by the complex numbers ab and $\dfrac{a}{b^*}$, where b is a non-zero complex number. A circle in the Argand diagram has centre C and passes through the point A, and is such that OA is a tangent to the circle. Show that the point P lies on the circle if and only if the point P' lies on the circle.

Conversely, show that if the points represented by the complex numbers ab and $\dfrac{a}{b^*}$, for some non-zero complex number b with $bb^* \neq 1$, both lie on a circle centre C in the Argand diagram which passes through A, then OA is a tangent to the circle.

Section B: Mechanics

9 Two particles, A and B, move without friction along a horizontal line which is perpendicular to a vertical wall. The coefficient of restitution between the two particles is e and the coefficient of restitution between particle B and the wall is also e, where $0 < e < 1$. The mass of particle A is $4em$ (with $m > 0$), and the mass of particle B is $(1-e)^2 m$.

Initially, A is moving towards the wall with speed $(1-e)v$ (where $v > 0$) and B is moving away from the wall and towards A with speed $2ev$. The two particles collide at a distance d from the wall. Find the speeds of A and B after the collision.

When B strikes the wall, it rebounds along the same line. Show that a second collision will take place, at a distance de from the wall.

Deduce that further collisions will take place. Find the distance from the wall at which the nth collision takes place, and show that the times between successive collisions are equal.

10 Two thin discs, each of radius r and mass m, are held on a rough horizontal surface with their centres a distance $6r$ apart. A thin light elastic band, of natural length $2\pi r$ and modulus $\dfrac{\pi mg}{12}$, is wrapped once round the discs, its straight sections being parallel. The contact between the elastic band and the discs is smooth. The coefficient of static friction between each disc and the horizontal surface is μ, and each disc experiences a force due to friction equal to μmg when it is sliding.

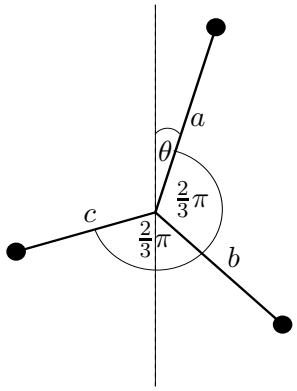
The discs are released simultaneously. If the discs collide, they rebound and a half of their total kinetic energy is lost in the collision.

(i) Show that the discs start sliding, but come to rest before colliding, if and only if $\frac{2}{3} < \mu < 1$.

(ii) Show that, if the discs collide at least once, their total kinetic energy just before the first collision is $\frac{4}{3}mgr(2-3\mu)$.

(iii) Show that if $\frac{4}{9} > \mu^2 > \frac{5}{27}$ the discs come to rest exactly once after the first collision.

11 A horizontal spindle rotates freely in a fixed bearing. Three light rods are each attached by one end to the spindle so that they rotate in a vertical plane. A particle of mass m is fixed to the other end of each of the three rods. The rods have lengths a, b and c, with $a > b > c$ and the angle between any pair of rods is $\frac{2}{3}\pi$. The angle between the rod of length a and the vertical is θ, as shown in the diagram.



Find an expression for the energy of the system and show that, if the system is in equilibrium, then

$$\tan\theta = -\frac{(b-c)\sqrt{3}}{2a-b-c}\,.$$

Deduce that there are exactly two equilibrium positions and determine which of the two equilibrium positions is stable.

Show that, for the system to make complete revolutions, it must pass through its position of stable equilibrium with an angular velocity of at least

$$\sqrt{\frac{4gR}{a^2+b^2+c^2}}\,,$$

where $2R^2 = (a-b)^2 + (b-c)^2 + (c-a)^2$.

Section C: Probability and Statistics

12 Five independent timers time a runner as she runs four laps of a track. Four of the timers measure the individual lap times, the results of the measurements being the random variables T_1 to T_4, each of which has variance σ^2 and expectation equal to the true time for the lap. The fifth timer measures the total time for the race, the result of the measurement being the random variable T which has variance σ^2 and expectation equal to the true race time (which is equal to the sum of the four true lap times).

Find a random variable X of the form $aT + b(T_1 + T_2 + T_3 + T_4)$, where a and b are constants independent of the true lap times, with the two properties:

(1) whatever the true lap times, the expectation of X is equal to the true race time;

(2) the variance of X is as small as possible.

Find also a random variable Y of the form $cT + d(T_1 + T_2 + T_3 + T_4)$, where c and d are constants independent of the true lap times, with the property that, whatever the true lap times, the expectation of Y^2 is equal to σ^2.

In one particular race, T takes the value 220 seconds and $(T_1 + T_2 + T_3 + T_4)$ takes the value 220.5 seconds. Use the random variables X and Y to estimate an interval in which the true race time lies.

13 A pack of cards consists of $n + 1$ cards, which are printed with the integers from 0 to n. A game consists of drawing cards repeatedly at random from the pack until the card printed with 0 is drawn, at which point the game ends. After each draw, the player receives £1 if the card drawn shows any of the integers from 1 to w inclusive but receives nothing if the card drawn shows any of the integers from $w + 1$ to n inclusive.

(i) In one version of the game, each card drawn is replaced immediately and randomly in the pack. Explain clearly why the probability that the player wins a total of exactly £3 is equal to the probability of the following event occurring: out of the first four cards drawn which show numbers in the range 0 to w, the numbers on the first three are non-zero and the number on the fourth is zero. Hence show that the probability that the player wins a total of exactly £3 is equal to $\dfrac{w^3}{(w+1)^4}$.

Write down the probability that the player wins a total of exactly £r and hence find the expected total win.

(ii) In another version of the game, each card drawn is removed from the pack. Show that the expected total win in this version is half of the expected total win in the other version.

14 In this question, you may use the result

$$\int_0^\infty \frac{t^m}{(t+k)^{n+2}}\,\mathrm{d}t = \frac{m!\,(n-m)!}{(n+1)!\,k^{n-m+1}}\,,$$

where m and n are positive integers with $n \geqslant m$, and where $k > 0$.

The random variable V has density function

$$\mathrm{f}(x) = \frac{C\,k^{a+1}\,x^a}{(x+k)^{2a+2}} \qquad (0 \leqslant x < \infty)\,,$$

where a is a positive integer. Show that $C = \dfrac{(2a+1)!}{a!\,a!}$.

Show, by means of a suitable substitution, that

$$\int_0^v \frac{x^a}{(x+k)^{2a+2}}\,\mathrm{d}x = \int_{\frac{k^2}{v}}^\infty \frac{u^a}{(u+k)^{2a+2}}\,\mathrm{d}u$$

and deduce that the median value of V is k. Find the expected value of V.

The random variable V represents the speed of a randomly chosen gas molecule. The time taken for such a particle to travel a fixed distance s is given by the random variable $T = \dfrac{s}{V}$.

Show that

$$\mathrm{P}(T < t) = \int_{\frac{s}{t}}^\infty \frac{C\,k^{a+1}\,x^a}{(x+k)^{2a+2}}\,\mathrm{d}x \qquad (*)$$

and hence find the density function of T. You may find it helpful to make the substitution $u = \dfrac{s}{x}$ in the integral $(*)$.

Hence show that the product of the median time and the median speed is equal to the distance s, but that the product of the expected time and the expected speed is greater than s.

Section A: Pure Mathematics

1 Sketch the curve with cartesian equation

$$y = \frac{2x(x^2-5)}{x^2-4}$$

and give the equations of the asymptotes and of the tangent to the curve at the origin.

Hence determine the number of real roots of the following equations:

(i) $3x(x^2-5) = (x^2-4)(x+3)$;

(ii) $4x(x^2-5) = (x^2-4)(5x-2)$;

(iii) $4x^2(x^2-5)^2 = (x^2-4)^2(x^2+1)$.

2 Let

$$I = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\theta}{1-\sin\theta\sin 2\alpha}\,\mathrm{d}\theta \quad \text{and} \quad J = \int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\sec^2\theta}{1+\tan^2\theta\cos^2 2\alpha}\,\mathrm{d}\theta$$

where $0 < \alpha < \frac{1}{4}\pi$.

(i) Show that $I = \displaystyle\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{\cos^2\theta}{1+\sin\theta\sin 2\alpha}\,\mathrm{d}\theta$ and hence that $2I = \displaystyle\int_{-\frac{1}{2}\pi}^{\frac{1}{2}\pi} \frac{2}{1+\tan^2\theta\cos^2 2\alpha}\,\mathrm{d}\theta$.

(ii) Find J.

(iii) By considering $I\sin^2 2\alpha + J\cos^2 2\alpha$, or otherwise, show that $I = \frac{1}{2}\pi\sec^2\alpha$.

(iv) Evaluate I in the case $\frac{1}{4}\pi < \alpha < \frac{1}{2}\pi$.

3 (i) Let

$$\tan x = \sum_{n=0}^{\infty} a_n x^n \quad \text{and} \quad \cot x = \frac{1}{x} + \sum_{n=0}^{\infty} b_n x^n$$

for $0 < x < \frac{1}{2}\pi$. Explain why $a_n = 0$ for even n.

Prove the identity

$$\cot x - \tan x \equiv 2\cot 2x$$

and show that

$$a_n = (1 - 2^{n+1})b_n\,.$$

(ii) Let $\operatorname{cosec} x = \dfrac{1}{x} + \displaystyle\sum_{n=0}^{\infty} c_n x^n$ for $0 < x < \frac{1}{2}\pi$. By considering $\cot x + \tan x$, or otherwise, show that

$$c_n = (2^{-n} - 1)b_n\,.$$

(iii) Show that

$$\left(1 + x\sum_{n=0}^{\infty} b_n x^n\right)^2 + x^2 = \left(1 + x\sum_{n=0}^{\infty} c_n x^n\right)^2\,.$$

Deduce from this and the previous results that $a_1 = 1$, and find a_3.

4 The function f satisfies the identity

$$\mathrm{f}(x) + \mathrm{f}(y) \equiv \mathrm{f}(x + y) \qquad (*)$$

for all x and y. Show that $2\mathrm{f}(x) \equiv \mathrm{f}(2x)$ and deduce that $\mathrm{f}''(0) = 0$. By considering the Maclaurin series for $\mathrm{f}(x)$, find the most general function that satisfies $(*)$.
[*Do not consider issues of existence or convergence of Maclaurin series in this question.*]

(i) By considering the function G, defined by $\ln\big(\mathrm{g}(x)\big) = \mathrm{G}(x)$, find the most general function that, for all x and y, satisfies the identity

$$\mathrm{g}(x)\mathrm{g}(y) \equiv \mathrm{g}(x + y)\,.$$

(ii) By considering the function H, defined by $\mathrm{h}(\mathrm{e}^u) = \mathrm{H}(u)$, find the most general function that satisfies, for all positive x and y, the identity

$$\mathrm{h}(x) + \mathrm{h}(y) \equiv \mathrm{h}(xy)\,.$$

(iii) Find the most general function t that, for all x and y, satisfies the identity

$$\mathrm{t}(x) + \mathrm{t}(y) \equiv \mathrm{t}(z)\,,$$

where $z = \dfrac{x + y}{1 - xy}$.

5 Show that the distinct complex numbers α, β and γ represent the vertices of an equilateral triangle (in clockwise or anti-clockwise order) if and only if

$$\alpha^2 + \beta^2 + \gamma^2 - \beta\gamma - \gamma\alpha - \alpha\beta = 0\,.$$

Show that the roots of the equation

$$z^3 + az^2 + bz + c = 0 \qquad (*)$$

represent the vertices of an equilateral triangle if and only if $a^2 = 3b$.

Under the transformation $z = pw + q$, where p and q are given complex numbers with $p \neq 0$, the equation $(*)$ becomes

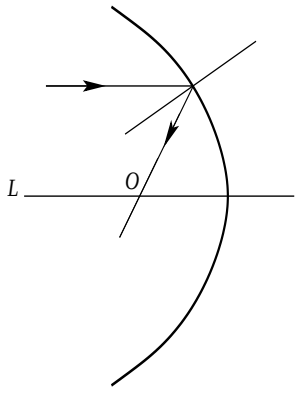
$$w^3 + Aw^2 + Bw + C = 0\,. \qquad (**)$$

Show that if the roots of equation $(*)$ represent the vertices of an equilateral triangle, then the roots of equation $(**)$ also represent the vertices of an equilateral triangle.

6 Show that in polar coordinates the gradient of any curve at the point (r, θ) is

$$\frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\tan\theta + r}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta} - r\tan\theta}\,.$$

A mirror is designed so that if an incident ray of light is parallel to a fixed line L the reflected ray passes through a fixed point O on L. Prove that the mirror intersects any plane containing L in a parabola. You should assume that the angle between the incident ray and the normal to the mirror is the same as the angle between the reflected ray and the normal.



7 (i) Solve the equation $u^2 + 2u\sinh x - 1 = 0$ giving u in terms of x.

Find the solution of the differential equation

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2\frac{\mathrm{d}y}{\mathrm{d}x}\sinh x - 1 = 0$$

that satisfies $y = 0$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} > 0$ at $x = 0$.

(ii) Find the solution, not identically zero, of the differential equation

$$\sinh y\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2\frac{\mathrm{d}y}{\mathrm{d}x} - \sinh y = 0$$

that satisfies $y = 0$ at $x = 0$, expressing your solution in the form $\cosh y = \mathrm{f}(x)$. Show that the asymptotes to the solution curve are $y = \pm(-x + \ln 4)$.

8 $\triangle$ is an operation that takes polynomials in x to polynomials in x; that is, given any polynomial $\mathrm{h}(x)$, there is a polynomial called $\triangle\mathrm{h}(x)$ which is obtained from $\mathrm{h}(x)$ using the rules that define $\triangle$. These rules are as follows:

(i) $\triangle x = 1$;

(ii) $\triangle\big(\mathrm{f}(x) + \mathrm{g}(x)\big) = \triangle\mathrm{f}(x) + \triangle\mathrm{g}(x)$ for any polynomials $\mathrm{f}(x)$ and $\mathrm{g}(x)$;

(iii) $\triangle\big(\lambda\mathrm{f}(x)\big) = \lambda\triangle\mathrm{f}(x)$ for any constant λ and any polynomial $\mathrm{f}(x)$;

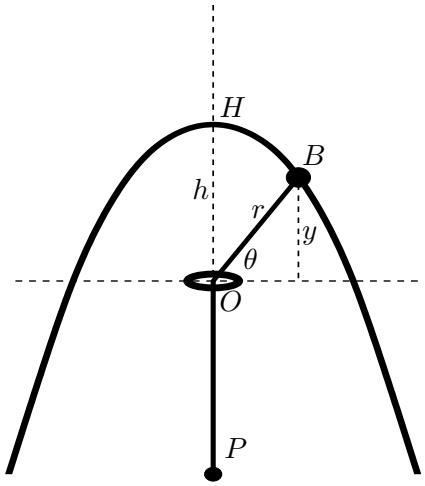
(iv) $\triangle\big(\mathrm{f}(x)\mathrm{g}(x)\big) = \mathrm{f}(x)\triangle\mathrm{g}(x) + \mathrm{g}(x)\triangle\mathrm{f}(x)$ for any polynomials $\mathrm{f}(x)$ and $\mathrm{g}(x)$.

Using these rules show that, if $\mathrm{f}(x)$ is a polynomial of degree zero (that is, a constant), then $\triangle\mathrm{f}(x) = 0$. Calculate $\triangle x^2$ and $\triangle x^3$.

Prove that $\triangle\mathrm{h}(x) \equiv \dfrac{\mathrm{dh}(x)}{\mathrm{d}x}$ for any polynomial $\mathrm{h}(x)$. You should make it clear whenever you use one of the above rules in your proof.

Section B: Mechanics

9 A long, light, inextensible string passes through a small, smooth ring fixed at the point O. One end of the string is attached to a particle P of mass m which hangs freely below O. The other end is attached to a bead, B, also of mass m, which is threaded on a smooth rigid wire fixed in the same vertical plane as O. The distance OB is r, the distance OH is h and the height of the bead above the horizontal plane through O is y, as shown in the diagram.



The shape of the wire is such that the system can be in static equilibrium for all positions of the bead. By considering potential energy, show that the equation of the wire is $y + r = 2h$.

The bead is initially at H. It is then projected along the wire with initial speed V. Show that, in the subsequent motion,

$$\dot{\theta} = -\frac{h\dot{r}}{r\sqrt{rh - h^2}}$$

where θ is given by $\theta = \arcsin(y/r)$.

Hence show that the speed of the particle P is $V\left(\dfrac{r-h}{2r-h}\right)^{\frac{1}{2}}$.

[*Note that* $\arcsin\theta$ *is another notation for* $\sin^{-1}\theta$.]

10 A disc rotates freely in a horizontal plane about a vertical axis through its centre. The moment of inertia of the disc about this axis is mk^2 (where $k > 0$). Along one diameter is a smooth narrow groove in which a particle of mass m slides freely. At time $t = 0$, the disc is rotating with angular speed Ω, and the particle is a distance a from the axis and is moving with speed V along the groove, towards the axis, where $k^2V^2 = \Omega^2a^2(k^2 + a^2)$.

Show that, at a later time t, while the particle is still moving towards the axis, the angular speed ω of the disc and the distance r of the particle from the axis are related by

$$\omega = \frac{\Omega(k^2 + a^2)}{k^2 + r^2} \quad \text{and} \quad \left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 = \frac{\Omega^2 r^2(k^2 + a^2)^2}{k^2(k^2 + r^2)} .$$

Deduce that

$$k\frac{\mathrm{d}r}{\mathrm{d}\theta} = -r(k^2 + r^2)^{\frac{1}{2}} ,$$

where θ is the angle through which the disc has turned by time t.

By making the substitution $u = k/r$, or otherwise, show that $r\sinh(\theta + \alpha) = k$, where $\sinh\alpha = k/a$. Deduce that the particle never reaches the axis.

11 A lift of mass M and its counterweight of mass M are connected by a light inextensible cable which passes over a fixed frictionless pulley. The lift is constrained to move vertically between smooth guides. The distance between the floor and the ceiling of the lift is h. Initially, the lift is at rest, and the distance between the top of the lift and the pulley is greater than h. A small tile of mass m becomes detached from the ceiling of the lift and falls to the floor of the lift. Show that the speed of the tile just before the impact is

$$\sqrt{\frac{(2M - m)gh}{M}} .$$

The coefficient of restitution between the tile and the floor of the lift is e. Given that the magnitude of the impulsive force on the lift due to tension in the cable is equal to the magnitude of the impulsive force on the counterweight due to tension in the cable, show that the loss of energy of the system due to the impact is $mgh(1 - e^2)$. Comment on this result.

Section C: Probability and Statistics

12 Fifty times a year, 1024 tourists disembark from a cruise liner at a port. From there they must travel to the city centre either by bus or by taxi. Tourists are equally likely to be directed to the bus station or to the taxi rank. Each bus of the bus company holds 32 passengers, and the company currently runs 15 buses. The company makes a profit of £1 for each passenger carried. It carries as many passengers as it can, with any excess being (eventually) transported by taxi. Show that the largest annual licence fee, in pounds, that the company should consider paying to be allowed to run an extra bus is approximately

$$1600\Phi(2) - \frac{800}{\sqrt{2\pi}}\left(1 - \mathrm{e}^{-2}\right),$$

where $\Phi(x) = \dfrac{1}{\sqrt{2\pi}} \displaystyle\int_{-\infty}^{x} \mathrm{e}^{-\frac{1}{2}t^2}\mathrm{d}t$.

[*You should not consider continuity corrections.*]

13 Two points are chosen independently at random on the perimeter (including the diameter) of a semicircle of unit radius. The area of the triangle whose vertices are these two points and the midpoint of the diameter is denoted by the random variable A. Show that the expected value of A is $(2+\pi)^{-1}$.

14 For any random variables X_1 and X_2, state the relationship between $\mathrm{E}(aX_1+bX_2)$ and $\mathrm{E}(X_1)$ and $\mathrm{E}(X_2)$, where a and b are constants. If X_1 and X_2 are independent, state the relationship between $\mathrm{E}(X_1X_2)$ and $\mathrm{E}(X_1)$ and $\mathrm{E}(X_2)$.

An industrial process produces rectangular plates. The length and the breadth of the plates are modelled by independent random variables X_1 and X_2 with non-zero means μ_1 and μ_2 and non-zero standard deviations σ_1 and σ_2, respectively. Using the results in the paragraph above, and without quoting a formula for $\mathrm{Var}\,(aX_1 + bX_2)$, find the means and standard deviations of the perimeter P and area A of the plates. Show that P and A are not independent.

The random variable Z is defined by $Z = P - \alpha A$, where α is a constant. Show that Z and A are not independent if

$$\alpha \neq \frac{2(\mu_1\sigma_2^2 + \mu_2\sigma_1^2)}{\mu_1^2\sigma_2^2 + \mu_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2}\,.$$

Given that X_1 and X_2 can each take values 1 and 3 only, and that they each take these values with probability $\frac{1}{2}$, show that Z and A are not independent for any value of α.

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MATHEMATICS III

WEDNESDAY 27 JUNE 2007

Afternoon

Time: 3 hours

Additional materials: Answer paper
Graph paper
Formulae booklet

Candidates may not use electronic calculators

*C U P / T A S / I I I *

INSTRUCTIONS TO CANDIDATES

- Write your name, Centre number and candidate number in the spaces on the answer paper/ answer booklet.
- Begin each answer on a new page.

INFORMATION FOR CANDIDATES

- Each question is marked out of 20. There is no restriction of choice.
- You will be assessed on the **six** questions for which you gain the highest marks.
- You are advised to concentrate on no more than **six** questions. Little credit will be given to fragmentary answers.
- You are provided with Mathematical Formulae and Tables.
- **Electronic calculators are not permitted.**

This document consists of **10** printed pages and **2** blank pages.

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Section A: Pure Mathematics

1 *In this question, do not consider the special cases in which the denominators of any of your expressions are zero.*

Express $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4)$ in terms of t_i, where $t_1 = \tan\theta_1$, etc.

Given that $\tan\theta_1$, $\tan\theta_2$, $\tan\theta_3$ and $\tan\theta_4$ are the four roots of the equation

$$at^4 + bt^3 + ct^2 + dt + e = 0$$

(where $a \neq 0$), find an expression in terms of a, b, c, d and e for $\tan(\theta_1 + \theta_2 + \theta_3 + \theta_4)$.

The four real numbers θ_1, θ_2, θ_3 and θ_4 lie in the range $0 \leqslant \theta_i < 2\pi$ and satisfy the equation

$$p\cos 2\theta + \cos(\theta - \alpha) + p = 0\,,$$

where p and α are independent of θ. Show that $\theta_1 + \theta_2 + \theta_3 + \theta_4 = n\pi$ for some integer n.

2 **(i)** Show that $1.3.5.7. \ldots .(2n-1) = \dfrac{(2n)!}{2^n n!}$ and that, for $|x| < \frac{1}{4}$,

$$\frac{1}{\sqrt{1-4x}} = 1 + \sum_{n=1}^{\infty} \frac{(2n)!}{(n!)^2}\, x^n\,.$$

(ii) By differentiating the above result, deduce that

$$\sum_{n=1}^{\infty} \frac{(2n)!}{n!\,(n-1)!} \left(\frac{6}{25}\right)^n = 60\,.$$

(iii) Show that

$$\sum_{n=1}^{\infty} \frac{2^{n+1}(2n)!}{3^{2n}(n+1)!\,n!} = 1\,.$$

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3 A sequence of numbers, F_1, F_2, ..., is defined by $F_1 = 1$, $F_2 = 1$, and

$$F_n = F_{n-1} + F_{n-2} \quad \text{for } n \geqslant 3.$$

(i) Write down the values of F_3, F_4, ... , F_8.

(ii) Prove that $F_{2k+3}F_{2k+1} - F_{2k+2}^2 = -F_{2k+2}F_{2k} + F_{2k+1}^2$.

(iii) Prove by induction or otherwise that $F_{2n+1}F_{2n-1} - F_{2n}^2 = 1$ and deduce that $F_{2n}^2 + 1$ is divisible by F_{2n+1}.

(iv) Prove that $F_{2n-1}^2 + 1$ is divisible by F_{2n+1}.

4 A curve is given parametrically by

$$x = a\big(\cos t + \ln \tan \tfrac{1}{2}t\big),$$
$$y = a \sin t,$$

where $0 < t < \frac{1}{2}\pi$ and a is a positive constant. Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \tan t$ and sketch the curve.

Let P be the point with parameter t and let Q be the point where the tangent to the curve at P meets the x-axis. Show that $PQ = a$.

The *radius of curvature*, ρ, at P is defined by

$$\rho = \frac{\left(\dot{x}^2 + \dot{y}^2\right)^{\frac{3}{2}}}{|\dot{x}\ddot{y} - \dot{y}\ddot{x}|},$$

where the dots denote differentiation with respect to t. Show that $\rho = a \cot t$.

The point C lies on the normal to the curve at P, a distance ρ from P and above the curve. Show that CQ is parallel to the y-axis.

5 Let $y = \ln(x^2 - 1)$, where $x > 1$, and let r and θ be functions of x determined by $r = \sqrt{x^2 - 1}$ and $\coth \theta = x$. Show that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2\cosh\theta}{r} \quad \text{and} \quad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{2\cosh 2\theta}{r^2},$$

and find an expression in terms of r and θ for $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3}$.

Find, with proof, a similar formula for $\dfrac{\mathrm{d}^ny}{\mathrm{d}x^n}$ in terms of r and θ.

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[Turn over

6 The distinct points P, Q, R and S in the Argand diagram lie on a circle of radius a centred at the origin and are represented by the complex numbers p, q, r and s, respectively. Show that

$$pq = -a^2 \frac{p-q}{p^*-q^*}.$$

Deduce that, if the chords PQ and RS are perpendicular, then $pq + rs = 0$.

The distinct points A_1, A_2, ..., A_n (where $n \geqslant 3$) lie on a circle. The points B_1, B_2, ..., B_n lie on the same circle and are chosen so that the chords B_1B_2, B_2B_3, ..., B_nB_1 are perpendicular, respectively, to the chords A_1A_2, A_2A_3, ..., A_nA_1. Show that, for $n = 3$, there are only two choices of B_1 for which this is possible. What is the corresponding result for $n = 4$? State the corresponding results for values of n greater than 4.

7 The functions $\mathrm{s}(x)$ $(0 \leqslant x < 1)$ and $\mathrm{t}(x)$ $(x \geqslant 0)$, and the real number p, are defined by

$$\mathrm{s}(x) = \int_0^x \frac{1}{\sqrt{1-u^2}}\,\mathrm{d}u\,, \quad \mathrm{t}(x) = \int_0^x \frac{1}{1+u^2}\,\mathrm{d}u\,, \quad p = 2\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u\,.$$

For this question, do not evaluate any of the above integrals explicitly in terms of inverse trigonometric functions or the number π.

(i) Use the substitution $u = v^{-1}$ to show that $\mathrm{t}(x) = \displaystyle\int_{1/x}^\infty \frac{1}{1+v^2}\,\mathrm{d}v$. Hence evaluate $\mathrm{t}(1/x) + \mathrm{t}(x)$ in terms of p and deduce that $2\mathrm{t}(1) = \frac{1}{2}p$.

(ii) Let $y = \dfrac{u}{\sqrt{1+u^2}}$. Express u in terms of y, and show that $\dfrac{\mathrm{d}u}{\mathrm{d}y} = \dfrac{1}{\sqrt{(1-y^2)^3}}$.

By making a substitution in the integral for $\mathrm{t}(x)$, show that

$$\mathrm{t}(x) = \mathrm{s}\left(\frac{x}{\sqrt{1+x^2}}\right).$$

Deduce that $\mathrm{s}\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{4}p$.

(iii) Let $z = \dfrac{u + \frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}u}$. Show that $\mathrm{t}\left(\frac{1}{\sqrt{3}}\right) = \displaystyle\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{1}{1+z^2}\,\mathrm{d}z$, and hence that $3\mathrm{t}\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{2}p$.

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8 **(i)** Find functions $\mathrm{a}(x)$ and $\mathrm{b}(x)$ such that $u = x$ and $u = \mathrm{e}^{-x}$ both satisfy the equation

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \mathrm{a}(x)\frac{\mathrm{d}u}{\mathrm{d}x} + \mathrm{b}(x)u = 0\,.$$

For these functions $\mathrm{a}(x)$ and $\mathrm{b}(x)$, write down the general solution of the equation.

Show that the substitution $y = \dfrac{1}{3u}\dfrac{\mathrm{d}u}{\mathrm{d}x}$ transforms the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 3y^2 + \frac{x}{1+x}y = \frac{1}{3(1+x)} \qquad (*)$$

into

$$\frac{\mathrm{d}^2u}{\mathrm{d}x^2} + \frac{x}{1+x}\frac{\mathrm{d}u}{\mathrm{d}x} - \frac{1}{1+x}u = 0$$

and hence show that the solution of equation $(*)$ that satisfies $y = 0$ at $x = 0$ is given by $y = \dfrac{1-\mathrm{e}^{-x}}{3(x+\mathrm{e}^{-x})}$.

(ii) Find the solution of the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y^2 + \frac{x}{1-x}y = \frac{1}{1-x}$$

that satisfies $y = 2$ at $x = 0$.

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Section B: Mechanics

9 Two small beads, A and B, each of mass m, are threaded on a smooth horizontal circular hoop of radius a and centre O. The angle θ is the acute angle determined by $2\theta = \angle AOB$.

The beads are connected by a light straight spring. The energy stored in the spring is

$$mk^2a^2(\theta - \alpha)^2,$$

where k and α are constants satisfying $k > 0$ and $\frac{\pi}{4} < \alpha < \frac{\pi}{2}$.

The spring is held in compression with $\theta = \beta$ and then released. Find the period of oscillations in the two cases that arise according to the value of β and state the value of β for which oscillations do not occur.

10 A particle is projected from a point on a plane that is inclined at an angle ϕ to the horizontal. The position of the particle at time t after it is projected is (x, y), where $(0, 0)$ is the point of projection, x measures distance up the line of greatest slope and y measures perpendicular distance from the plane. Initially, the velocity of the particle is given by $(\dot{x}, \dot{y}) = (V\cos\theta, V\sin\theta)$, where $V > 0$ and $\phi + \theta < \pi/2$. Write down expressions for x and y.

The particle bounces on the plane and returns along the same path to the point of projection. Show that

$$2\tan\phi\tan\theta = 1$$

and that

$$R = \frac{V^2\cos^2\theta}{2g\sin\phi},$$

where R is the range along the plane.

Show further that

$$\frac{2V^2}{gR} = 3\sin\phi + \operatorname{cosec}\phi$$

and deduce that the largest possible value of R is $V^2/(\sqrt{3}\,g)$.

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11 (i) A wheel consists of a thin light circular rim attached by light spokes of length a to a small hub of mass m. The wheel rolls without slipping on a rough horizontal table directly towards a straight edge of the table. The plane of the wheel is vertical throughout the motion. The speed of the wheel is u, where $u^2 < ag$.

Show that, after the wheel reaches the edge of the table and while it is still in contact with the table, the frictional force on the wheel is zero. Show also that the hub will fall a vertical distance $(ag - u^2)/(3g)$ before the rim loses contact with the table.

(ii) Two particles, each of mass $m/2$, are attached to a light circular hoop of radius a, at the ends of a diameter. The hoop rolls without slipping on a rough horizontal table directly towards a straight edge of the table. The plane of the hoop is vertical throughout the motion. When the centre of the hoop is vertically above the edge of the table it has speed u, where $u^2 < ag$, and one particle is vertically above the other.

Show that, after the hoop reaches the edge of the table and while it is still in contact with the table, the frictional force on the hoop is non-zero and deduce that the hoop will slip before it loses contact with the table.

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[Turn over

Section C: Probability and Statistics

12 I choose a number from the integers 1, 2, ..., $(2n-1)$ and the outcome is the random variable N. Calculate $\mathrm{E}(N)$ and $\mathrm{E}(N^2)$.

I then repeat a certain experiment N times, the outcome of the ith experiment being the random variable X_i $(1 \leqslant i \leqslant N)$. For each i, the random variable X_i has mean μ and variance σ^2, and X_i is independent of X_j for $i \neq j$ and also independent of N. The random variable Y is defined by $Y = \sum_{i=1}^{N} X_i$. Show that $\mathrm{E}(Y) = n\mu$ and that $\mathrm{Cov}(Y, N) = \frac{1}{3}n(n-1)\mu$. Find $\mathrm{Var}(Y)$ in terms of n, σ^2 and μ.

13 A frog jumps towards a large pond. Each jump takes the frog either 1 m or 2 m nearer to the pond. The probability of a 1 m jump is p and the probability of a 2 m jump is q, where $p+q=1$, the occurence of long and short jumps being independent.

(i) Let $p_n(j)$ be the probability that the frog, starting at a point $(n-\frac{1}{2})$ m away from the edge of the pond, lands in the pond for the first time on its jth jump. Show that $p_2(2) = p$.

(ii) Let u_n be the expected number of jumps, starting at a point $(n-\frac{1}{2})$ m away from the edge of the pond, required to land in the pond for the first time. Write down the value of u_1. By finding first the relevant values of $p_n(m)$, calculate u_2 and show that $u_3 = 3 - 2q + q^2$.

(iii) Given that u_n can be expressed in the form $u_n = A(-q)^{n-1} + B + Cn$, where A, B and C are constants (independent of n), show that $C = (1+q)^{-1}$ and find A and B in terms of q. Hence show that, for large n, $u_n \approx \dfrac{n}{p+2q}$ and explain carefully why this result is to be expected.

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14 (i) My favourite dartboard is a disc of unit radius and centre O. I never miss the board, and the probability of my hitting any given area of the dartboard is proportional to the area. Each throw is independent of any other throw. I throw a dart n times (where $n > 1$). Find the expected area of the smallest circle, with centre O, that encloses all the n holes made by my dart.

Find also the expected area of the smallest circle, with centre O, that encloses all the $(n-1)$ holes nearest to O.

(ii) My other dartboard is a square of side 2 units, with centre Q. I never miss the board, and the probability of my hitting any given area of the dartboard is proportional to the area. Each throw is independent of any other throw. I throw a dart n times (where $n > 1$). Find the expected area of the smallest square, with centre Q, that encloses all the n holes made by my dart.

(iii) Determine, without detailed calculations, whether the expected area of the smallest circle, with centre Q, on my square dartboard that encloses all the n holes made by my darts is larger or smaller than that for my circular dartboard.

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CAMBRIDGE ASSESSMENT

Sixth Term Examination Papers

MATHEMATICS 3 **9475**

MONDAY 23 JUNE 2008 Afternoon

Time: 3 hours

Additional materials: Answer paper
Graph paper
Formulae booklet

Candidates may not use electronic calculators

5038753228

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Write your name, Centre number and candidate number in the spaces on the answer booklet.

Begin each answer on a new page.

INFORMATION FOR CANDIDATES

Each question is marked out of 20. There is no restriction of choice.

You will be assessed on the **six** questions for which you gain the highest marks.

You are advised to concentrate on no more than **six** questions. Little credit will be given for fragmentary answers.

You are provided with Mathematical Formulae and Tables.

Electronic calculators are not permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 7 printed pages and 1 blank page.

SP (SLM) T70357
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[Turn over

Section A: Pure Mathematics

1 Find all values of a, b, x and y that satisfy the simultaneous equations

$$\begin{aligned} a + b &= 1 \\ ax + by &= \tfrac{1}{3} \\ ax^2 + by^2 &= \tfrac{1}{5} \\ ax^3 + by^3 &= \tfrac{1}{7}\,. \end{aligned}$$

[**Hint**: you may wish to start by multiplying the second equation by $x + y$.]

2 Let $S_k(n) \equiv \sum_{r=0}^{n} r^k$, where k is a positive integer, so that

$$S_1(n) \equiv \tfrac{1}{2}n(n+1) \qquad \text{and} \qquad S_2(n) \equiv \tfrac{1}{6}n(n+1)(2n+1)\,.$$

(i) By considering $\sum_{r=0}^{n} \left[(r+1)^k - r^k\right]$, show that

$$kS_{k-1}(n) = (n+1)^k - (n+1) - \binom{k}{2}S_{k-2}(n) - \binom{k}{3}S_{k-3}(n) - \cdots - \binom{k}{k-1}S_1(n)\,. \quad (*)$$

Obtain simplified expressions for $S_3(n)$ and $S_4(n)$.

(ii) Explain, using $(*)$, why $S_k(n)$ is a polynomial of degree $k+1$ in n. Show that in this polynomial the constant term is zero and the sum of the coefficients is 1.

3 The point $P(a\cos\theta\,,\, b\sin\theta)$, where $a > b > 0$, lies on the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1\,.$$

The point $S(-ea\,,\,0)$, where $b^2 = a^2(1-e^2)$, is a focus of the ellipse. The point N is the foot of the perpendicular from the origin, O, to the tangent to the ellipse at P. The lines SP and ON intersect at T. Show that the y-coordinate of T is

$$\frac{b\sin\theta}{1 + e\cos\theta}\,.$$

Show that T lies on the circle with centre S and radius a.

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4 **(i)** Show, with the aid of a sketch, that $y > \tanh(y/2)$ for $y > 0$ and deduce that

$$\operatorname{arcosh} x > \frac{x-1}{\sqrt{x^2-1}} \quad \text{for} \quad x > 1. \qquad (*)$$

(ii) By integrating $(*)$, show that $\operatorname{arcosh} x > 2\dfrac{x-1}{\sqrt{x^2-1}}$ for $x > 1$.

(iii) Show that $\operatorname{arcosh} x > 3\dfrac{\sqrt{x^2-1}}{x+2}$ for $x > 1$.

[**Note:** $\operatorname{arcosh} x$ is another notation for $\cosh^{-1} x$.]

5 The functions $\mathrm{T}_n(x)$, for $n = 0, 1, 2, \ldots$, satisfy the recurrence relation

$$\mathrm{T}_{n+1}(x) - 2x\mathrm{T}_n(x) + \mathrm{T}_{n-1}(x) = 0 \qquad (n \geqslant 1). \qquad (*)$$

Show by induction that

$$(\mathrm{T}_n(x))^2 - \mathrm{T}_{n-1}(x)\mathrm{T}_{n+1}(x) = \mathrm{f}(x)\,,$$

where $\mathrm{f}(x) = (\mathrm{T}_1(x))^2 - \mathrm{T}_0(x)\mathrm{T}_2(x)$.

In the case $\mathrm{f}(x) \equiv 0$, determine (with proof) an expression for $\mathrm{T}_n(x)$ in terms of $\mathrm{T}_0(x)$ (assumed to be non-zero) and $\mathrm{r}(x)$, where $\mathrm{r}(x) = \mathrm{T}_1(x)/\mathrm{T}_0(x)$. Find the two possible expressions for $\mathrm{r}(x)$ in terms of x.

6 In this question, p denotes $\dfrac{\mathrm{d}y}{\mathrm{d}x}$.

(i) Given that

$$y = p^2 + 2xp\,,$$

show by differentiating with respect to x that

$$\frac{\mathrm{d}x}{\mathrm{d}p} = -2 - \frac{2x}{p}.$$

Hence show that $x = -\frac{2}{3}p + Ap^{-2}$, where A is an arbitrary constant.

Find y in terms of x if $p = -3$ when $x = 2$.

(ii) Given instead that

$$y = 2xp + p\ln p\,,$$

and that $p = 1$ when $x = -\frac{1}{4}$, show that $x = -\frac{1}{2}\ln p - \frac{1}{4}$ and find y in terms of x.

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[Turn over

7 The points A, B and C in the Argand diagram are the vertices of an equilateral triangle described anticlockwise. Show that the complex numbers a, b and c representing A, B and C satisfy

$$2c = (a + b) + i\sqrt{3}(b - a).$$

Find a similar relation in the case that A, B and C are the vertices of an equilateral triangle described clockwise.

(i) The quadrilateral $DEFG$ lies in the Argand diagram. Show that points P, Q, R and S can be chosen so that PDE, QEF, RFG and SGD are equilateral triangles and $PQRS$ is a parallelogram.

(ii) The triangle LMN lies in the Argand diagram. Show that the centroids U, V and W of the equilateral triangles drawn externally on the sides of LMN are the vertices of an equilateral triangle.

[**Note:** The *centroid* of a triangle with vertices represented by the complex numbers x, y and z is the point represented by $\frac{1}{3}(x + y + z)$.]

8 **(i)** The coefficients in the series

$$S = \tfrac{1}{3}x + \tfrac{1}{6}x^2 + \tfrac{1}{12}x^3 + \cdots + a_r x^r + \cdots$$

satisfy a recurrence relation of the form $a_{r+1} + pa_r = 0$. Write down the value of p.

By considering $(1 + px)S$, find an expression for the sum to infinity of S (assuming that it exists). Find also an expression for the sum of the first $n + 1$ terms of S.

(ii) The coefficients in the series

$$T = 2 + 8x + 18x^2 + 37x^3 + \cdots + a_r x^r + \cdots$$

satisfy a recurrence relation of the form $a_{r+2} + pa_{r+1} + qa_r = 0$. Find an expression for the sum to infinity of T (assuming that it exists). By expressing T in partial fractions, or otherwise, find an expression for the sum of the first $n + 1$ terms of T.

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Section B: Mechanics

9 A particle of mass m is initially at rest on a rough horizontal surface. The particle experiences a force $mg\sin\pi t$, where t is time, acting in a fixed horizontal direction. The coefficient of friction between the particle and the surface is μ. Given that the particle starts to move first at $t = T_0$, state the relation between T_0 and μ.

(i) For $\mu = \mu_0$, the particle comes to rest for the first time at $t = 1$. Sketch the acceleration-time graph for $0 \leqslant t \leqslant 1$. Show that

$$1 + \left(1 - \mu_0^2\right)^{\frac{1}{2}} - \mu_0\pi + \mu_0 \arcsin \mu_0 = 0\,.$$

(ii) For $\mu = \mu_0$ sketch the acceleration-time graph for $0 \leqslant t \leqslant 3$. Describe the motion of the particle in this case and in the case $\mu = 0$.

[**Note:** $\arcsin x$ is another notation for $\sin^{-1} x$.]

10 A long string consists of n short light strings joined together, each of natural length ℓ and modulus of elasticity λ. It hangs vertically at rest, suspended from one end. Each of the short strings has a particle of mass m attached to its lower end. The short strings are numbered 1 to n, the nth short string being at the top. By considering the tension in the rth short string, determine the length of the long string. Find also the elastic energy stored in the long string.

A uniform heavy rope of mass M and natural length L_0 has modulus of elasticity λ. The rope hangs vertically at rest, suspended from one end. Show that the length, L, of the rope is given by

$$L = L_0\left(1 + \frac{Mg}{2\lambda}\right),$$

and find an expression in terms of L, L_0 and λ for the elastic energy stored in the rope.

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[Turn over

11 A circular wheel of radius r has moment of inertia I about its axle, which is fixed in a horizontal position. A light string is wrapped around the circumference of the wheel and a particle of mass m hangs from the free end. The system is released from rest and the particle descends. The string does not slip on the wheel.

As the particle descends, the wheel turns through n_1 revolutions, and the string then detaches from the wheel. At this moment, the angular speed of the wheel is ω_0. The wheel then turns through a further n_2 revolutions, in time T, before coming to rest. The couple on the wheel due to resistance is constant.

Show that

$$\frac{1}{2}\omega_0 T = 2\pi n_2$$

and

$$I = \frac{mgrn_1T^2 - 4\pi mr^2n_2^2}{4\pi n_2(n_1 + n_2)} .$$

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Section C: Probability and Statistics

12 Let X be a random variable with a Laplace distribution, so that its probability density function is given by

$$\mathrm{f}(x) = \frac{1}{2}\mathrm{e}^{-|x|}\,, \qquad -\infty < x < \infty\,. \qquad (*)$$

Sketch $\mathrm{f}(x)$. Show that its moment generating function $\mathrm{M}_X(\theta)$ is given by $\mathrm{M}_X(\theta) = (1-\theta^2)^{-1}$ and hence find the variance of X.

A frog is jumping up and down, attempting to land on the same spot each time. In fact, in each of n successive jumps he always lands on a fixed straight line but when he lands from the ith jump ($i = 1, 2, \ldots, n$) his displacement from the point from which he jumped is X_i cm, where X_i has the distribution $(*)$. His displacement from his starting point after n jumps is Y cm (so that $Y = \sum_{i=1}^{n} X_i$). Each jump is independent of the others.

Obtain the moment generating function for $Y/\sqrt{2n}$ and, by considering its logarithm, show that this moment generating function tends to $\exp(\frac{1}{2}\theta^2)$ as $n \to \infty$.

Given that $\exp(\frac{1}{2}\theta^2)$ is the moment generating function of the standard Normal random variable, estimate the least number of jumps such that there is a 5% chance that the frog lands 25 cm or more from his starting point.

13 A box contains n pieces of string, each of which has two ends. I select two string ends at random and tie them together. This creates either a ring (if the two ends are from the same string) or a longer piece of string. I repeat the process of tying together string ends chosen at random until there are none left.

Find the expected number of rings created at the first step and hence obtain an expression for the expected number of rings created by the end of the process. Find also an expression for the variance of the number of rings created.

Given that $\ln 20 \approx 3$ and that $1 + \frac{1}{2} + \cdots + \frac{1}{n} \approx \ln n$ for large n, determine approximately the expected number of rings created in the case $n = 40\,000$.

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Section A: Pure Mathematics

1 The points S, T, U and V have coordinates (s, ms), (t, mt), (u, nu) and (v, nv), respectively. The lines SV and UT meet the line $y = 0$ at the points with coordinates $(p, 0)$ and $(q, 0)$, respectively. Show that

$$p = \frac{(m-n)sv}{ms - nv}\,,$$

and write down a similar expression for q.

Given that S and T lie on the circle $x^2 + (y-c)^2 = r^2$, find a quadratic equation satisfied by s and by t, and hence determine st and $s + t$ in terms of m, c and r.

Given that S, T, U and V lie on the above circle, show that $p + q = 0$.

2 **(i)** Let $y = \sum_{n=0}^{\infty} a_n x^n$, where the coefficients a_n are independent of x and are such that this series and all others in this question converge. Show that

$$y' = \sum_{n=1}^{\infty} n a_n x^{n-1}\,,$$

and write down a similar expression for y''.

Write out explicitly each of the three series as far as the term containing a_3.

(ii) It is given that y satisfies the differential equation

$$xy'' - y' + 4x^3 y = 0\,.$$

By substituting the series of part (i) into the differential equation and comparing coefficients, show that $a_1 = 0$.

Show that, for $n \geqslant 4$,

$$a_n = -\frac{4}{n(n-2)}\, a_{n-4}\,,$$

and that, if $a_0 = 1$ and $a_2 = 0$, then $y = \cos(x^2)$.

Find the corresponding result when $a_0 = 0$ and $a_2 = 1$.

3 The function $\mathrm{f}(t)$ is defined, for $t \neq 0$, by

$$\mathrm{f}(t) = \frac{t}{\mathrm{e}^t - 1}\,.$$

(i) By expanding e^t, show that $\lim_{t\to 0} \mathrm{f}(t) = 1$. Find $\mathrm{f}'(t)$ and evaluate $\lim_{t\to 0} \mathrm{f}'(t)$.

(ii) Show that $\mathrm{f}(t) + \frac{1}{2}t$ is an even function. [**Note:** A function $\mathrm{g}(t)$ is said to be *even* if $\mathrm{g}(t) \equiv \mathrm{g}(-t)$.]

(iii) Show with the aid of a sketch that $\mathrm{e}^t(1-t) \leqslant 1$ and deduce that $\mathrm{f}'(t) \neq 0$ for $t \neq 0$.

Sketch the graph of $\mathrm{f}(t)$.

4 For any given (suitable) function f, the *Laplace transform* of f is the function F defined by

$$\mathrm{F}(s) = \int_0^\infty \mathrm{e}^{-st}\mathrm{f}(t)\mathrm{d}t \qquad (s > 0)\,.$$

(i) Show that the Laplace transform of $\mathrm{e}^{-bt}\mathrm{f}(t)$, where $b > 0$, is $\mathrm{F}(s+b)$.

(ii) Show that the Laplace transform of $\mathrm{f}(at)$, where $a > 0$, is $a^{-1}\mathrm{F}(\frac{s}{a})$.

(iii) Show that the Laplace transform of $\mathrm{f}'(t)$ is $s\mathrm{F}(s) - \mathrm{f}(0)$.

(iv) In the case $\mathrm{f}(t) = \sin t$, show that $\mathrm{F}(s) = \dfrac{1}{s^2+1}$.

Using only these four results, find the Laplace transform of $\mathrm{e}^{-pt}\cos qt$, where $p > 0$ and $q > 0$.

5 The numbers x, y and z satisfy

$$\begin{aligned} x+y+z &= 1 \\ x^2+y^2+z^2 &= 2 \\ x^3+y^3+z^3 &= 3\,. \end{aligned}$$

Show that

$$yz+zx+xy = -\frac{1}{2}\,.$$

Show also that $x^2y+x^2z+y^2z+y^2x+z^2x+z^2y = -1$, and hence that

$$xyz = \frac{1}{6}\,.$$

Let $S_n = x^n+y^n+z^n$. Use the above results to find numbers a, b and c such that the relation

$$S_{n+1} = aS_n + bS_{n-1} + cS_{n-2}\,,$$

holds for all n.

6 Show that $\left|\mathrm{e}^{\mathrm{i}\beta}-\mathrm{e}^{\mathrm{i}\alpha}\right| = 2\sin\frac{1}{2}(\beta-\alpha)$ for $0<\alpha<\beta<2\pi$. Hence show that

$$\left|\mathrm{e}^{\mathrm{i}\alpha}-\mathrm{e}^{\mathrm{i}\beta}\right|\left|\mathrm{e}^{\mathrm{i}\gamma}-\mathrm{e}^{\mathrm{i}\delta}\right| + \left|\mathrm{e}^{\mathrm{i}\beta}-\mathrm{e}^{\mathrm{i}\gamma}\right|\left|\mathrm{e}^{\mathrm{i}\alpha}-\mathrm{e}^{\mathrm{i}\delta}\right| = \left|\mathrm{e}^{\mathrm{i}\alpha}-\mathrm{e}^{\mathrm{i}\gamma}\right|\left|\mathrm{e}^{\mathrm{i}\beta}-\mathrm{e}^{\mathrm{i}\delta}\right|,$$

where $0<\alpha<\beta<\gamma<\delta<2\pi$.

Interpret this result as a theorem about cyclic quadrilaterals.

7 **(i)** The functions $\mathrm{f}_n(x)$ are defined for $n = 0,\ 1,\ 2,\ \ldots$, by

$$\mathrm{f}_0(x) = \frac{1}{1+x^2} \qquad \text{and} \qquad \mathrm{f}_{n+1}(x) = \frac{\mathrm{d}\mathrm{f}_n(x)}{\mathrm{d}x}\,.$$

Prove, for $n \geqslant 1$, that

$$(1+x^2)\mathrm{f}_{n+1}(x) + 2(n+1)x\mathrm{f}_n(x) + n(n+1)\mathrm{f}_{n-1}(x) = 0\,.$$

(ii) The functions $\mathrm{P}_n(x)$ are defined for $n = 0,\ 1,\ 2,\ \ldots$, by

$$\mathrm{P}_n(x) = (1+x^2)^{n+1}\mathrm{f}_n(x)\,.$$

Find expressions for $\mathrm{P}_0(x)$, $\mathrm{P}_1(x)$ and $\mathrm{P}_2(x)$.

Prove, for $n \geqslant 0$, that

$$\mathrm{P}_{n+1}(x) - (1+x^2)\frac{\mathrm{d}\mathrm{P}_n(x)}{\mathrm{d}x} + 2(n+1)x\mathrm{P}_n(x) = 0\,,$$

and that $\mathrm{P}_n(x)$ is a polynomial of degree n.

8 Let m be a positive integer and let n be a non-negative integer.

(i) Use the result $\lim\limits_{t\to\infty} \mathrm{e}^{-mt}t^n = 0$ to show that

$$\lim_{x\to 0} x^m(\ln x)^n = 0\,.$$

By writing x^x as $\mathrm{e}^{x\ln x}$ show that

$$\lim_{x\to 0} x^x = 1\,.$$

(ii) Let $I_n = \displaystyle\int_0^1 x^m(\ln x)^n \mathrm{d}x$. Show that

$$I_{n+1} = -\frac{n+1}{m+1}I_n$$

and hence evaluate I_n.

(iii) Show that

$$\int_0^1 x^x \mathrm{d}x = 1 - \left(\tfrac{1}{2}\right)^2 + \left(\tfrac{1}{3}\right)^3 - \left(\tfrac{1}{4}\right)^4 + \cdots\,.$$

Section B: Mechanics

9 A particle is projected under gravity from a point P and passes through a point Q. The angles of the trajectory with the positive horizontal direction at P and at Q are θ and ϕ, respectively. The angle of elevation of Q from P is α.

(i) Show that $\tan\theta + \tan\phi = 2\tan\alpha$.

(ii) It is given that there is a second trajectory from P to Q with the same speed of projection. The angles of this trajectory with the positive horizontal direction at P and at Q are θ' and ϕ', respectively. By considering a quadratic equation satisfied by $\tan\theta$, show that $\tan(\theta + \theta') = -\cot\alpha$. Show also that $\theta + \theta' = \pi + \phi + \phi'$.

10 A light spring is fixed at its lower end and its axis is vertical. When a certain particle P rests on the top of the spring, the compression is d. When, instead, P is dropped onto the top of the spring from a height h above it, the compression at time t after P hits the top of the spring is x. Obtain a second-order differential equation relating x and t for $0 \leqslant t \leqslant T$, where T is the time at which P first loses contact with the spring.

Find the solution of this equation in the form

$$x = A + B\cos(\omega t) + C\sin(\omega t)\,,$$

where the constants A, B, C and ω are to be given in terms of d, g and h as appropriate.

Show that

$$T = \sqrt{d/g}\,\left(2\pi - 2\arctan\sqrt{2h/d}\,\right)\,.$$

11 A comet in deep space picks up mass as it travels through a large stationary dust cloud. It is subject to a gravitational force of magnitude Mf acting in the direction of its motion. When it entered the cloud, the comet had mass M and speed V. After a time t, it has travelled a distance x through the cloud, its mass is $M(1 + bx)$, where b is a positive constant, and its speed is v.

(i) In the case when $f = 0$, write down an equation relating V, x, v and b. Hence find an expression for x in terms of b, V and t.

(ii) In the case when f is a non-zero constant, use Newton's second law in the form

$$\text{force} = \text{rate of change of momentum}$$

to show that

$$v = \frac{ft + V}{1 + bx}\,.$$

Hence find an expression for x in terms of b, V, f and t.

Show that it is possible, if b, V and f are suitably chosen, for the comet to move with constant speed. Show also that, if the comet does not move with constant speed, its speed tends to a constant as $t \to \infty$.

Section C: Probability and Statistics

12 **(i)** Albert tosses a fair coin k times, where k is a given positive integer. The number of heads he gets is X_1. He then tosses the coin X_1 times, getting X_2 heads. He then tosses the coin X_2 times, getting X_3 heads. The random variables X_4, X_5, ... are defined similarly. Write down $\mathrm{E}(X_1)$.

By considering $\mathrm{E}(X_2 \mid X_1 = x_1)$, or otherwise, show that $\mathrm{E}(X_2) = \frac{1}{4}k$.

Find $\displaystyle\sum_{i=1}^{\infty} \mathrm{E}(X_i)$.

(ii) Bertha has k fair coins. She tosses the first coin until she gets a tail. The number of heads she gets before the first tail is Y_1. She then tosses the second coin until she gets a tail and the number of heads she gets with this coin before the first tail is Y_2. The random variables Y_3, Y_4, ... , Y_k are defined similarly, and $Y = \displaystyle\sum_{i=1}^{k} Y_i$.

Obtain the probability generating function of Y, and use it to find $\mathrm{E}(Y)$, $\mathrm{Var}\,(Y)$ and $\mathrm{P}(Y = r)$.

13 **(i)** The point P lies on the circumference of a circle of unit radius and centre O. The angle, θ, between OP and the positive x-axis is a random variable, uniformly distributed on the interval $0 \leqslant \theta < 2\pi$. The cartesian coordinates of P with respect to O are (X, Y). Find the probability density function for X, and calculate $\mathrm{Var}\,(X)$.

Show that X and Y are uncorrelated and discuss briefly whether they are independent.

(ii) The points P_i ($i = 1, 2, \ldots, n$) are chosen independently on the circumference of the circle, as in part (i), and have cartesian coordinates (X_i, Y_i). The point $\overline{P}$ has coordinates $(\overline{X}, \overline{Y})$, where $\overline{X} = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} X_i$ and $\overline{Y} = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} Y_i$. Show that $\overline{X}$ and $\overline{Y}$ are uncorrelated.

Show that, for large n, $\mathrm{P}\left(|\overline{X}| \leqslant \sqrt{\dfrac{2}{n}}\right) \approx 0.95$.

CAMBRIDGE ASSESSMENT

Sixth Term Examination Papers **9475**

MATHEMATICS 3 Afternoon

MONDAY 21 JUNE 2010 Time: 3 hours

9309140880

Additional Materials: Answer Paper
Formulae Booklet

Candidates may **not** use a calculator

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Write your name, centre number and candidate number in the spaces on the answer booklet.

Begin each answer on a new page.

INFORMATION FOR CANDIDATES

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You will be assessed on the **six** questions for which you gain the highest marks.

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You are provided with a Mathematical Formulae Booklet.

Calculators are not permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 8 printed pages and 4 blank pages.

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Section A: Pure Mathematics

1 Let x_1, x_2, …, x_n and x_{n+1} be any fixed real numbers. The numbers A and B are defined by

$$A = \frac{1}{n}\sum_{k=1}^{n} x_k\,, \quad B = \frac{1}{n}\sum_{k=1}^{n}(x_k - A)^2\,,$$

and the numbers C and D are defined by

$$C = \frac{1}{n+1}\sum_{k=1}^{n+1} x_k\,, \quad D = \frac{1}{n+1}\sum_{k=1}^{n+1}(x_k - C)^2\,.$$

(i) Express C in terms of A, x_{n+1} and n.

(ii) Show that $B = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} x_k^2 - A^2$.

(iii) Express D in terms of B, A, x_{n+1} and n.
Hence show that $(n+1)D \geqslant nB$ for all values of x_{n+1}, but that $D < B$ if and only if

$$A - \sqrt{\frac{(n+1)B}{n}} < x_{n+1} < A + \sqrt{\frac{(n+1)B}{n}}\,.$$

2 In this question, a is a positive constant.

(i) Express $\cosh a$ in terms of exponentials.
By using partial fractions, prove that

$$\int_0^1 \frac{1}{x^2 + 2x\cosh a + 1}\,\mathrm{d}x = \frac{a}{2\sinh a}\,.$$

(ii) Find, expressing your answers in terms of hyperbolic functions,

$$\int_1^\infty \frac{1}{x^2 + 2x\sinh a - 1}\,\mathrm{d}x$$

and

$$\int_0^\infty \frac{1}{x^4 + 2x^2\cosh a + 1}\,\mathrm{d}x\,.$$

3 For any given positive integer n, a number a (which may be complex) is said to be a *primitive nth root of unity* if $a^n = 1$ and there is no integer m such that $0 < m < n$ and $a^m = 1$. Write down the two primitive 4th roots of unity.

Let $\mathrm{C}_n(x)$ be the polynomial such that the roots of the equation $\mathrm{C}_n(x) = 0$ are the primitive nth roots of unity, the coefficient of the highest power of x is one and the equation has no repeated roots. Show that $\mathrm{C}_4(x) = x^2 + 1$.

(i) Find $\mathrm{C}_1(x)$, $\mathrm{C}_2(x)$, $\mathrm{C}_3(x)$, $\mathrm{C}_5(x)$ and $\mathrm{C}_6(x)$, giving your answers as unfactorised polynomials.

(ii) Find the value of n for which $\mathrm{C}_n(x) = x^4 + 1$.

(iii) Given that p is prime, find an expression for $\mathrm{C}_p(x)$, giving your answer as an unfactorised polynomial.

(iv) Prove that there are no positive integers q, r and s such that $\mathrm{C}_q(x) \equiv \mathrm{C}_r(x)\mathrm{C}_s(x)$.

4 **(i)** The number α is a common root of the equations $x^2 + ax + b = 0$ and $x^2 + cx + d = 0$ (that is, α satisfies both equations). Given that $a \neq c$, show that

$$\alpha = -\frac{b-d}{a-c}.$$

Hence, or otherwise, show that the equations have at least one common root if and only if

$$(b-d)^2 - a(b-d)(a-c) + b(a-c)^2 = 0.$$

Does this result still hold if the condition $a \neq c$ is not imposed?

(ii) Show that the equations $x^2 + ax + b = 0$ and $x^3 + (a+1)x^2 + qx + r = 0$ have at least one common root if and only if

$$(b-r)^2 - a(b-r)(a+b-q) + b(a+b-q)^2 = 0.$$

Hence, or otherwise, find the values of b for which the equations $2x^2 + 5x + 2b = 0$ and $2x^3 + 7x^2 + 5x + 1 = 0$ have at least one common root.

[Turn over

5 The vertices A, B, C and D of a square have coordinates $(0, 0)$, $(a, 0)$, (a, a) and $(0, a)$, respectively. The points P and Q have coordinates $(an, 0)$ and $(0, am)$ respectively, where $0 < m < n < 1$. The line CP produced meets DA produced at R and the line CQ produced meets BA produced at S. The line PQ produced meets the line RS produced at T. Show that TA is perpendicular to AC.

Explain how, given a square of area a^2, a square of area $2a^2$ may be constructed using only a straight-edge.

[**Note**: a straight-edge is a ruler with no markings on it; no measurements (and no use of compasses) are allowed in the construction.]

6 The points P, Q and R lie on a sphere of unit radius centred at the origin, O, which is fixed. Initially, P is at $P_0(1, 0, 0)$, Q is at $Q_0(0, 1, 0)$ and R is at $R_0(0, 0, 1)$.

(i) The sphere is then rotated about the z-axis, so that the line OP turns directly towards the positive y-axis through an angle ϕ. The position of P after this rotation is denoted by P_1. Write down the coordinates of P_1.

(ii) The sphere is now rotated about the line in the x-y plane perpendicular to OP_1, so that the line OP turns directly towards the positive z-axis through an angle λ. The position of P after this rotation is denoted by P_2. Find the coordinates of P_2. Find also the coordinates of the points Q_2 and R_2, which are the positions of Q and R after the two rotations.

(iii) The sphere is now rotated for a third time, so that P returns from P_2 to its original position P_0. During the rotation, P remains in the plane containing P_0, P_2 and O. Show that the angle of this rotation, θ, satisfies

$$\cos\theta = \cos\phi\cos\lambda\,,$$

and find a vector in the direction of the axis about which this rotation takes place.

7 Given that $y = \cos(m \arcsin x)$, for $|x| < 1$, prove that

$$(1 - x^2)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - x\frac{\mathrm{d}y}{\mathrm{d}x} + m^2y = 0\,.$$

Obtain a similar equation relating $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3}$, $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, and a similar equation relating $\dfrac{\mathrm{d}^4y}{\mathrm{d}x^4}$, $\dfrac{\mathrm{d}^3y}{\mathrm{d}x^3}$ and $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2}$.

Conjecture and prove a relation between $\dfrac{\mathrm{d}^{n+2}y}{\mathrm{d}x^{n+2}}$, $\dfrac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}}$ and $\dfrac{\mathrm{d}^ny}{\mathrm{d}x^n}$.

Obtain the first three non-zero terms of the Maclaurin series for y. Show that, if m is an even integer, $\cos m\theta$ may be written as a polynomial in $\sin\theta$ beginning

$$1 - \frac{m^2\sin^2\theta}{2!} + \frac{m^2(m^2 - 2^2)\sin^4\theta}{4!} - \cdots\,. \qquad (|\theta| < \tfrac{1}{2}\pi)$$

State the degree of the polynomial.

8 Given that $\mathrm{P}(x) = \mathrm{Q}(x)\mathrm{R}'(x) - \mathrm{Q}'(x)\mathrm{R}(x)$, write down an expression for

$$\int \frac{\mathrm{P}(x)}{\big(\mathrm{Q}(x)\big)^2}\,\mathrm{d}x\,.$$

(i) By choosing the function $\mathrm{R}(x)$ to be of the form $a + bx + cx^2$, find

$$\int \frac{5x^2 - 4x - 3}{(1 + 2x + 3x^2)^2}\,\mathrm{d}x\,.$$

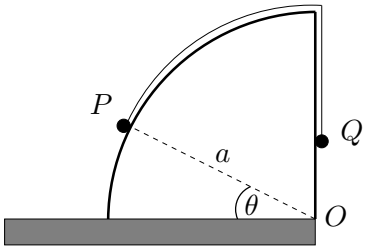
Show that the choice of $\mathrm{R}(x)$ is not unique and, by comparing the two functions $\mathrm{R}(x)$ corresponding to two different values of a, explain how the different choices are related.

(ii) Find the general solution of

$$(1 + \cos x + 2\sin x)\frac{\mathrm{d}y}{\mathrm{d}x} + (\sin x - 2\cos x)y = 5 - 3\cos x + 4\sin x\,.$$

Section B: Mechanics

9



The diagram shows two particles, P and Q, connected by a light inextensible string which passes over a smooth block fixed to a horizontal table. The cross-section of the block is a quarter circle with centre O, which is at the edge of the table, and radius a. The angle between OP and the table is θ. The masses of P and Q are m and M, respectively, where $m < M$.

Initially, P is held at rest on the table and in contact with the block, Q is vertically above O, and the string is taut. Then P is released. Given that, in the subsequent motion, P remains in contact with the block as θ increases from 0 to $\frac{1}{2}\pi$, find an expression, in terms of m, M, θ and g, for the normal reaction of the block on P and show that

$$\frac{m}{M} \geqslant \frac{\pi - 1}{3}\,.$$

10 A small bead B, of mass m, slides without friction on a fixed horizontal ring of radius a. The centre of the ring is at O. The bead is attached by a light elastic string to a fixed point P in the plane of the ring such that $OP = b$, where $b > a$. The natural length of the elastic string is c, where $c < b - a$, and its modulus of elasticity is λ. Show that the equation of motion of the bead is

$$ma\ddot{\phi} = -\lambda\left(\frac{a\sin\phi}{c\sin\theta} - 1\right)\sin(\theta + \phi)\,,$$

where $\theta = \angle BPO$ and $\phi = \angle BOP$.

Given that θ and ϕ are small, show that $a(\theta + \phi) \approx b\theta$. Hence find the period of small oscillations about the equilibrium position $\theta = \phi = 0$.

11 A bullet of mass m is fired horizontally with speed u into a wooden block of mass M at rest on a horizontal surface. The coefficient of friction between the block and the surface is μ. While the bullet is moving through the block, it experiences a constant force of resistance to its motion of magnitude R, where $R > (M+m)\mu g$. The bullet moves horizontally in the block and does not emerge from the other side of the block.

(i) Show that the magnitude, a, of the deceleration of the bullet relative to the block while the bullet is moving through the block is given by

$$a = \frac{R}{m} + \frac{R - (M+m)\mu g}{M}.$$

(ii) Show that the common speed, v, of the block and bullet when the bullet stops moving through the block satisfies

$$av = \frac{Ru - (M+m)\mu g u}{M}.$$

(iii) Obtain an expression, in terms of u, v and a, for the distance moved by the block while the bullet is moving through the block.

(iv) Show that the total distance moved by the block is

$$\frac{muv}{2(M+m)\mu g}.$$

Describe briefly what happens if $R < (M+m)\mu g$.

Section C: Probability and Statistics

12 The infinite series S is given by

$$S = 1 + (1 + d)r + (1 + 2d)r^2 + \cdots + (1 + nd)r^n + \cdots ,$$

for $|r| < 1$. By considering $S - rS$, or otherwise, prove that

$$S = \frac{1}{1 - r} + \frac{rd}{(1 - r)^2} .$$

Arthur and Boadicea shoot arrows at a target. The probability that an arrow shot by Arthur hits the target is a; the probability that an arrow shot by Boadicea hits the target is b. Each shot is independent of all others. Prove that the expected number of shots it takes Arthur to hit the target is $1/a$.

Arthur and Boadicea now have a contest. They take alternate shots, with Arthur going first. The winner is the one who hits the target first. The probability that Arthur wins the contest is α and the probability that Boadicea wins is β. Show that

$$\alpha = \frac{a}{1 - a'b'} ,$$

where $a' = 1 - a$ and $b' = 1 - b$, and find β.

Show that the expected number of shots in the contest is $\dfrac{\alpha}{a} + \dfrac{\beta}{b}$.

13 In this question, $\mathrm{Corr}(U, V)$ denotes the product moment correlation coefficient between the random variables U and V, defined by

$$\mathrm{Corr}(U, V) \equiv \frac{\mathrm{Cov}(U, V)}{\sqrt{\mathrm{Var}\,(U)\mathrm{Var}\,(V)}} .$$

The independent random variables Z_1, Z_2 and Z_3 each have expectation 0 and variance 1. What is the value of $\mathrm{Corr}(Z_1, Z_2)$?

Let $Y_1 = Z_1$ and let

$$Y_2 = \rho_{12}Z_1 + (1 - \rho_{12}^2)^{\frac{1}{2}} Z_2 ,$$

where ρ_{12} is a given constant with $-1 < \rho_{12} < 1$. Find $\mathrm{E}(Y_2)$, $\mathrm{Var}\,(Y_2)$ and $\mathrm{Corr}(Y_1, Y_2)$.

Now let $Y_3 = aZ_1 + bZ_2 + cZ_3$, where a, b and c are real constants and $c \geqslant 0$. Given that $\mathrm{E}(Y_3) = 0$, $\mathrm{Var}\,(Y_3) = 1$, $\mathrm{Corr}(Y_1, Y_3) = \rho_{13}$ and $\mathrm{Corr}(Y_2, Y_3) = \rho_{23}$, express a, b and c in terms of ρ_{23}, ρ_{13} and ρ_{12}.

Given constants μ_i and σ_i, for $i = 1$, 2 and 3, give expressions in terms of the Y_i for random variables X_i such that $\mathrm{E}(X_i) = \mu_i$, $\mathrm{Var}\,(X_i) = \sigma_i^2$ and $\mathrm{Corr}(X_i, X_j) = \rho_{ij}$.

CAMBRIDGE ASSESSMENT

Sixth Term Examination Papers **9475**

MATHEMATICS 3 Afternoon

WEDNESDAY 22 JUNE 2011 Time: 3 hours

7926693564

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Write your name, centre number and candidate number in the spaces on the answer booklet.

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Please wait to be told you may begin before turning this page.

This question paper consists of 8 printed pages and 4 blank pages.

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Section A: Pure Mathematics

1 **(i)** Find the general solution of the differential equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} - \left(\frac{x+2}{x+1}\right) u = 0\,.$$

(ii) Show that substituting $y = z\mathrm{e}^{-x}$ (where z is a function of x) into the second order differential equation

$$(x+1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - y = 0 \qquad (*)$$

leads to a first order differential equation for $\dfrac{\mathrm{d}z}{\mathrm{d}x}$. Find z and hence show that the general solution of $(*)$ is

$$y = Ax + B\mathrm{e}^{-x}\,,$$

where A and B are arbitrary constants.

(iii) Find the general solution of the differential equation

$$(x+1)\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + x\frac{\mathrm{d}y}{\mathrm{d}x} - y = (x+1)^2.$$

2 The polynomial $\mathrm{f}(x)$ is defined by

$$\mathrm{f}(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_2x^2 + a_1x + a_0\,,$$

where $n \geqslant 2$ and the coefficients $a_0, \ldots, a_{n-1}$ are integers, with $a_0 \neq 0$. Suppose that the equation $\mathrm{f}(x) = 0$ has a rational root p/q, where p and q are integers with no common factor greater than 1, and $q > 0$. By considering $q^{n-1}\mathrm{f}(p/q)$, find the value of q and deduce that any rational root of the equation $\mathrm{f}(x) = 0$ must be an integer.

(i) Show that the nth root of 2 is irrational for $n \geqslant 2$.

(ii) Show that the cubic equation

$$x^3 - x + 1 = 0$$

has no rational roots.

(iii) Show that the polynomial equation

$$x^n - 5x + 7 = 0$$

has no rational roots for $n \geqslant 2$.

3 Show that, provided $q^2 \neq 4p^3$, the polynomial

$$x^3 - 3px + q \qquad\qquad (p \neq 0,\ q \neq 0)$$

can be written in the form

$$a(x-\alpha)^3 + b(x-\beta)^3\,,$$

where α and β are the roots of the quadratic equation $pt^2 - qt + p^2 = 0$, and a and b are constants which you should express in terms of α and β.

Hence show that one solution of the equation $x^3 - 24x + 48 = 0$ is

$$x = \frac{2(2 - 2^{\frac{1}{3}})}{1 - 2^{\frac{1}{3}}}$$

and obtain similar expressions for the other two solutions in terms of ω, where $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$.

Find also the roots of $x^3 - 3px + q = 0$ when $p = r^2$ and $q = 2r^3$ for some non-zero constant r.

4 The following result applies to any function f which is continuous, has positive gradient and satisfies $\mathrm{f}(0) = 0$:

$$ab \leqslant \int_0^a \mathrm{f}(x)\,\mathrm{d}x + \int_0^b \mathrm{f}^{-1}(y)\,\mathrm{d}y\,, \qquad (*)$$

where f^{-1} denotes the inverse function of f, and $a \geqslant 0$ and $b \geqslant 0$.

(i) By considering the graph of $y = \mathrm{f}(x)$, explain briefly why the inequality $(*)$ holds.

In the case $a > 0$ and $b > 0$, state a condition on a and b under which equality holds.

(ii) By taking $\mathrm{f}(x) = x^{p-1}$ in $(*)$, where $p > 1$, show that if $\dfrac{1}{p} + \dfrac{1}{q} = 1$ then

$$ab \leqslant \frac{a^p}{p} + \frac{b^q}{q}\,.$$

Verify that equality holds under the condition you stated above.

(iii) Show that, for $0 \leqslant a \leqslant \frac{1}{2}\pi$ and $0 \leqslant b \leqslant 1$,

$$ab \leqslant b \arcsin b + \sqrt{1 - b^2} - \cos a\,.$$

Deduce that, for $t \geqslant 1$,

$$\arcsin(t^{-1}) \geqslant t - \sqrt{t^2 - 1}\,.$$

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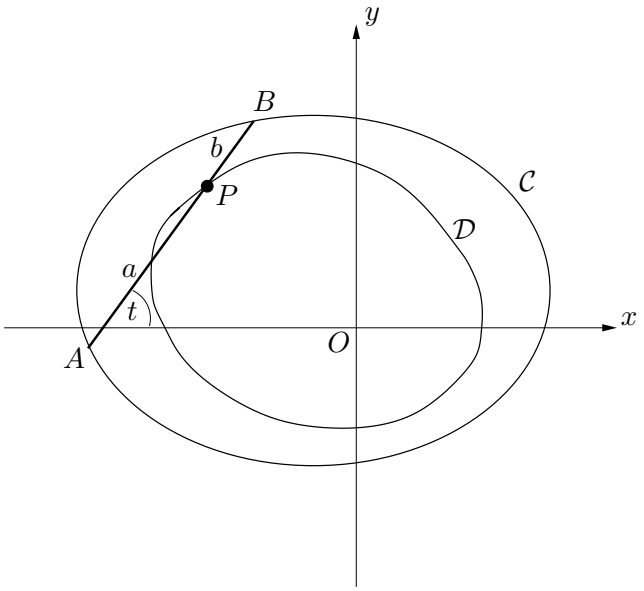
5 A movable point P has cartesian coordinates (x, y), where x and y are functions of t. The polar coordinates of P with respect to the origin O are r and θ. Starting with the expression

$$\tfrac{1}{2}\int r^2\,\mathrm{d}\theta$$

for the area swept out by OP, obtain the equivalent expression

$$\tfrac{1}{2}\int\left(x\frac{\mathrm{d}y}{\mathrm{d}t}-y\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t\,. \qquad (*)$$

The ends of a thin straight rod AB lie on a closed convex curve $\mathcal{C}$. The point P on the rod is a fixed distance a from A and a fixed distance b from B. The angle between AB and the positive x direction is t. As A and B move anticlockwise round $\mathcal{C}$, the angle t increases from 0 to 2π and P traces a closed convex curve $\mathcal{D}$ inside $\mathcal{C}$, with the origin O lying inside $\mathcal{D}$, as shown in the diagram.



Let (x, y) be the coordinates of P. Write down the coordinates of A and B in terms of a, b, x, y and t.

The areas swept out by OA, OB and OP are denoted by $[A]$, $[B]$ and $[P]$, respectively. Show, using $(*)$, that

$$[A] = [P] + \pi a^2 - af$$

where

$$f = \tfrac{1}{2}\int_0^{2\pi}\left(\left(x+\frac{\mathrm{d}y}{\mathrm{d}t}\right)\cos t+\left(y-\frac{\mathrm{d}x}{\mathrm{d}t}\right)\sin t\right)\mathrm{d}t\,.$$

Obtain a corresponding expression for $[B]$ involving b. Hence show that the area between the curves $\mathcal{C}$ and $\mathcal{D}$ is πab.

6 The definite integrals T, U, V and X are defined by

$$T = \int_{\frac{1}{3}}^{\frac{1}{2}} \frac{\operatorname{artanh} t}{t}\,\mathrm{d}t\,, \qquad U = \int_{\ln 2}^{\ln 3} \frac{u}{2\sinh u}\,\mathrm{d}u\,,$$

$$V = -\int_{\frac{1}{3}}^{\frac{1}{2}} \frac{\ln v}{1-v^2}\,\mathrm{d}v\,, \qquad X = \int_{\frac{1}{2}\ln 2}^{\frac{1}{2}\ln 3} \ln(\coth x)\,\mathrm{d}x\,.$$

Show, without evaluating any of them, that T, U, V and X are all equal.

7 Let

$$T_n = \left(\sqrt{a+1} + \sqrt{a}\right)^n\,,$$

where n is a positive integer and a is any given positive integer.

(i) In the case when n is even, show by induction that T_n can be written in the form

$$A_n + B_n\sqrt{a(a+1)}\,,$$

where A_n and B_n are integers (depending on a and n) and $A_n^2 = a(a+1)B_n^2 + 1$.

(ii) In the case when n is odd, show by considering $(\sqrt{a+1} + \sqrt{a})T_m$ where m is even, or otherwise, that T_n can be written in the form

$$C_n\sqrt{a+1} + D_n\sqrt{a}\,,$$

where C_n and D_n are integers (depending on a and n) and $(a+1)C_n^2 = aD_n^2 + 1$.

(iii) Deduce that, for each n, T_n can be written as the sum of the square roots of two consecutive integers.

8 The complex numbers z and w are related by

$$w = \frac{1 + \mathrm{i}z}{\mathrm{i} + z}\,.$$

Let $z = x + \mathrm{i}y$ and $w = u + \mathrm{i}v$, where x, y, u and v are real. Express u and v in terms of x and y.

(i) By setting $x = \tan(\theta/2)$, or otherwise, show that if the locus of z is the real axis $y = 0$, $-\infty < x < \infty$, then the locus of w is the circle $u^2 + v^2 = 1$ with one point omitted.

(ii) Find the locus of w when the locus of z is the line segment $y = 0$, $-1 < x < 1$.

(iii) Find the locus of w when the locus of z is the line segment $x = 0$, $-1 < y < 1$.

(iv) Find the locus of w when the locus of z is the line $y = 1$, $-\infty < x < \infty$.

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Section B: Mechanics

9 Particles P and Q have masses $3m$ and $4m$, respectively. They lie on the outer curved surface of a smooth circular cylinder of radius a which is fixed with its axis horizontal. They are connected by a light inextensible string of length $\frac{1}{2}\pi a$, which passes over the surface of the cylinder. The particles and the string all lie in a vertical plane perpendicular to the axis of the cylinder, and the axis intersects this plane at O. Initially, the particles are in equilibrium.

Equilibrium is slightly disturbed and Q begins to move downwards. Show that while the two particles are still in contact with the cylinder the angle θ between OQ and the vertical satisfies

$$7a\dot{\theta}^2 + 8g\cos\theta + 6g\sin\theta = 10g\,.$$

(i) Given that Q loses contact with the cylinder first, show that it does so when $\theta = \beta$, where β satisfies

$$15\cos\beta + 6\sin\beta = 10.$$

(ii) Show also that while P and Q are still in contact with the cylinder the tension in the string is $\frac{12}{7}mg(\sin\theta + \cos\theta)$.

10 Particles P and Q, each of mass m, lie initially at rest a distance a apart on a smooth horizontal plane. They are connected by a light elastic string of natural length a and modulus of elasticity $\frac{1}{2}ma\omega^2$, where ω is a constant.

Then P receives an impulse which gives it a velocity u directly away from Q. Show that when the string next returns to length a, the particles have travelled a distance $\frac{1}{2}\pi u/\omega$, and find the speed of each particle.

Find also the total time between the impulse and the subsequent collision of the particles.

11 A thin uniform circular disc of radius a and mass m is held in equilibrium in a horizontal plane a distance b below a horizontal ceiling, where $b > 2a$. It is held in this way by n light inextensible vertical strings, each of length b; one end of each string is attached to the edge of the disc and the other end is attached to a point on the ceiling. The strings are equally spaced around the edge of the disc. One of the strings is attached to the point P on the disc which has coordinates $(a, 0, -b)$ with respect to cartesian axes with origin on the ceiling directly above the centre of the disc.

The disc is then rotated through an angle θ (where $\theta < \pi$) about its vertical axis of symmetry and held at rest by a couple acting in the plane of the disc. Show that the string attached to P now makes an angle ϕ with the vertical, where

$$b \sin \phi = 2a \sin \tfrac{1}{2}\theta \,.$$

Show further that the magnitude of the couple is

$$\frac{mga^2 \sin \theta}{\sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta}} \,.$$

The disc is now released from rest. Show that its angular speed, ω, when the strings are vertical is given by

$$\frac{a^2\omega^2}{4g} = b - \sqrt{b^2 - 4a^2 \sin^2 \tfrac{1}{2}\theta} \,.$$

Section C: Probability and Statistics

12 The random variable N takes positive integer values and has pgf (probability generating function) $\mathrm{G}(t)$. The random variables X_i, where $i = 1, 2, 3, \ldots$, are independently and identically distributed, each with pgf $\mathrm{H}(t)$. The random variables X_i are also independent of N. The random variable Y is defined by

$$Y = \sum_{i=1}^{N} X_i \, .$$

Given that the pgf of Y is $\mathrm{G}(\mathrm{H}(t))$, show that

$$\mathrm{E}(Y) = \mathrm{E}(N)\mathrm{E}(X_i) \qquad \text{and} \qquad \mathrm{Var}\,(Y) = \mathrm{Var}\,(N)\big(\mathrm{E}(X_i)\big)^2 + \mathrm{E}(N)\mathrm{Var}\,(X_i)\,.$$

A fair coin is tossed until a head occurs. The total number of tosses is N. The coin is then tossed a further N times and the total number of heads in these N tosses is Y. Find in this particular case the pgf of Y, $\mathrm{E}(Y)$, $\mathrm{Var}\,(Y)$ and $\mathrm{P}(Y = r)$.

13 In this question, the notation $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x, so for example $\lfloor \pi \rfloor = 3$ and $\lfloor 3 \rfloor = 3$.

(i) A bag contains n balls, of which b are black. A sample of k balls is drawn, one after another, at random *with* replacement. The random variable X denotes the number of black balls in the sample. By considering

$$\frac{\mathrm{P}(X = r+1)}{\mathrm{P}(X = r)}\,,$$

show that, in the case that it is unique, the most probable number of black balls in the sample is

$$\left\lfloor \frac{(k+1)b}{n} \right\rfloor .$$

Under what circumstances is the answer not unique?

(ii) A bag contains n balls, of which b are black. A sample of k balls (where $k \leqslant b$) is drawn, one after another, at random *without* replacement. Find, in the case that it is unique, the most probable number of black balls in the sample.

Under what circumstances is the answer not unique?

CAMBRIDGE ASSESSMENT

Sixth Term Examination Papers **9475**

MATHEMATICS 3 Afternoon

WEDNESDAY 27 JUNE 2012 Time: 3 hours

3159333053

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

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Section A: Pure Mathematics

1 Given that $z = y^n \left(\dfrac{\mathrm{d}y}{\mathrm{d}x}\right)^2$, show that

$$\frac{\mathrm{d}z}{\mathrm{d}x} = y^{n-1}\frac{\mathrm{d}y}{\mathrm{d}x}\left(n\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2}\right).$$

(i) Use the above result to show that the solution to the equation

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + 2y\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \sqrt{y} \qquad (y > 0)$$

that satisfies $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ when $x = 0$ is $y = \left(\frac{3}{8}x^2 + 1\right)^{\frac{2}{3}}$.

(ii) Find the solution to the equation

$$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 - y\frac{\mathrm{d}^2y}{\mathrm{d}x^2} + y^2 = 0$$

that satisfies $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ when $x = 0$.

2 In this question, $|x| < 1$ and you may ignore issues of convergence.

(i) Simplify

$$(1-x)(1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n}),$$

where n is a positive integer, and deduce that

$$\frac{1}{1-x} = (1+x)(1+x^2)(1+x^4)\cdots(1+x^{2^n}) + \frac{x^{2^{n+1}}}{1-x}.$$

Deduce further that

$$\ln(1-x) = -\sum_{r=0}^{\infty} \ln\left(1+x^{2^r}\right),$$

and hence that

$$\frac{1}{1-x} = \frac{1}{1+x} + \frac{2x}{1+x^2} + \frac{4x^3}{1+x^4} + \cdots.$$

(ii) Show that

$$\frac{1+2x}{1+x+x^2} = \frac{1-2x}{1-x+x^2} + \frac{2x-4x^3}{1-x^2+x^4} + \frac{4x^3-8x^7}{1-x^4+x^8} + \cdots.$$

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3 It is given that the two curves

$$y = 4 - x^2 \quad \text{and} \quad mx = k - y^2\,,$$

where $m > 0$, touch exactly once.

(i) In each of the following four cases, sketch the two curves on a single diagram, noting the coordinates of any intersections with the axes:

(a) $k < 0$;

(b) $0 < k < 16$, $k/m < 2$;

(c) $k > 16$, $k/m > 2$;

(d) $k > 16$, $k/m < 2$.

(ii) Now set $m = 12$.
Show that the x-coordinate of any point at which the two curves meet satisfies

$$x^4 - 8x^2 + 12x + 16 - k = 0\,.$$

Let a be the value of x at the point where the curves touch. Show that a satisfies

$$a^3 - 4a + 3 = 0$$

and hence find the three possible values of a.

Derive also the equation

$$k = -4a^2 + 9a + 16\,.$$

Which of the four sketches in part (i) arise?

4 **(i)** Show that

$$\sum_{n=1}^{\infty} \frac{n+1}{n!} = 2\mathrm{e} - 1$$

and

$$\sum_{n=1}^{\infty} \frac{(n+1)^2}{n!} = 5\mathrm{e} - 1\,.$$

Sum the series $\displaystyle\sum_{n=1}^{\infty} \frac{(2n-1)^3}{n!}$.

(ii) Sum the series $\displaystyle\sum_{n=0}^{\infty} \frac{(n^2+1)2^{-n}}{(n+1)(n+2)}$, giving your answer in terms of natural logarithms.

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5 **(i)** The point with coordinates (a, b), where a and b are rational numbers, is called:
an *integer rational point* if both a and b are integers;
a *non-integer rational point* if neither a nor b is an integer.

(a) Write down an integer rational point and a non-integer rational point on the circle $x^2 + y^2 = 1$.

(b) Write down an integer rational point on the circle $x^2 + y^2 = 2$. Simplify

$$(\cos\theta + \sqrt{m}\sin\theta)^2 + (\sin\theta - \sqrt{m}\cos\theta)^2$$

and hence obtain a non-integer rational point on the circle $x^2 + y^2 = 2$.

(ii) The point with coordinates $(p+\sqrt{2}\,q\,,\ r+\sqrt{2}\,s)$, where p, q, r and s are rational numbers, is called:
an *integer 2-rational point* if all of p, q, r and s are integers;
a *non-integer 2-rational point* if none of p, q, r and s is an integer.

(a) Write down an integer 2-rational point, and obtain a non-integer 2-rational point, on the circle $x^2 + y^2 = 3$.

(b) Obtain a non-integer 2-rational point on the circle $x^2 + y^2 = 11$.

(c) Obtain a non-integer 2-rational point on the hyperbola $x^2 - y^2 = 7$.

6 Let $x + \mathrm{i}y$ be a root of the quadratic equation $z^2 + pz + 1 = 0$, where p is a real number. Show that $x^2 - y^2 + px + 1 = 0$ and $(2x + p)y = 0$. Show further that

$$\text{either} \quad p = -2x \quad \text{or} \quad p = -(x^2 + 1)/x \quad \text{with} \quad x \neq 0\,.$$

Hence show that the set of points in the Argand diagram that can (as p varies) represent roots of the quadratic equation consists of the real axis with one point missing and a circle. This set of points is called the *root locus* of the quadratic equation.

Obtain and sketch in the Argand diagram the root locus of the equation

$$pz^2 + z + 1 = 0$$

and the root locus of the equation

$$pz^2 + p^2 z + 2 = 0\,.$$

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7 A pain-killing drug is injected into the bloodstream. It then diffuses into the brain, where it is absorbed. The quantities at time t of the drug in the blood and the brain respectively are $y(t)$ and $z(t)$. These satisfy

$$\dot{y} = -2(y - z)\,, \qquad \dot{z} = -\dot{y} - 3z\,,$$

where the dot denotes differentiation with respect to t.

Obtain a second order differential equation for y and hence derive the solution

$$y = A\mathrm{e}^{-t} + B\mathrm{e}^{-6t}\,, \qquad z = \tfrac{1}{2}A\mathrm{e}^{-t} - 2B\mathrm{e}^{-6t}\,,$$

where A and B are arbitrary constants.

(i) Obtain the solution that satisfies $z(0) = 0$ and $y(0) = 5$. The quantity of the drug in the brain for this solution is denoted by $z_1(t)$.

(ii) Obtain the solution that satisfies $z(0) = z(1) = c$, where c is a given constant. The quantity of the drug in the brain for this solution is denoted by $z_2(t)$.

(iii) Show that for $0 \leqslant t \leqslant 1$,

$$z_2(t) = \sum_{n=-\infty}^{0} z_1(t - n)\,,$$

provided c takes a particular value that you should find.

8 The sequence F_0, F_1, F_2, ... is defined by $F_0 = 0$, $F_1 = 1$ and, for $n \geqslant 0$,

$$F_{n+2} = F_{n+1} + F_n\,.$$

(i) Show that $F_0F_3 - F_1F_2 = F_2F_5 - F_3F_4\,.$

(ii) Find the values of $F_nF_{n+3} - F_{n+1}F_{n+2}$ in the two cases that arise.

(iii) Prove that, for $r = 1, 2, 3, \ldots,$

$$\arctan\left(\frac{1}{F_{2r}}\right) = \arctan\left(\frac{1}{F_{2r+1}}\right) + \arctan\left(\frac{1}{F_{2r+2}}\right)$$

and hence evaluate the following sum (which you may assume converges):

$$\sum_{r=1}^{\infty} \arctan\left(\frac{1}{F_{2r+1}}\right)\,.$$

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Section B: Mechanics

9 A pulley consists of a disc of radius r with centre O and a light thin axle through O perpendicular to the plane of the disc. The disc is non-uniform, its mass is M and its centre of mass is at O. The axle is fixed and horizontal.

Two particles, of masses m_1 and m_2 where $m_1 > m_2$, are connected by a light inextensible string which passes over the pulley. The contact between the string and the pulley is rough enough to prevent the string sliding. The pulley turns and the vertical force on the axle is found, by measurement, to be $P + Mg$.

(i) The moment of inertia of the pulley about its axle is calculated assuming that the pulley rotates without friction about its axle. Show that the calculated value is

$$\frac{((m_1 + m_2)P - 4m_1m_2g)r^2}{(m_1 + m_2)g - P}\,. \qquad (*)$$

(ii) Instead, the moment of inertia of the pulley about its axle is calculated assuming that a couple of magnitude C due to friction acts on the axle of the pulley. Determine whether this calculated value is greater or smaller than $(*)$.

Show that $C < (m_1 - m_2)rg$.

10 A small ring of mass m is free to slide without friction on a hoop of radius a. The hoop is fixed in a vertical plane. The ring is connected by a light elastic string of natural length a to the highest point of the hoop. The ring is initially at rest at the lowest point of the hoop and is then slightly displaced. In the subsequent motion the angle of the string to the downward vertical is ϕ. Given that the ring first comes to rest just as the string becomes slack, find an expression for the modulus of elasticity of the string in terms of m and g.

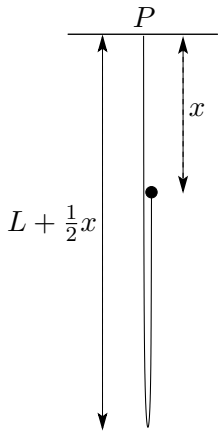
Show that, throughout the motion, the magnitude R of the reaction between the ring and the hoop is given by

$$R = (12\cos^2\phi - 15\cos\phi + 5)mg$$

and that R is non-zero throughout the motion.

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11 One end of a thin heavy uniform inextensible perfectly flexible rope of length $2L$ and mass $2M$ is attached to a fixed point P. A particle of mass m is attached to the other end. Initially, the particle is held at P and the rope hangs vertically in a loop below P. The particle is then released so that it and a section of the rope (of decreasing length) fall vertically as shown in the diagram.



You may assume that each point on the moving section of the rope falls at the same speed as the particle. Given that energy is conserved, show that, when the particle has fallen a distance x (where $x < 2L$), its speed v is given by

$$v^2 = \frac{2gx\left(mL + ML - \frac{1}{4}Mx\right)}{mL + ML - \frac{1}{2}Mx}.$$

Hence show that the acceleration of the particle is

$$g + \frac{Mgx\left(mL + ML - \frac{1}{4}Mx\right)}{2\left(mL + ML - \frac{1}{2}Mx\right)^2}.$$

Deduce that the acceleration of the particle after it is released is greater than g.

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Section C: Probability and Statistics

12 **(i)** A point P lies in an equilateral triangle ABC of height 1. The perpendicular distances from P to the sides AB, BC and CA are x_1, x_2 and x_3, respectively. By considering the areas of triangles with one vertex at P, show that $x_1 + x_2 + x_3 = 1$.

Suppose now that P is placed at random in the equilateral triangle (so that the probability of it lying in any given region of the triangle is proportional to the area of that region). The perpendicular distances from P to the sides AB, BC and CA are random variables X_1, X_2 and X_3, respectively. In the case $X_1 = \min(X_1, X_2, X_3)$, give a sketch showing the region of the triangle in which P lies.

Let $X = \min(X_1, X_2, X_3)$. Show that the probability density function for X is given by

$$\mathrm{f}(x) = \begin{cases} 6(1-3x) & 0 \leqslant x \leqslant \frac{1}{3}\,, \\ 0 & \text{otherwise}\,. \end{cases}$$

Find the expected value of X.

(ii) A point is chosen at random in a regular tetrahedron of height 1. Find the expected value of the distance from the point to the closest face.
[The volume of a tetrahedron is $\frac{1}{3}\times$ area of base $\times$ height and its centroid is a distance $\frac{1}{4}\times$ height from the base.]

13 **(i)** The random variable Z has a Normal distribution with mean 0 and variance 1. Show that the expectation of Z given that $a < Z < b$ is

$$\frac{\exp(-\frac{1}{2}a^2) - \exp(-\frac{1}{2}b^2)}{\sqrt{2\pi}\,\big(\Phi(b) - \Phi(a)\big)},$$

where Φ denotes the cumulative distribution function for Z.

(ii) The random variable X has a Normal distribution with mean μ and variance σ^2. Show that

$$\mathrm{E}(X \,|\, X > 0) = \mu + \sigma \mathrm{E}(Z \,|\, Z > -\mu/\sigma).$$

Hence, or otherwise, show that the expectation, m, of $|X|$ is given by

$$m = \mu\big(1 - 2\Phi(-\mu/\sigma)\big) + \sigma\sqrt{2/\pi}\,\exp(-\tfrac{1}{2}\mu^2/\sigma^2)\,.$$

Obtain an expression for the variance of $|X|$ in terms of μ, σ and m.

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Admissions Testing Service

Sixth Term Examination Papers **9475**

MATHEMATICS 3 Morning

WEDNESDAY 26 JUNE 2013 Time: 3 hours

8165781507

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Write your name, centre number and candidate number in the spaces on the answer booklet.

Begin each answer on a new page.

Write the numbers of the questions you answer in the order attempted on the front of the answer booklet.

INFORMATION FOR CANDIDATES

Each question is marked out of 20. There is no restriction of choice.

All questions attempted will be marked.

Your final mark will be based on the **six** questions for which you gain the highest marks.

You are advised to concentrate on no more than **six** questions. Little credit will be given for fragmentary answers.

You are provided with a Mathematical Formulae Booklet.

Calculators are not permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 10 printed pages and 2 blank pages.

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Section A: Pure Mathematics

1 Given that $t = \tan \frac{1}{2}x$, show that $\dfrac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{2}(1 + t^2)$ and $\sin x = \dfrac{2t}{1 + t^2}$.

Hence show that

$$\int_0^{\frac{1}{2}\pi} \frac{1}{1 + a\sin x}\,\mathrm{d}x = \frac{2}{\sqrt{1 - a^2}} \arctan \frac{\sqrt{1 - a}}{\sqrt{1 + a}} \qquad (0 < a < 1).$$

Let

$$I_n = \int_0^{\frac{1}{2}\pi} \frac{\sin^n x}{2 + \sin x}\,\mathrm{d}x \qquad (n \geqslant 0).$$

By considering $I_{n+1} + 2I_n$, or otherwise, evaluate I_3.

2 In this question, you may ignore questions of convergence.

Let $y = \dfrac{\arcsin x}{\sqrt{1 - x^2}}$. Show that

$$(1 - x^2)\frac{\mathrm{d}y}{\mathrm{d}x} - xy - 1 = 0$$

and prove that, for any positive integer n,

$$(1 - x^2)\frac{\mathrm{d}^{n+2}y}{\mathrm{d}x^{n+2}} - (2n + 3)x\frac{\mathrm{d}^{n+1}y}{\mathrm{d}x^{n+1}} - (n + 1)^2\frac{\mathrm{d}^n y}{\mathrm{d}x^n} = 0\,.$$

Hence obtain the Maclaurin series for $\dfrac{\arcsin x}{\sqrt{1 - x^2}}$, giving the general term for odd and for even powers of x.

Evaluate the infinite sum

$$1 + \frac{1}{3!} + \frac{2^2}{5!} + \frac{2^2 \times 3^2}{7!} + \cdots + \frac{2^2 \times 3^2 \times \cdots \times n^2}{(2n + 1)!} + \cdots\,.$$

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3 The four vertices P_i $(i = 1, 2, 3, 4)$ of a regular tetrahedron lie on the surface of a sphere with centre at O and of radius 1. The position vector of P_i with respect to O is $\mathbf{p}_i$ $(i = 1, 2, 3, 4)$. Use the fact that $\mathbf{p}_1 + \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 = \mathbf{0}$ to show that $\mathbf{p}_i \,.\, \mathbf{p}_j = -\frac{1}{3}$ for $i \neq j$.

Let X be any point on the surface of the sphere, and let XP_i denote the length of the line joining X and P_i $(i = 1, 2, 3, 4)$.

(i) By writing $(XP_i)^2$ as $(\mathbf{p}_i - \mathbf{x}) \,.\, (\mathbf{p}_i - \mathbf{x})$, where $\mathbf{x}$ is the position vector of X with respect to O, show that

$$\sum_{i=1}^{4} (XP_i)^2 = 8 \,.$$

(ii) Given that P_1 has coordinates $(0, 0, 1)$ and that the coordinates of P_2 are of the form $(a, 0, b)$, where $a > 0$, show that $a = 2\sqrt{2}/3$ and $b = -1/3$, and find the coordinates of P_3 and P_4.

(iii) Show that

$$\sum_{i=1}^{4} (XP_i)^4 = 4 \sum_{i=1}^{4} (1 - \mathbf{x} \,.\, \mathbf{p}_i)^2 \,.$$

By letting the coordinates of X be (x, y, z), show further that $\sum_{i=1}^{4} (XP_i)^4$ is independent of the position of X.

4 Show that $(z - \mathrm{e}^{\mathrm{i}\theta})(z - \mathrm{e}^{-\mathrm{i}\theta}) = z^2 - 2z\cos\theta + 1 \,.$

Write down the $(2n)$th roots of -1 in the form $\mathrm{e}^{\mathrm{i}\theta}$, where $-\pi < \theta \leqslant \pi$, and deduce that

$$z^{2n} + 1 = \prod_{k=1}^{n} \left(z^2 - 2z \cos\left(\tfrac{(2k-1)\pi}{2n}\right) + 1 \right) .$$

Here, n is a positive integer, and the $\prod$ notation denotes the product.

(i) By substituting $z = \mathrm{i}$ show that, when n is even,

$$\cos\left(\tfrac{\pi}{2n}\right) \cos\left(\tfrac{3\pi}{2n}\right) \cos\left(\tfrac{5\pi}{2n}\right) \cdots \cos\left(\tfrac{(2n-1)\pi}{2n}\right) = (-1)^{\frac{1}{2}n} 2^{1-n} \,.$$

(ii) Show that, when n is odd,

$$\cos^2\left(\tfrac{\pi}{2n}\right) \cos^2\left(\tfrac{3\pi}{2n}\right) \cos^2\left(\tfrac{5\pi}{2n}\right) \cdots \cos^2\left(\tfrac{(n-2)\pi}{2n}\right) = n 2^{1-n} \,.$$

You may use without proof the fact that $1 + z^{2n} = (1 + z^2)(1 - z^2 + z^4 - \cdots + z^{2n-2})$ when n is odd.

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5 In this question, you may assume that, if a, b and c are positive integers such that a and b are coprime and a divides bc, then a divides c. (Two positive integers are said to be *coprime* if their highest common factor is 1.)

(i) Suppose that there are positive integers p, q, n and N such that p and q are coprime and $q^n N = p^n$. Show that $N = kp^n$ for some positive integer k and deduce the value of q.

Hence prove that, for any positive integers n and N, $\sqrt[n]{N}$ is either a positive integer or irrational.

(ii) Suppose that there are positive integers a, b, c and d such that a and b are coprime and c and d are coprime, and $a^a d^b = b^a c^b$. Prove that $d^b = b^a$ and deduce that, if p is a prime factor of d, then p is also a prime factor of b.

If p^m and p^n are the highest powers of the prime number p that divide d and b, respectively, express b in terms of a, m and n and hence show that $p^n \leqslant n$. Deduce the value of b. (You may assume that if $x > 0$ and $y \geqslant 2$ then $y^x > x$.)

Hence prove that, if r is a positive rational number such that r^r is rational, then r is a positive integer.

6 Let z and w be complex numbers. Use a diagram to show that $|z - w| \leqslant |z| + |w|$.

For any complex numbers z and w, E is defined by

$$E = zw^* + z^*w + 2|zw|\,.$$

(i) Show that $|z - w|^2 = (|z| + |w|)^2 - E$, and deduce that E is real and non-negative.

(ii) Show that $|1 - zw^*|^2 = (1 + |zw|)^2 - E$.

Hence show that, if both $|z| > 1$ and $|w| > 1$, then

$$\frac{|z - w|}{|1 - zw^*|} \leqslant \frac{|z| + |w|}{1 + |zw|}\,.$$

Does this inequality also hold if both $|z| < 1$ and $|w| < 1$?

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7 **(i)** Let $y(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2y}{\mathrm{d}x^2} + y^3 = 0$ with $y = 1$ and $\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 + \tfrac{1}{2}y^4 .$$

Show by differentiation that E is constant and deduce that $|y(x)| \leqslant 1$ for all x.

(ii) Let $v(x)$ be a solution of the differential equation $\dfrac{\mathrm{d}^2v}{\mathrm{d}x^2} + x\dfrac{\mathrm{d}v}{\mathrm{d}x} + \sinh v = 0$ with $v = \ln 3$ and $\dfrac{\mathrm{d}v}{\mathrm{d}x} = 0$ at $x = 0$, and let

$$\mathrm{E}(x) = \left(\frac{\mathrm{d}v}{\mathrm{d}x}\right)^2 + 2\cosh v .$$

Show that $\dfrac{\mathrm{dE}}{\mathrm{d}x} \leqslant 0$ for $x \geqslant 0$ and deduce that $\cosh v(x) \leqslant \frac{5}{3}$ for $x \geqslant 0$.

(iii) Let $w(x)$ be a solution of the differential equation

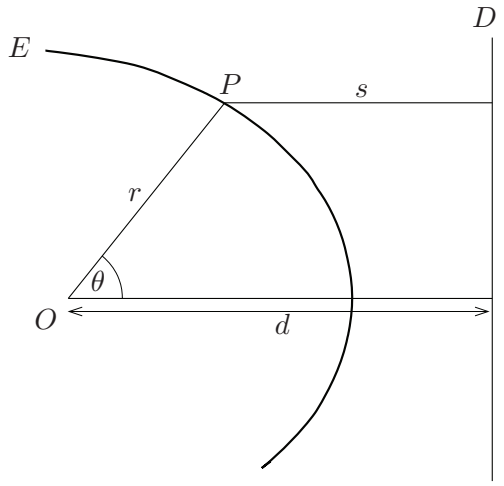
$$\frac{\mathrm{d}^2w}{\mathrm{d}x^2} + (5\cosh x - 4\sinh x - 3)\frac{\mathrm{d}w}{\mathrm{d}x} + (w\cosh w + 2\sinh w) = 0$$

with $\dfrac{\mathrm{d}w}{\mathrm{d}x} = \dfrac{1}{\sqrt{2}}$ and $w = 0$ at $x = 0$. Show that $\cosh w(x) \leqslant \frac{5}{4}$ for $x \geqslant 0$.

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8 Evaluate $\displaystyle\sum_{r=0}^{n-1} \mathrm{e}^{2\mathrm{i}(\alpha+r\pi/n)}$ where α is a fixed angle and $n \geqslant 2$.

The fixed point O is a distance d from a fixed line D. For any point P, let s be the distance from P to D and let r be the distance from P to O. Write down an expression for s in terms of d, r and the angle θ, where θ is as shown in the diagram below.



The curve E shown in the diagram is such that, for any point P on E, the relation $r = ks$ holds, where k is a fixed number with $0 < k < 1$.

Each of the n lines $L_1, \ldots, L_n$ passes through O and the angle between adjacent lines is $\frac{\pi}{n}$. The line L_j $(j = 1, \ldots, n)$ intersects E in two points forming a chord of length l_j. Show that, for $n \geqslant 2$,

$$\sum_{j=1}^{n} \frac{1}{l_j} = \frac{(2-k^2)n}{4kd}\,.$$

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Section B: Mechanics

9 A sphere of radius R and uniform density ρ_s is floating in a large tank of liquid of uniform density ρ. Given that the centre of the sphere is a distance x above the level of the liquid, where $x < R$, show that the volume of liquid displaced is

$$\frac{\pi}{3}(2R^3 - 3R^2x + x^3)\,.$$

The sphere is acted upon by two forces only: its weight and an upward force equal in magnitude to the weight of the liquid it has displaced. Show that

$$4R^3\rho_s(g + \ddot{x}) = (2R^3 - 3R^2x + x^3)\rho g\,.$$

Given that the sphere is in equilibrium when $x = \frac{1}{2}R$, find ρ_s in terms of ρ. Find, in terms of R and g, the period of small oscillations about this equilibrium position.

10 A uniform rod AB has mass M and length $2a$. The point P lies on the rod a distance $a - x$ from A. Show that the moment of inertia of the rod about an axis through P and perpendicular to the rod is

$$\tfrac{1}{3}M(a^2 + 3x^2)\,.$$

The rod is free to rotate, in a horizontal plane, about a fixed vertical axis through P. Initially the rod is at rest. The end B is struck by a particle of mass m moving horizontally with speed u in a direction perpendicular to the rod. The coefficient of restitution between the rod and the particle is e. Show that the angular velocity of the rod immediately after impact is

$$\frac{3mu(1+e)(a+x)}{M(a^2+3x^2)+3m(a+x)^2}\,.$$

In the case $m = 2M$, find the value of x for which the angular velocity is greatest and show that this angular velocity is $u(1+e)/a\,.$

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11 An equilateral triangle, comprising three light rods each of length $\sqrt{3}a$, has a particle of mass m attached to each of its vertices. The triangle is suspended horizontally from a point vertically above its centre by three identical springs, so that the springs and rods form a tetrahedron. Each spring has natural length a and modulus of elasticity kmg, and is light. Show that when the springs make an angle θ with the horizontal the tension in each spring is

$$\frac{kmg(1-\cos\theta)}{\cos\theta}\,.$$

Given that the triangle is in equilibrium when $\theta = \frac{1}{6}\pi$, show that $k = 4\sqrt{3} + 6$.

The triangle is released from rest from the position at which $\theta = \frac{1}{3}\pi$. Show that when it passes through the equilibrium position its speed V satisfies

$$V^2 = \frac{4ag}{3}(6+\sqrt{3})\,.$$

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Section C: Probability and Statistics

12 A list consists only of letters A and B arranged in a row. In the list, there are a letter As and b letter Bs, where $a \geqslant 2$ and $b \geqslant 2$, and $a + b = n$. Each possible ordering of the letters is equally probable. The random variable X_1 is defined by

$$X_1 = \begin{cases} 1 & \text{if the first letter in the row is } A; \\ 0 & \text{otherwise.} \end{cases}$$

The random variables X_k $(2 \leqslant k \leqslant n)$ are defined by

$$X_k = \begin{cases} 1 & \text{if the } (k-1)\text{th letter is } B \text{ and the } k\text{th is } A; \\ 0 & \text{otherwise.} \end{cases}$$

The random variable S is defined by $S = \sum_{i=1}^{n} X_i$.

(i) Find expressions for $\mathrm{E}(X_i)$, distinguishing between the cases $i = 1$ and $i \neq 1$, and show that $\mathrm{E}(S) = \dfrac{a(b+1)}{n}$.

(ii) Show that:

(a) for $j \geqslant 3$, $\mathrm{E}(X_1 X_j) = \dfrac{a(a-1)b}{n(n-1)(n-2)}$;

(b) $\displaystyle\sum_{i=2}^{n-2} \Big(\sum_{j=i+2}^{n} \mathrm{E}(X_i X_j) \Big) = \frac{a(a-1)b(b-1)}{2n(n-1)}$;

(c) $\mathrm{Var}\,(S) = \dfrac{a(a-1)b(b+1)}{n^2(n-1)}$.

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13 **(a)** The continuous random variable X satisfies $0 \leqslant X \leqslant 1$, and has probability density function $\mathrm{f}(x)$ and cumulative distribution function $\mathrm{F}(x)$. The greatest value of $\mathrm{f}(x)$ is M, so that $0 \leqslant \mathrm{f}(x) \leqslant M$.

(i) Show that $0 \leqslant \mathrm{F}(x) \leqslant Mx$ for $0 \leqslant x \leqslant 1$.

(ii) For any function $\mathrm{g}(x)$, show that

$$\int_0^1 2\mathrm{g}(x)\mathrm{F}(x)\mathrm{f}(x)\mathrm{d}x = \mathrm{g}(1) - \int_0^1 \mathrm{g}'(x)\big(\mathrm{F}(x)\big)^2\mathrm{d}x\,.$$

(b) The continuous random variable Y satisfies $0 \leqslant Y \leqslant 1$, and has probability density function $k\mathrm{F}(y)\mathrm{f}(y)$, where f and F are as above.

(i) Determine the value of the constant k.

(ii) Show that

$$1 + \frac{nM}{n+1}\mu_{n+1} - \frac{nM}{n+1} \leqslant \mathrm{E}(Y^n) \leqslant 2M\mu_{n+1}\,,$$

where $\mu_{n+1} = \mathrm{E}(X^{n+1})$ and $n \geqslant 0$.

(iii) Hence show that, for $n \geqslant 1$,

$$\mu_n \geqslant \frac{n}{(n+1)M} - \frac{n-1}{n+1}\,.$$

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Sixth Term Examination Papers **9475**

MATHEMATICS 3 Morning

FRIDAY 27 JUNE 2014 Time: 3 hours

6244203071

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

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Write your name, centre number and candidate number in the spaces on the answer booklet.

Begin each answer on a new page.

Write the numbers of the questions you answer in the order attempted on the front of the answer booklet.

INFORMATION FOR CANDIDATES

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Calculators are not permitted.

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This question paper consists of 9 printed pages and 3 blank pages.

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Section A: Pure Mathematics

1 Let a, b and c be real numbers such that $a+b+c=0$ and let

$$(1+ax)(1+bx)(1+cx) = 1+qx^2+rx^3$$

for all real x. Show that $q = bc+ca+ab$ and $r = abc$.

(i) Show that the coefficient of x^n in the series expansion (in ascending powers of x) of $\ln(1+qx^2+rx^3)$ is $(-1)^{n+1}S_n$ where

$$S_n = \frac{a^n+b^n+c^n}{n}\,, \qquad (n \geqslant 1).$$

(ii) Find, in terms of q and r, the coefficients of x^2, x^3 and x^5 in the series expansion (in ascending powers of x) of $\ln(1+qx^2+rx^3)$ and hence show that $S_2S_3 = S_5$.

(iii) Show that $S_2S_5 = S_7$.

(iv) Give a proof of, or find a counterexample to, the claim that $S_2S_7 = S_9$.

2 **(i)** Show, by means of the substitution $u = \cosh x$, that

$$\int \frac{\sinh x}{\cosh 2x}\,\mathrm{d}x = \frac{1}{2\sqrt{2}} \ln\left|\frac{\sqrt{2}\cosh x - 1}{\sqrt{2}\cosh x + 1}\right| + C\,.$$

(ii) Use a similar substitution to find an expression for

$$\int \frac{\cosh x}{\cosh 2x}\,\mathrm{d}x\,.$$

(iii) Using parts (i) and (ii) above, show that

$$\int_0^1 \frac{1}{1+u^4}\,\mathrm{d}u = \frac{\pi + 2\ln(\sqrt{2}+1)}{4\sqrt{2}}\,.$$

3 **(i)** The line L has equation $y = mx + c$, where $m > 0$ and $c > 0$. Show that, in the case $mc > a > 0$, the shortest distance between L and the parabola $y^2 = 4ax$ is

$$\frac{mc - a}{m\sqrt{m^2 + 1}}\,.$$

What is the shortest distance in the case that $mc \leqslant a$?

(ii) Find the shortest distance between the point $(p, 0)$, where $p > 0$, and the parabola $y^2 = 4ax$, where $a > 0$, in the different cases that arise according to the value of p/a. [*You may wish to use the parametric coordinates* $(at^2, 2at)$ *of points on the parabola.*]

Hence find the shortest distance between the circle $(x - p)^2 + y^2 = b^2$, where $p > 0$ and $b > 0$, and the parabola $y^2 = 4ax$, where $a > 0$, in the different cases that arise according to the values of p, a and b.

4 **(i)** Let

$$I = \int_0^1 \left((y')^2 - y^2\right) \mathrm{d}x \qquad \text{and} \qquad I_1 = \int_0^1 (y' + y\tan x)^2 \,\mathrm{d}x\,,$$

where y is a given function of x satisfying $y = 0$ at $x = 1$. Show that $I - I_1 = 0$ and deduce that $I \geqslant 0$. Show further that $I = 0$ only if $y = 0$ for all x $(0 \leqslant x \leqslant 1)$.

(ii) Let

$$J = \int_0^1 \left((y')^2 - a^2y^2\right) \mathrm{d}x\,,$$

where a is a given positive constant and y is a given function of x, not identically zero, satisfying $y = 0$ at $x = 1$. By considering an integral of the form

$$\int_0^1 (y' + ay\tan bx)^2 \,\mathrm{d}x\,,$$

where b is suitably chosen, show that $J \geqslant 0$. You should state the range of values of a, in the form $a < k$, for which your proof is valid.

In the case $a = k$, find a function y (not everywhere zero) such that $J = 0$.

5 A quadrilateral drawn in the complex plane has vertices A, B, C and D, labelled anticlockwise. These vertices are represented, respectively, by the complex numbers a, b, c and d. Show that $ABCD$ is a parallelogram (defined as a quadrilateral in which opposite sides are parallel and equal in length) if and only if $a + c = b + d$. Show further that, in this case, $ABCD$ is a square if and only if $\mathrm{i}(a - c) = b - d$.

Let $PQRS$ be a quadrilateral in the complex plane, with vertices labelled anticlockwise, the internal angles of which are all less than $180°$. Squares with centres X, Y, Z and T are constructed externally to the quadrilateral on the sides PQ, QR, RS and SP, respectively.

(i) If P and Q are represented by the complex numbers p and q, respectively, show that X can be represented by

$$\tfrac{1}{2}\big(p(1+\mathrm{i}) + q(1-\mathrm{i})\big)\,.$$

(ii) Show that $XYZT$ is a square if and only if $PQRS$ is a parallelogram.

6 Starting from the result that

$$\mathrm{h}(t) > 0 \text{ for } 0 < t < x \Longrightarrow \int_0^x \mathrm{h}(t)\,\mathrm{d}t > 0\,,$$

show that, if $\mathrm{f}''(t) > 0$ for $0 < t < x_0$ and $\mathrm{f}(0) = \mathrm{f}'(0) = 0$, then $\mathrm{f}(t) > 0$ for $0 < t < x_0$.

(i) Show that, for $0 < x < \frac{1}{2}\pi$,

$$\cos x \cosh x < 1\,.$$

(ii) Show that, for $0 < x < \frac{1}{2}\pi$,

$$\frac{1}{\cosh x} < \frac{\sin x}{x} < \frac{x}{\sinh x}\,.$$

7 The four distinct points P_i $(i = 1, 2, 3, 4)$ are the vertices, labelled anticlockwise, of a cyclic quadrilateral. The lines P_1P_3 and P_2P_4 intersect at Q.

(i) By considering the triangles P_1QP_4 and P_2QP_3 show that $(P_1Q)(QP_3) = (P_2Q)(QP_4)$.

(ii) Let $\mathbf{p}_i$ be the position vector of the point P_i $(i = 1, 2, 3, 4)$. Show that there exist numbers a_i, not all zero, such that

$$\sum_{i=1}^{4} a_i = 0 \qquad \text{and} \qquad \sum_{i=1}^{4} a_i \mathbf{p}_i = \mathbf{0}\,. \tag{$*$}$$

(iii) Let a_i $(i = 1, 2, 3, 4)$ be any numbers, not all zero, that satisfy $(*)$. Show that $a_1 + a_3 \neq 0$ and that the lines P_1P_3 and P_2P_4 intersect at the point with position vector

$$\frac{a_1\mathbf{p}_1 + a_3\mathbf{p}_3}{a_1 + a_3}\,.$$

Deduce that $a_1a_3(P_1P_3)^2 = a_2a_4(P_2P_4)^2$.

8 The numbers $\mathrm{f}(r)$ satisfy $\mathrm{f}(r) > \mathrm{f}(r+1)$ for $r = 1, 2, \ldots$. Show that, for any non-negative integer n,

$$k^n(k-1)\,\mathrm{f}(k^{n+1}) \leqslant \sum_{r=k^n}^{k^{n+1}-1} \mathrm{f}(r) \leqslant k^n(k-1)\,\mathrm{f}(k^n)$$

where k is an integer greater than 1.

(i) By taking $\mathrm{f}(r) = 1/r$, show that

$$\frac{N+1}{2} \leqslant \sum_{r=1}^{2^{N+1}-1} \frac{1}{r} \leqslant N+1\,.$$

Deduce that the sum $\sum_{r=1}^{\infty} \frac{1}{r}$ does not converge.

(ii) By taking $\mathrm{f}(r) = 1/r^3$, show that

$$\sum_{r=1}^{\infty} \frac{1}{r^3} \leqslant 1\tfrac{1}{3}\,.$$

(iii) Let $S(n)$ be the set of positive integers less than n which do not have a 2 in their decimal representation and let $\sigma(n)$ be the sum of the reciprocals of the numbers in $S(n)$, so for example $\sigma(5) = 1 + \frac{1}{3} + \frac{1}{4}$. Show that $S(1000)$ contains $9^3 - 1$ distinct numbers.

Show that $\sigma(n) < 80$ for all n.

Section B: Mechanics

9 A particle of mass m is projected with velocity $\mathbf{u}$. It is acted upon by the force $m\mathbf{g}$ due to gravity and by a resistive force $-mk\mathbf{v}$, where $\mathbf{v}$ is its velocity and k is a positive constant.

Given that, at time t after projection, its position $\mathbf{r}$ relative to the point of projection is given by

$$\mathbf{r} = \frac{kt - 1 + \mathrm{e}^{-kt}}{k^2}\,\mathbf{g} + \frac{1 - \mathrm{e}^{-kt}}{k}\,\mathbf{u}\,,$$

find an expression for $\mathbf{v}$ in terms of k, t, $\mathbf{g}$ and $\mathbf{u}$. Verify that the equation of motion and the initial conditions are satisfied.

Let $\mathbf{u} = u\cos\alpha\,\mathbf{i} + u\sin\alpha\,\mathbf{j}$ and $\mathbf{g} = -g\,\mathbf{j}$, where $0 < \alpha < 90°$, and let T be the time after projection at which $\mathbf{r}\,.\,\mathbf{j} = 0$. Show that

$$uk\sin\alpha = \left(\frac{kT}{1 - \mathrm{e}^{-kT}} - 1\right) g\,.$$

Let β be the acute angle between $\mathbf{v}$ and $\mathbf{i}$ at time T. Show that

$$\tan\beta = \frac{(\mathrm{e}^{kT} - 1)g}{uk\cos\alpha} - \tan\alpha\,.$$

Show further that $\tan\beta > \tan\alpha$ (you may assume that $\sinh kT > kT$) and deduce that $\beta > \alpha$.

10 Two particles X and Y, of equal mass m, lie on a smooth horizontal table and are connected by a light elastic spring of natural length a and modulus of elasticity λ. Two more springs, identical to the first, connect X to a point P on the table and Y to a point Q on the table. The distance between P and Q is $3a$.

Initially, the particles are held so that $XP = a$, $YQ = \frac{1}{2}a$, and $PXYQ$ is a straight line. The particles are then released.

At time t, the particle X is a distance $a + x$ from P and the particle Y is a distance $a + y$ from Q. Show that

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = -\frac{\lambda}{a}(2x + y)$$

and find a similar expression involving $\dfrac{\mathrm{d}^2y}{\mathrm{d}t^2}$. Deduce that

$$x - y = A\cos\omega t + B\sin\omega t$$

where A and B are constants to be determined and $ma\omega^2 = \lambda$. Find a similar expression for $x + y$.

Show that Y will never return to its initial position.

11 A particle P of mass m is connected by two light inextensible strings to two fixed points A and B, with A vertically above B. The string AP has length x. The particle is rotating about the vertical through A and B with angular velocity ω, and both strings are taut. Angles PAB and PBA are α and β, respectively.

Find the tensions T_A and T_B in the strings AP and BP (respectively), and hence show that $\omega^2 x \cos\alpha \geqslant g$.

Consider now the case that $\omega^2 x \cos\alpha = g$. Given that $AB = h$ and $BP = d$, where $h > d$, show that $h\cos\alpha \geqslant \sqrt{h^2 - d^2}$. Show further that

$$mg < T_A \leqslant \frac{mgh}{\sqrt{h^2 - d^2}}\,.$$

Describe the geometry of the strings when T_A attains its upper bound.

Section C: Probability and Statistics

12 The random variable X has probability density function $\mathrm{f}(x)$ (which you may assume is differentiable) and cumulative distribution function $\mathrm{F}(x)$ where $-\infty < x < \infty$. The random variable Y is defined by $Y = \mathrm{e}^X$. You may assume throughout this question that X and Y have unique modes.

(i) Find the median value y_m of Y in terms of the median value x_m of X.

(ii) Show that the probability density function of Y is $\mathrm{f}(\ln y)/y$, and deduce that the mode λ of Y satisfies $\mathrm{f}'(\ln \lambda) = \mathrm{f}(\ln \lambda)$.

(iii) Suppose now that $X \sim \mathrm{N}(\mu, \sigma^2)$, so that

$$\mathrm{f}(x) = \frac{1}{\sigma\sqrt{2\pi}} \mathrm{e}^{-(x-\mu)^2/(2\sigma^2)} .$$

Explain why

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{e}^{-(x-\mu-\sigma^2)^2/(2\sigma^2)} \, \mathrm{d}x = 1$$

and hence show that $\mathrm{E}(Y) = \mathrm{e}^{\mu+\frac{1}{2}\sigma^2}$.

(iv) Show that, when $X \sim \mathrm{N}(\mu, \sigma^2)$,

$$\lambda < y_m < \mathrm{E}(Y) .$$

13 I play a game which has repeated rounds. Before the first round, my score is 0. Each round can have three outcomes:

1. my score is unchanged and the game ends;
2. my score is unchanged and I continue to the next round;
3. my score is increased by one and I continue to the next round.

The probabilities of these outcomes are a, b and c, respectively (the same in each round), where $a + b + c = 1$ and $abc \neq 0$. The random variable N represents my score at the end of a randomly chosen game.

Let $\mathrm{G}(t)$ be the probability generating function of N.

(i) Suppose in the first round, the game ends. Show that the probability generating function conditional on this happening is 1.

(ii) Suppose in the first round, the game continues to the next round with no change in score. Show that the probability generating function conditional on this happening is $\mathrm{G}(t)$.

(iii) By comparing the coefficients of t^n, show that $\mathrm{G}(t) = a + b\mathrm{G}(t) + ct\mathrm{G}(t)$. Deduce that, for $n \geqslant 0$,

$$P(N = n) = \frac{ac^n}{(1-b)^{n+1}}\,.$$

(iv) Show further that, for $n \geqslant 0$,

$$P(N = n) = \frac{\mu^n}{(1+\mu)^{n+1}}\,,$$

where $\mu = \mathrm{E}(N)$.

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Sixth Term Examination Papers **9475**

MATHEMATICS 3 Morning

FRIDAY 19 JUNE 2015 Time: 3 hours

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Additional Materials: Answer Booklet
Formulae Booklet

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Section A: Pure Mathematics

1 **(i)** Let

$$I_n = \int_0^\infty \frac{1}{(1+u^2)^n}\,\mathrm{d}u\,,$$

where n is a positive integer. Show that

$$I_n - I_{n+1} = \frac{1}{2n} I_n$$

and deduce that

$$I_{n+1} = \frac{(2n)!\,\pi}{2^{2n+1}(n!)^2}\,.$$

(ii) Let

$$J = \int_0^\infty \mathrm{f}\big((x-x^{-1})^2\big)\,\mathrm{d}x\,,$$

where f is any function for which the integral exists. Show that

$$J = \int_0^\infty x^{-2}\mathrm{f}\big((x-x^{-1})^2\big)\,\mathrm{d}x = \frac{1}{2}\int_0^\infty (1+x^{-2})\mathrm{f}\big((x-x^{-1})^2\big)\,\mathrm{d}x = \int_0^\infty \mathrm{f}\big(u^2\big)\,\mathrm{d}u\,.$$

(iii) Hence evaluate

$$\int_0^\infty \frac{x^{2n-2}}{(x^4-x^2+1)^n}\,\mathrm{d}x\,,$$

where n is a positive integer.

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2 If s_1, s_2, s_3, ... and t_1, t_2, t_3, ... are sequences of positive numbers, we write

$$(s_n) \leqslant (t_n)$$

to mean

"there exists a positive integer m such that $s_n \leqslant t_n$ whenever $n \geqslant m$".

Determine whether each of the following statements is true or false. In the case of a true statement, you should give a proof which includes an explicit determination of an appropriate m; in the case of a false statement, you should give a counterexample.

(i) $(1000n) \leqslant (n^2)$.

(ii) If it is not the case that $(s_n) \leqslant (t_n)$, then it is the case that $(t_n) \leqslant (s_n)$.

(iii) If $(s_n) \leqslant (t_n)$ and $(t_n) \leqslant (u_n)$, then $(s_n) \leqslant (u_n)$.

(iv) $(n^2) \leqslant (2^n)$.

3 In this question, r and θ are polar coordinates with $r \geqslant 0$ and $-\pi < \theta \leqslant \pi$, and a and b are positive constants.

Let L be a fixed line and let A be a fixed point not lying on L. Then the locus of points that are a fixed distance (call it d) from L measured along lines through A is called a *conchoid of Nicomedes*.

(i) Show that if

$$|r - a\sec\theta| = b\,, \qquad (*)$$

where $a > b$, then $\sec\theta > 0$. Show that all points with coordinates satisfying $(*)$ lie on a certain conchoid of Nicomedes (you should identify L, d and A). Sketch the locus of these points.

(ii) In the case $a < b$, sketch the curve (including the loop for which $\sec\theta < 0$) given by

$$|r - a\sec\theta| = b\,.$$

Find the area of the loop in the case $a = 1$ and $b = 2$.

[Note: $\int \sec\theta\,\mathrm{d}\theta = \ln|\sec\theta + \tan\theta| + C$.]

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4 **(i)** If a, b and c are all real, show that the equation

$$z^3 + az^2 + bz + c = 0 \qquad (*)$$

has at least one real root.

(ii) Let

$$S_1 = z_1 + z_2 + z_3, \quad S_2 = z_1^2 + z_2^2 + z_3^2, \quad S_3 = z_1^3 + z_2^3 + z_3^3\,,$$

where z_1, z_2 and z_3 are the roots of the equation $(*)$. Express a and b in terms of S_1 and S_2, and show that

$$6c = -S_1^3 + 3S_1S_2 - 2S_3\,.$$

(iii) The six real numbers r_k and θ_k $(k = 1,\ 2,\ 3)$, where $r_k > 0$ and $-\pi < \theta_k < \pi$, satisfy

$$\sum_{k=1}^{3} r_k \sin(\theta_k) = 0\,, \quad \sum_{k=1}^{3} r_k^2 \sin(2\theta_k) = 0\,, \quad \sum_{k=1}^{3} r_k^3 \sin(3\theta_k) = 0\,.$$

Show that $\theta_k = 0$ for at least one value of k.

Show further that if $\theta_1 = 0$ then $\theta_2 = -\theta_3$.

5 **(i)** In the following argument to show that $\sqrt{2}$ is irrational, give proofs appropriate for steps 3, 5 and 6.

1. Assume that $\sqrt{2}$ is rational.
2. Define the set S to be the set of positive integers with the following property:

   n is in S if and only if $n\sqrt{2}$ is an integer.

3. Show that the set S contains at least one positive integer.
4. Define the integer k to be the smallest positive integer in S.
5. Show that $(\sqrt{2} - 1)k$ is in S.
6. Show that steps 4 and 5 are contradictory and hence that $\sqrt{2}$ is irrational.

(ii) Prove that $2^{\frac{1}{3}}$ is rational if and only if $2^{\frac{2}{3}}$ is rational.

Use an argument similar to that of part (i) to prove that $2^{\frac{1}{3}}$ and $2^{\frac{2}{3}}$ are irrational.

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6 **(i)** Let w and z be complex numbers, and let $u = w + z$ and $v = w^2 + z^2$. Prove that w and z are real if and only if u and v are real and $u^2 \leqslant 2v$.

(ii) The complex numbers u, w and z satisfy the equations

$$\begin{aligned} w + z - u &= 0 \\ w^2 + z^2 - u^2 &= -\tfrac{2}{3} \\ w^3 + z^3 - \lambda u &= -\lambda \end{aligned}$$

where λ is a positive real number. Show that for all values of λ except one (which you should find) there are three possible values of u, all real.

Are w and z necessarily real? Give a proof or counterexample.

7 An operator D is defined, for any function f, by

$$\mathrm{Df}(x) = x\frac{\mathrm{df}(x)}{\mathrm{d}x}.$$

The notation D^n means that D is applied n times; for example

$$\mathrm{D}^2\mathrm{f}(x) = x\frac{\mathrm{d}}{\mathrm{d}x}\left(x\frac{\mathrm{df}(x)}{\mathrm{d}x}\right).$$

Show that, for any constant a, $\mathrm{D}^2 x^a = a^2 x^a$.

(i) Show that if $\mathrm{P}(x)$ is a polynomial of degree r (where $r \geqslant 1$) then, for any positive integer n, $\mathrm{D}^n\mathrm{P}(x)$ is also a polynomial of degree r.

(ii) Show that if n and m are positive integers with $n < m$, then $\mathrm{D}^n(1-x)^m$ is divisible by $(1-x)^{m-n}$.

(iii) Deduce that, if m and n are positive integers with $n < m$, then

$$\sum_{r=0}^{m}(-1)^r\binom{m}{r}r^n = 0\,.$$

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8 **(i)** Show that under the changes of variable $x = r\cos\theta$ and $y = r\sin\theta$, where r is a function of θ with $r > 0$, the differential equation

$$(y + x)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x$$

becomes

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} + r = 0\,.$$

Sketch a solution in the x-y plane.

(ii) Show that the solutions of

$$\left(y + x - x(x^2 + y^2)\right)\frac{\mathrm{d}y}{\mathrm{d}x} = y - x - y(x^2 + y^2)$$

can be written in the form

$$r^2 = \frac{1}{1 + A\mathrm{e}^{2\theta}}$$

and sketch the different forms of solution that arise according to the value of A.

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Section B: Mechanics

9 A particle P of mass m moves on a smooth fixed straight horizontal rail and is attached to a fixed peg Q by a light elastic string of natural length a and modulus λ. The peg Q is a distance a from the rail. Initially P is at rest with $PQ = a$.

An impulse imparts to P a speed v along the rail. Let x be the displacement at time t of P from its initial position. Obtain the equation

$$\dot{x}^2 = v^2 - k^2\left(\sqrt{x^2 + a^2} - a\right)^2$$

where $k^2 = \lambda/(ma)$, $k > 0$ and the dot denotes differentiation with respect to t.

Find, in terms of k, a and v, the greatest value, x_0, attained by x. Find also the acceleration of P at $x = x_0$.

Obtain, in the form of an integral, an expression for the period of the motion. Show that, in the case $v \ll ka$ (that is, v is much less than ka), this is approximately

$$\sqrt{\frac{32a}{kv}} \int_0^1 \frac{1}{\sqrt{1 - u^4}}\,\mathrm{d}u\,.$$

10 A light rod of length $2a$ has a particle of mass m attached to each end and it moves in a vertical plane. The midpoint of the rod has coordinates (x, y), where the x-axis is horizontal (within the plane of motion) and y is the height above a horizontal table. Initially, the rod is vertical, and at time t later it is inclined at an angle θ to the vertical.

Show that the velocity of one particle can be written in the form

$$\begin{pmatrix} \dot{x} + a\dot{\theta}\cos\theta \\ \dot{y} - a\dot{\theta}\sin\theta \end{pmatrix}$$

and that

$$m\begin{pmatrix} \ddot{x} + a\ddot{\theta}\cos\theta - a\dot{\theta}^2\sin\theta \\ \ddot{y} - a\ddot{\theta}\sin\theta - a\dot{\theta}^2\cos\theta \end{pmatrix} = -T\begin{pmatrix} \sin\theta \\ \cos\theta \end{pmatrix} - mg\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

where the dots denote differentiation with respect to time t and T is the tension in the rod. Obtain the corresponding equations for the other particle.

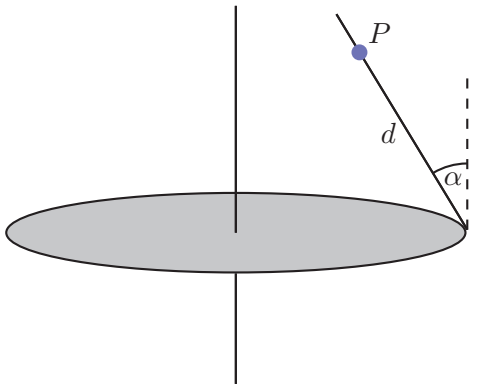
Deduce that $\ddot{x} = 0$, $\ddot{y} = -g$ and $\ddot{\theta} = 0$.

Initially, the midpoint of the rod is a height h above the table, the velocity of the higher particle is $\begin{pmatrix} u \\ v \end{pmatrix}$, and the velocity of the lower particle is $\begin{pmatrix} 0 \\ v \end{pmatrix}$. Given that the two particles hit the table for the first time simultaneously, when the rod has rotated by $\frac{1}{2}\pi$, show that

$$2hu^2 = \pi^2 a^2 g - 2\pi uva\,.$$

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11 **(i)** A horizontal disc of radius r rotates about a vertical axis through its centre with angular speed ω. One end of a light rod is fixed by a smooth hinge to the edge of the disc so that it can rotate freely in a vertical plane through the centre of the disc. A particle P of mass m is attached to the rod at a distance d from the hinge. The rod makes a constant angle α with the upward vertical, as shown in the diagram, and $d\sin\alpha < r$.



By considering moments about the hinge for the (light) rod, show that the force exerted on the rod by P is parallel to the rod.

Show also that

$$r\cot\alpha = a + d\cos\alpha\,,$$

where $a = \dfrac{g}{\omega^2}$. State clearly the direction of the force exerted by the hinge on the rod, and find an expression for its magnitude in terms of m, g and α.

(ii) The disc and rod rotate as in part (i), but two particles (instead of P) are attached to the rod. The masses of the particles are m_1 and m_2 and they are attached to the rod at distances d_1 and d_2 from the hinge, respectively. The rod makes a constant angle β with the upward vertical and $d_1\sin\beta < d_2\sin\beta < r$. Show that β satisfies an equation of the form

$$r\cot\beta = a + b\cos\beta\,,$$

where b should be expressed in terms of d_1, d_2, m_1 and m_2.

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Section C: Probability and Statistics

12 A 6-sided fair die has the numbers 1, 2, 3, 4, 5, 6 on its faces. The die is thrown n times, the outcome (the number on the top face) of each throw being independent of the outcome of any other throw. The random variable S_n is the sum of the outcomes.

(i) The random variable R_n is the remainder when S_n is divided by 6. Write down the probability generating function, $\mathrm{G}(x)$, of R_1 and show that the probability generating function of R_2 is also $\mathrm{G}(x)$. Use a generating function to find the probability that S_n is divisible by 6.

(ii) The random variable T_n is the remainder when S_n is divided by 5. Write down the probability generating function, $\mathrm{G}_1(x)$, of T_1 and show that $\mathrm{G}_2(x)$, the probability generating function of T_2, is given by

$$\mathrm{G}_2(x) = \tfrac{1}{36}(x^2 + 7y)$$

where $y = 1 + x + x^2 + x^3 + x^4$.

Obtain the probability generating function of T_n and hence show that the probability that S_n is divisible by 5 is

$$\frac{1}{5}\left(1 - \frac{1}{6^n}\right)$$

if n is not divisible by 5. What is the corresponding probability if n is divisible by 5?

13 Each of the two independent random variables X and Y is uniformly distributed on the interval $[0, 1]$.

(i) By considering the lines $x + y =$ constant in the x-y plane, find the cumulative distribution function of $X + Y$.

Hence show that the probability density function f of $(X + Y)^{-1}$ is given by

$$\mathrm{f}(t) = \begin{cases} 2t^{-2} - t^{-3} & \text{for } \frac{1}{2} \leqslant t \leqslant 1 \\ t^{-3} & \text{for } 1 \leqslant t < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Evaluate $\mathrm{E}\left(\dfrac{1}{X+Y}\right)$.

(ii) Find the cumulative distribution function of Y/X and use this result to find the probability density function of $\dfrac{X}{X+Y}$.

Write down $\mathrm{E}\left(\dfrac{X}{X+Y}\right)$ and verify your result by integration.

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Sixth Term Examination Papers **9475**

MATHEMATICS 3 Afternoon

THURSDAY 23 JUNE 2016 Time: 3 hours

4156028566

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Write your name, centre number and candidate number in the spaces on the answer booklet.

Begin each answer on a new page.

Write the numbers of the questions you answer in the order attempted on the front of the answer booklet.

INFORMATION FOR CANDIDATES

Each question is marked out of 20. There is no restriction of choice.

All questions attempted will be marked.

Your final mark will be based on the **six** questions for which you gain the highest marks.

You are advised to concentrate on no more than **six** questions. Little credit will be given for fragmentary answers.

You are provided with a Mathematical Formulae Booklet.

Calculators are not permitted.

Please wait to be told you may begin before turning this page.

This question paper consists of 8 printed pages and 4 blank pages.

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Section A: Pure Mathematics

1 Let

$$I_n = \int_{-\infty}^{\infty} \frac{1}{(x^2 + 2ax + b)^n} \, \mathrm{d}x \,,$$

where a and b are constants with $b > a^2$, and n is a positive integer.

(i) By using the substitution $x + a = \sqrt{b - a^2} \tan u$, or otherwise, show that

$$I_1 = \frac{\pi}{\sqrt{b - a^2}} \,.$$

(ii) Show that $2n(b - a^2)\, I_{n+1} = (2n - 1)\, I_n$.

(iii) Hence prove by induction that

$$I_n = \frac{\pi}{2^{2n-2}(b - a^2)^{n - \frac{1}{2}}} \binom{2n-2}{n-1} \,.$$

2 The distinct points $P(ap^2, 2ap)$, $Q(aq^2, 2aq)$ and $R(ar^2, 2ar)$ lie on the parabola $y^2 = 4ax$, where $a > 0$. The points are such that the normal to the parabola at Q and the normal to the parabola at R both pass through P.

(i) Show that $q^2 + qp + 2 = 0$.

(ii) Show that QR passes through a certain point that is independent of the choice of P.

(iii) Let T be the point of intersection of OP and QR, where O is the coordinate origin. Show that T lies on a line that is independent of the choice of P.

Show further that the distance from the x-axis to T is less than $\dfrac{a}{\sqrt{2}}$.

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3 **(i)** Given that

$$\int \frac{x^3-2}{(x+1)^2}\,\mathrm{e}^x \mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,\mathrm{e}^x + \text{constant}\,,$$

where $\mathrm{P}(x)$and $\mathrm{Q}(x)$ are polynomials, show that $\mathrm{Q}(x)$ has a factor of $x+1$.

Show also that the degree of $\mathrm{P}(x)$ is exactly one more than the degree of $\mathrm{Q}(x)$, and find $\mathrm{P}(x)$ in the case $\mathrm{Q}(x) = x+1$.

(ii) Show that there are no polynomials $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ such that

$$\int \frac{1}{x+1}\,\mathrm{e}^x \mathrm{d}x = \frac{\mathrm{P}(x)}{\mathrm{Q}(x)}\,\mathrm{e}^x + \text{constant}\,.$$

You need consider only the case when $\mathrm{P}(x)$ and $\mathrm{Q}(x)$ have no common factors.

4 **(i)** By considering $\dfrac{1}{1+x^r} - \dfrac{1}{1+x^{r+1}}$ for $|x| \neq 1$, simplify

$$\sum_{r=1}^{N} \frac{x^r}{(1+x^r)(1+x^{r+1})}\,.$$

Show that, for $|x| < 1$,

$$\sum_{r=1}^{\infty} \frac{x^r}{(1+x^r)(1+x^{r+1})} = \frac{x}{1-x^2}\,.$$

(ii) Deduce that

$$\sum_{r=1}^{\infty} \operatorname{sech}(ry)\operatorname{sech}((r+1)y) = 2\mathrm{e}^{-y}\operatorname{cosech}(2y)$$

for $y > 0$.

Hence simplify

$$\sum_{r=-\infty}^{\infty} \operatorname{sech}(ry)\operatorname{sech}((r+1)y)\,,$$

for $y > 0$.

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5 **(i)** By considering the binomial expansion of $(1+x)^{2m+1}$, prove that

$$\binom{2m+1}{m} < 2^{2m},$$

for any positive integer m.

(ii) For any positive integers r and s with $r < s$, $P_{r,s}$ is defined as follows: $P_{r,s}$ is the product of all the prime numbers greater than r and less than or equal to s, if there are any such primes numbers; if there are no such primes numbers, then $P_{r,s} = 1$.

For example, $P_{3,7} = 35$, $P_{7,10} = 1$ and $P_{14,18} = 17$.

Show that, for any positive integer m, $P_{m+1\,,\,2m+1}$ divides $\binom{2m+1}{m}$, and deduce that

$$P_{m+1\,,\,2m+1} < 2^{2m}.$$

(iii) Show that, if $P_{1,k} < 4^k$ for $k = 2, 3, \ldots, 2m$, then $P_{1,2m+1} < 4^{2m+1}$.

(iv) Prove that $\mathrm{P}_{1,n} < 4^n$ for $n \geqslant 2$.

6 Show, by finding R and γ, that $A\sinh x + B\cosh x$ can be written in the form $R\cosh(x+\gamma)$ if $B > A > 0$. Determine the corresponding forms in the other cases that arise, for $A > 0$, according to the value of B.

Two curves have equations $y = \operatorname{sech} x$ and $y = a\tanh x + b$, where $a > 0$.

(i) In the case $b > a$, show that if the curves intersect then the x-coordinates of the points of intersection can be written in the form

$$\pm \operatorname{arcosh}\left(\frac{1}{\sqrt{b^2 - a^2}}\right) - \operatorname{artanh}\frac{a}{b}.$$

(ii) Find the corresponding result in the case $a > b > 0$.

(iii) Find necessary and sufficient conditions on a and b for the curves to intersect at two distinct points.

(iv) Find necessary and sufficient conditions on a and b for the curves to touch and, given that they touch, express the y-coordinate of the point of contact in terms of a.

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7 Let $\omega = \mathrm{e}^{2\pi\mathrm{i}/n}$, where n is a positive integer. Show that, for any complex number z,

$$(z-1)(z-\omega)\cdots(z-\omega^{n-1}) = z^n - 1\,.$$

The points X_0, X_1, ... , X_{n-1} lie on a circle with centre O and radius 1, and are the vertices of a regular polygon.

(i) The point P is equidistant from X_0 and X_1. Show that, if n is even,

$$|PX_0| \times |PX_1| \times \cdots \times |PX_{n-1}| = |OP|^n + 1\,,$$

where $|PX_k|$ denotes the distance from P to X_k.

Give the corresponding result when n is odd. (There are two cases to consider.)

(ii) Show that

$$|X_0X_1| \times |X_0X_2| \times \cdots \times |X_0X_{n-1}| = n\,.$$

8 **(i)** The function f satisfies, for all x, the equation

$$\mathrm{f}(x) + (1-x)\mathrm{f}(-x) = x^2\,.$$

Show that $\mathrm{f}(-x) + (1+x)\mathrm{f}(x) = x^2$. Hence find $\mathrm{f}(x)$ in terms of x. You should verify that your function satisfies the original equation.

(ii) The function K is defined, for $x \neq 1$, by

$$\mathrm{K}(x) = \frac{x+1}{x-1}\,.$$

Show that, for $x \neq 1$, $\mathrm{K}(\mathrm{K}(x)) = x$.

The function g satisfies the equation

$$\mathrm{g}(x) + x\,\mathrm{g}\Big(\frac{x+1}{x-1}\Big) = x \qquad (x \neq 1)\,.$$

Show that, for $x \neq 1$, $\mathrm{g}(x) = \dfrac{2x}{x^2+1}$.

(iii) Find $\mathrm{h}(x)$, for $x \neq 0$, $x \neq 1$, given that

$$\mathrm{h}(x) + \mathrm{h}\Big(\frac{1}{1-x}\Big) = 1 - x - \frac{1}{1-x} \qquad (x \neq 0,\quad x \neq 1)\,.$$

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Section B: Mechanics

9 Three pegs P, Q and R are fixed on a smooth horizontal table in such a way that they form the vertices of an equilateral triangle of side $2a$. A particle X of mass m lies on the table. It is attached to the pegs by three springs, PX, QX and RX, each of modulus of elasticity λ and natural length l, where $l < \frac{2}{\sqrt{3}}\,a$. Initially the particle is in equilibrium. Show that the extension in each spring is $\frac{2}{\sqrt{3}}\,a - l$.

The particle is then pulled a small distance directly towards P and released. Show that the tension T in the spring RX is given by

$$T = \frac{\lambda}{l}\left(\sqrt{\frac{4a^2}{3} + \frac{2ax}{\sqrt{3}} + x^2} \; - l\right),$$

where x is the displacement of X from its equilibrium position.

Show further that the particle performs approximate simple harmonic motion with period

$$2\pi\sqrt{\frac{4mla}{3(4a - \sqrt{3}\,l)\lambda}}\,.$$

10 A smooth plane is inclined at an angle α to the horizontal. A particle P of mass m is attached to a fixed point A above the plane by a light inextensible string of length a. The particle rests in equilibrium on the plane, and the string makes an angle β with the plane.

The particle is given a horizontal impulse parallel to the plane so that it has an initial speed of u. Show that the particle will not immediately leave the plane if $ag\cos(\alpha + \beta) > u^2 \tan\beta$.

Show further that a necessary condition for the particle to perform a complete circle whilst in contact with the plane is $6\tan\alpha\tan\beta < 1$.

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11 A car of mass m travels along a straight horizontal road with its engine working at a constant rate P. The resistance to its motion is such that the acceleration of the car is zero when it is moving with speed $4U$.

(i) Given that the resistance is proportional to the car's speed, show that the distance X_1 travelled by the car while it accelerates from speed U to speed $2U$, is given by

$$\lambda X_1 = 2 \ln \tfrac{9}{5} - 1 \,,$$

where $\lambda = P/(16mU^3)$.

(ii) Given instead that the resistance is proportional to the square of the car's speed, show that the distance X_2 travelled by the car while it accelerates from speed U to speed $2U$ is given by

$$\lambda X_2 = \tfrac{4}{3} \ln \tfrac{9}{8} \,.$$

(iii) Given that $3.17 < \ln 24 < 3.18$ and $1.60 < \ln 5 < 1.61$, determine which is the larger of X_1 and X_2.

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Section C: Probability and Statistics

12 Let X be a random variable with mean μ and standard deviation σ. *Chebyshev's inequality*, which you may use without proof, is

$$\mathrm{P}\left(|X-\mu| > k\sigma\right) \leqslant \frac{1}{k^2}\,,$$

where k is any positive number.

(i) The probability of a biased coin landing heads up is 0.2. It is thrown $100n$ times, where n is an integer greater than 1. Let α be the probability that the coin lands heads up N times, where $16n \leqslant N \leqslant 24n$.

Use Chebyshev's inequality to show that

$$\alpha \geqslant 1 - \frac{1}{n}\,.$$

(ii) Use Chebyshev's inequality to show that

$$1 + n + \frac{n^2}{2!} + \cdots + \frac{n^{2n}}{(2n)!} \geqslant \left(1 - \frac{1}{n}\right)\mathrm{e}^n\,.$$

13 Given a random variable X with mean μ and standard deviation σ, we define the *kurtosis*, κ, of X by

$$\kappa = \frac{\mathrm{E}\big((X-\mu)^4\big)}{\sigma^4} - 3\,.$$

Show that the random variable $X - a$, where a is a constant, has the same kurtosis as X.

(i) Show by integration that a random variable which is Normally distributed with mean 0 has kurtosis 0.

(ii) Let $Y_1, Y_2, \ldots, Y_n$ be n independent, identically distributed, random variables with mean 0, and let $T = \sum_{r=1}^{n} Y_r$. Show that

$$\mathrm{E}(T^4) = \sum_{r=1}^{n} \mathrm{E}(Y_r^4) + 6\sum_{r=1}^{n-1}\sum_{s=r+1}^{n} \mathrm{E}(Y_s^2)\mathrm{E}(Y_r^2)\,.$$

(iii) Let $X_1, X_2, \ldots, X_n$ be n independent, identically distributed, random variables each with kurtosis κ. Show that the kurtosis of their sum is $\dfrac{\kappa}{n}$.

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Sixth Term Examination Papers **9475**

MATHEMATICS 3 Morning

THURSDAY 22 JUNE 2017 Time: 3 hours

0707967579

Additional Materials: Answer Booklet
Formulae Booklet

INSTRUCTIONS TO CANDIDATES

Please read this page carefully, but do not open this question paper until you are told that you may do so.

Please also read the additional instructions on the front of the answer booklet.

Write your name, centre number, candidate number, date of birth, and circle the paper number in the spaces provided on the answer booklet.

Make sure you fill in page 1 **AND** page 3 of the answer booklet with your details.

INFORMATION FOR CANDIDATES

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All questions attempted will be marked.

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PV2

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Section A: Pure Mathematics

1 **(i)** Prove that, for any positive integers n and r,

$$\frac{1}{{}^{n+r}\mathrm{C}_{r+1}} = \frac{r+1}{r}\left(\frac{1}{{}^{n+r-1}\mathrm{C}_{r}} - \frac{1}{{}^{n+r}\mathrm{C}_{r}}\right).$$

Hence determine

$$\sum_{n=1}^{\infty} \frac{1}{{}^{n+r}\mathrm{C}_{r+1}},$$

and deduce that $\displaystyle\sum_{n=2}^{\infty} \frac{1}{{}^{n+2}\mathrm{C}_{3}} = \frac{1}{2}$.

(ii) Show that, for $n \geqslant 3$,

$$\frac{3!}{n^3} < \frac{1}{{}^{n+1}\mathrm{C}_{3}} \quad \text{and} \quad \frac{20}{{}^{n+1}\mathrm{C}_{3}} - \frac{1}{{}^{n+2}\mathrm{C}_{5}} < \frac{5!}{n^3}.$$

By summing these inequalities for $n \geqslant 3$, show that

$$\frac{115}{96} < \sum_{n=1}^{\infty} \frac{1}{n^3} < \frac{116}{96}.$$

Note: ${}^{n}\mathrm{C}_{r}$ is another notation for $\binom{n}{r}$.

2 The transformation R in the complex plane is a rotation (anticlockwise) by an angle θ about the point represented by the complex number a. The transformation S in the complex plane is a rotation (anticlockwise) by an angle ϕ about the point represented by the complex number b.

(i) The point P is represented by the complex number z. Show that the image of P under R is represented by

$$\mathrm{e}^{\mathrm{i}\theta} z + a(1 - \mathrm{e}^{\mathrm{i}\theta}).$$

(ii) Show that the transformation SR (equivalent to R followed by S) is a rotation about the point represented by c, where

$$c \sin \tfrac{1}{2}(\theta + \phi) = a\,\mathrm{e}^{\mathrm{i}\phi/2} \sin \tfrac{1}{2}\theta + b\,\mathrm{e}^{-\mathrm{i}\theta/2} \sin \tfrac{1}{2}\phi,$$

provided $\theta + \phi \neq 2n\pi$ for any integer n.

What is the transformation SR if $\theta + \phi = 2\pi$?

(iii) Under what circumstances is $RS = SR$?

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[Turn over

3 Let α, β, γ and δ be the roots of the quartic equation

$$x^4 + px^3 + qx^2 + rx + s = 0\,.$$

You are given that, for any such equation, $\alpha\beta + \gamma\delta\,,\ \ \alpha\gamma + \beta\delta$ and $\alpha\delta + \beta\gamma$ satisfy a cubic equation of the form

$$y^3 + Ay^2 + (pr - 4s)y + (4qs - p^2s - r^2) = 0\,.$$

Determine A.

Now consider the quartic equation given by $p = 0\,,\ q = 3\,,\ r = -6$ and $s = 10\,.$

(i) Find the value of $\alpha\beta + \gamma\delta$, given that it is the largest root of the corresponding cubic equation.

(ii) Hence, using the values of q and s, find the value of $(\alpha+\beta)(\gamma+\delta)$ and the value of $\alpha\beta$ given that $\alpha\beta > \gamma\delta\,.$

(iii) Using these results, and the values of p and r, solve the quartic equation.

4 For any function f satisfying $\mathrm{f}(x) > 0$, we define the *geometric mean*, F, by

$$\mathrm{F}(y) \ = \ \mathrm{e}^{\frac{1}{y}\int_0^y \ln \mathrm{f}(x)\,\mathrm{d}x} \qquad (y > 0)\,.$$

(i) The function f satisfies $\mathrm{f}(x) > 0$ and a is a positive number with $a \neq 1$. Prove that

$$\mathrm{F}(y) = \ a^{\frac{1}{y}\int_0^y \log_a \mathrm{f}(x)\,\mathrm{d}x}\,.$$

(ii) The functions f and g satisfy $\mathrm{f}(x) > 0$ and $\mathrm{g}(x) > 0$, and the function h is defined by $\mathrm{h}(x) = \mathrm{f}(x)\mathrm{g}(x)$. Their geometric means are F, G and H, respectively. Show that $\mathrm{H}(y) = \mathrm{F}(y)\mathrm{G}(y)\,.$

(iii) Prove that, for any positive number b, the geometric mean of b^x is $\sqrt{b^y}\,.$

(iv) Prove that, if $\mathrm{f}(x) > 0$ and the geometric mean of $\mathrm{f}(x)$ is $\sqrt{\mathrm{f}(y)}$, then $\mathrm{f}(x) = b^x$ for some positive number b.

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5 The point with cartesian coordinates (x, y) lies on a curve with polar equation $r = \mathrm{f}(\theta)$. Find an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ in terms of $\mathrm{f}(\theta)$, $\mathrm{f}'(\theta)$ and $\tan\theta$.

Two curves, with polar equations $r = \mathrm{f}(\theta)$ and $r = \mathrm{g}(\theta)$, meet at right angles. Show that where they meet

$$\mathrm{f}'(\theta)\mathrm{g}'(\theta) + \mathrm{f}(\theta)\mathrm{g}(\theta) = 0\,.$$

The curve C has polar equation $r = \mathrm{f}(\theta)$ and passes through the point given by $r = 4$, $\theta = -\frac{1}{2}\pi$. For each positive value of a, the curve with polar equation $r = a(1 + \sin\theta)$ meets C at right angles. Find $\mathrm{f}(\theta)$.

Sketch on a single diagram the three curves with polar equations $r = 1+\sin\theta$, $r = 4(1+\sin\theta)$ and $r = \mathrm{f}(\theta)$.

6 *In this question, you are not permitted to use any properties of trigonometric functions or inverse trigonometric functions.*

The function T is defined for $x > 0$ by

$$\mathrm{T}(x) = \int_0^x \frac{1}{1+u^2}\,\mathrm{d}u\,,$$

and $T_\infty = \displaystyle\int_0^\infty \frac{1}{1+u^2}\,\mathrm{d}u$ (which has a finite value).

(i) By making an appropriate substitution in the integral for $\mathrm{T}(x)$, show that

$$\mathrm{T}(x) = \mathrm{T}_\infty - \mathrm{T}(x^{-1})\,.$$

(ii) Let $v = \dfrac{u+a}{1-au}$, where a is a constant. Verify that, for $u \neq a^{-1}$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{1+v^2}{1+u^2}\,.$$

Hence show that, for $a > 0$ and $x < \dfrac{1}{a}$,

$$\mathrm{T}(x) = \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a)\,.$$

Deduce that

$$\mathrm{T}(x^{-1}) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{x+a}{1-ax}\right) - \mathrm{T}(a^{-1})$$

and hence that, for $b > 0$ and $y > \dfrac{1}{b}$,

$$\mathrm{T}(y) = 2\mathrm{T}_\infty - \mathrm{T}\left(\frac{y+b}{by-1}\right) - \mathrm{T}(b)\,.$$

(iii) Use the above results to show that $\mathrm{T}(\sqrt{3}) = \frac{2}{3}\mathrm{T}_\infty$ and $\mathrm{T}(\sqrt{2}-1) = \frac{1}{4}\mathrm{T}_\infty$.

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[Turn over

7 Show that the point T with coordinates

$$\left(\frac{a(1-t^2)}{1+t^2}\,,\ \frac{2bt}{1+t^2}\right) \qquad (*)$$

(where a and b are non-zero) lies on the ellipse

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1\,.$$

(i) The line L is the tangent to the ellipse at T. The point (X,Y) lies on L, and $X^2\neq a^2$. Show that

$$(a+X)bt^2-2aYt+b(a-X)=0\,.$$

Deduce that if $a^2Y^2>(a^2-X^2)b^2$, then there are two distinct lines through (X,Y) that are tangents to the ellipse. Interpret this result geometrically. Show, by means of a sketch, that the result holds also if $X^2=a^2$.

(ii) The distinct points P and Q are given by $(*)$, with $t=p$ and $t=q$, respectively. The tangents to the ellipse at P and Q meet at the point with coordinates (X,Y), where $X^2\neq a^2$. Show that

$$(a+X)pq=a-X$$

and find an expression for $p+q$ in terms of a, b, X and Y.

Given that the tangents meet the y-axis at points $(0,y_1)$ and $(0,y_2)$, where $y_1+y_2=2b$, show that

$$\frac{X^2}{a^2}+\frac{Y}{b}=1\,.$$

8 Prove that, for any numbers a_1, a_2, ..., and b_1, b_2, ..., and for $n\geqslant 1$,

$$\sum_{m=1}^{n}a_m(b_{m+1}-b_m)=a_{n+1}b_{n+1}-a_1b_1-\sum_{m=1}^{n}b_{m+1}(a_{m+1}-a_m)\,.$$

(i) By setting $b_m=\sin mx$, show that

$$\sum_{m=1}^{n}\cos(m+\tfrac{1}{2})x=\tfrac{1}{2}\big(\sin(n+1)x-\sin x\big)\operatorname{cosec}\tfrac{1}{2}x\,.$$

Note: $\sin A-\sin B=2\cos\big(\frac{A+B}{2}\big)\sin\big(\frac{A-B}{2}\big)$.

(ii) Show that

$$\sum_{m=1}^{n}m\sin mx=\big(p\sin(n+1)x+q\sin nx\big)\operatorname{cosec}^2\tfrac{1}{2}x\,,$$

where p and q are to be determined in terms of n.

Note: $2\sin A\sin B=\cos(A-B)-\cos(A+B)$;

$2\cos A\sin B=\sin(A+B)-\sin(A-B)$.

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Section B: Mechanics

9 Two particles A and B of masses m and $2m$, respectively, are connected by a light spring of natural length a and modulus of elasticity λ. They are placed on a smooth horizontal table with AB perpendicular to the edge of the table, and A is held on the edge of the table. Initially the spring is at its natural length.

Particle A is released. At a time t later, particle A has dropped a distance y and particle B has moved a distance x from its initial position (where $x < a$). Show that $y + 2x = \frac{1}{2}gt^2$.

The value of λ is such that particle B reaches the edge of the table at a time T given by $T = \sqrt{6a/g}$. By considering the total energy of the system (without solving any differential equations), show that the speed of particle B at this time is $\sqrt{2ag/3}$.

10 A uniform rod PQ of mass m and length $3a$ is freely hinged at P.

The rod is held horizontally and a particle of mass m is placed on top of the rod at a distance ℓ from P, where $\ell < 2a$. The coefficient of friction between the rod and the particle is μ.

The rod is then released. Show that, while the particle does not slip along the rod,

$$(3a^2 + \ell^2)\dot{\theta}^2 = g(3a + 2\ell)\sin\theta\,,$$

where θ is the angle through which the rod has turned, and the dot denotes the time derivative.

Hence, or otherwise, find an expression for $\ddot{\theta}$ and show that the normal reaction of the rod on the particle is non-zero when θ is acute.

Show further that, when the particle is on the point of slipping,

$$\tan\theta = \frac{\mu a(2a - \ell)}{2(\ell^2 + a\ell + a^2)}\,.$$

What happens at the moment the rod is released if, instead, $\ell > 2a$?

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[Turn over

11 A railway truck, initially at rest, can move forwards without friction on a long straight horizontal track. On the truck, n guns are mounted parallel to the track and facing backwards, where $n > 1$. Each of the guns is loaded with a single projectile of mass m. The mass of the truck and guns (but not including the projectiles) is M.

When a gun is fired, the projectile leaves its muzzle horizontally with a speed $v - V$ relative to the ground, where V is the speed of the truck immediately before the gun is fired.

(i) All n guns are fired simultaneously. Find the speed, u, with which the truck moves, and show that the kinetic energy, K, which is gained by the system (truck, guns and projectiles) is given by

$$K = \tfrac{1}{2}nmv^2\left(1 + \frac{nm}{M}\right).$$

(ii) Instead, the guns are fired one at a time. Let u_r be the speed of the truck when r guns have been fired, so that $u_0 = 0$. Show that, for $1 \leqslant r \leqslant n$,

$$u_r - u_{r-1} = \frac{mv}{M + (n-r)m} \qquad (*)$$

and hence that $u_n < u$.

(iii) Let K_r be the total kinetic energy of the system when r guns have been fired (one at a time), so that $K_0 = 0$. Using $(*)$, show that, for $1 \leqslant r \leqslant n$,

$$K_r - K_{r-1} = \tfrac{1}{2}mv^2 + \tfrac{1}{2}mv(u_r - u_{r-1})$$

and hence show that

$$K_n = \tfrac{1}{2}nmv^2 + \tfrac{1}{2}mvu_n.$$

Deduce that $K_n < K$.

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Section C: Probability and Statistics

12 The discrete random variables X and Y can each take the values $1, \ldots, n$ (where $n \geqslant 2$). Their joint probability distribution is given by

$$\mathrm{P}(X = x,\ Y = y) = k(x + y)\,,$$

where k is a constant.

(i) Show that

$$\mathrm{P}(X = x) = \frac{n + 1 + 2x}{2n(n + 1)}\,.$$

Hence determine whether X and Y are independent.

(ii) Show that the covariance of X and Y is negative.

13 The random variable X has mean μ and variance σ^2, and the function V is defined, for $-\infty < x < \infty$, by

$$\mathrm{V}(x) = \mathrm{E}\big((X - x)^2\big).$$

Express $\mathrm{V}(x)$ in terms of x, μ and σ.

The random variable Y is defined by $Y = \mathrm{V}(X)$. Show that

$$\mathrm{E}(Y) = 2\sigma^2. \qquad (*)$$

Now suppose that X is uniformly distributed on the interval $0 \leqslant x \leqslant 1$. Find $\mathrm{V}(x)$. Find also the probability density function of Y and use it to verify that $(*)$ holds in this case.

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Cambridge Assessment Admissions Testing

Sixth Term Examination Papers **9475**
MATHEMATICS 3 Morning
Thursday 21 June 2018 Time: 3 hours

2682044645

Additional Materials: Answer Booklet
Formulae Booklet

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INFORMATION FOR CANDIDATES

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PV1

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Section A: Pure Mathematics

1 **(i)** The function f is given by

$$\mathrm{f}(\beta) = \beta - \frac{1}{\beta} - \frac{1}{\beta^2} \qquad (\beta \neq 0)\,.$$

Find the stationary point of the curve $y = \mathrm{f}(\beta)$ and sketch the curve.

Sketch also the curve $y = \mathrm{g}(\beta)$, where

$$\mathrm{g}(\beta) = \beta + \frac{3}{\beta} - \frac{1}{\beta^2} \qquad (\beta \neq 0)\,.$$

(ii) Let u and v be the roots of the equation

$$x^2 + \alpha x + \beta = 0\,,$$

where $\beta \neq 0$. Obtain expressions in terms of α and β for $u + v + \dfrac{1}{uv}$ and $\dfrac{1}{u} + \dfrac{1}{v} + uv$.

(iii) Given that $u + v + \dfrac{1}{uv} = -1$, and that u and v are real, show that $\dfrac{1}{u} + \dfrac{1}{v} + uv \leqslant -1$.

(iv) Given instead that $u + v + \dfrac{1}{uv} = 3$, and that u and v are real, find the greatest value of $\dfrac{1}{u} + \dfrac{1}{v} + uv$.

2 The sequence of functions y_0, y_1, y_2, ... is defined by $y_0 = 1$ and, for $n \geqslant 1$,

$$y_n = (-1)^n \frac{1}{z} \frac{\mathrm{d}^n z}{\mathrm{d}x^n}\,,$$

where $z = \mathrm{e}^{-x^2}$.

(i) Show that $\dfrac{\mathrm{d}y_n}{\mathrm{d}x} = 2xy_n - y_{n+1}$ for $n \geqslant 1$.

(ii) Prove by induction that, for $n \geqslant 1$,

$$y_{n+1} = 2xy_n - 2ny_{n-1}\,.$$

Deduce that, for $n \geqslant 1$,

$$y_{n+1}^2 - y_n y_{n+2} = 2n\big(y_n^2 - y_{n-1}y_{n+1}\big) + 2y_n^2\,.$$

(iii) Hence show that $y_n^2 - y_{n-1}y_{n+1} > 0$ for $n \geqslant 1$.

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[Turn over

3 Show that the second-order differential equation

$$x^2y'' + (1-2p)x\,y' + (p^2-q^2)\,y = \mathrm{f}(x)\,,$$

where p and q are constants, can be written in the form

$$x^a\big(x^b(x^cy)'\big)' = \mathrm{f}(x)\,, \qquad (*)$$

where a, b and c are constants.

(i) Use $(*)$ to derive the general solution of the equation

$$x^2y'' + (1-2p)xy' + (p^2-q^2)y = 0$$

in the different cases that arise according to the values of p and q.

(ii) Use $(*)$ to derive the general solution of the equation

$$x^2y'' + (1-2p)xy' + p^2y = x^n$$

in the different cases that arise according to the values of p and n.

4 The point $P(a\sec\theta, b\tan\theta)$ lies on the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1\,,$$

where $a > b > 0$. Show that the equation of the tangent to the hyperbola at P can be written as

$$bx - ay\sin\theta = ab\cos\theta\,.$$

(i) This tangent meets the lines $\dfrac{x}{a} = \dfrac{y}{b}$ and $\dfrac{x}{a} = -\dfrac{y}{b}$ at S and T, respectively.

How is the mid-point of ST related to P?

(ii) The point $Q(a\sec\phi, b\tan\phi)$ also lies on the hyperbola and the tangents to the hyperbola at P and Q are perpendicular. These two tangents intersect at (x, y).

Obtain expressions for x^2 and y^2 in terms of a, θ and ϕ.

Hence, or otherwise, show that $x^2 + y^2 = a^2 - b^2$.

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5 The real numbers a_1, a_2, a_3, ... are all positive. For each positive integer n, A_n and G_n are defined by

$$A_n = \frac{a_1 + a_2 + \cdots + a_n}{n} \qquad \text{and} \qquad G_n = \left(a_1 a_2 \cdots a_n\right)^{1/n} .$$

(i) Show that, for any given positive integer k,

$$(k+1)(A_{k+1} - G_{k+1}) \geqslant k(A_k - G_k)$$

if and only if

$$\lambda_k^{k+1} - (k+1)\lambda_k + k \geqslant 0 ,$$

where $\lambda_k = \left(\dfrac{a_{k+1}}{G_k}\right)^{\frac{1}{k+1}}$.

(ii) Let

$$\mathrm{f}(x) = x^{k+1} - (k+1)x + k ,$$

where $x > 0$ and k is a positive integer. Show that $\mathrm{f}(x) \geqslant 0$ and that $\mathrm{f}(x) = 0$ if and only if $x = 1$.

(iii) Deduce that:

(a) $A_n \geqslant G_n$ for all n;

(b) if $A_n = G_n$ for some n, then $a_1 = a_2 = \cdots = a_n$.

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[Turn over

6 **(i)** The distinct points A, Q and C lie on a straight line in the Argand diagram, and represent the distinct complex numbers a, q and c, respectively. Show that $\dfrac{q-a}{c-a}$ is real and hence that $(c-a)(q^*-a^*) = (c^*-a^*)(q-a)$.

Given that $aa^* = cc^* = 1$, show further that

$$q + acq^* = a + c\,.$$

(ii) The distinct points A, B, C and D lie, in anticlockwise order, on the circle of unit radius with centre at the origin (so that, for example, $aa^* = 1$). The lines AC and BD meet at Q. Show that

$$(ac - bd)q^* = (a + c) - (b + d)\,,$$

where b and d are complex numbers represented by the points B and D respectively, and show further that

$$(ac - bd)(q + q^*) = (a - b)(1 + cd) + (c - d)(1 + ab)\,.$$

(iii) The lines AB and CD meet at P, which represents the complex number p. Given that p is real, show that $p(1 + ab) = a + b$. Given further that $ac - bd \neq 0$, show that

$$p(q + q^*) = 2\,.$$

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7 **(i)** Use De Moivre's theorem to show that, if $\sin\theta \neq 0$, then

$$\frac{(\cot\theta + \mathrm{i})^{2n+1} - (\cot\theta - \mathrm{i})^{2n+1}}{2\mathrm{i}} = \frac{\sin(2n+1)\theta}{\sin^{2n+1}\theta},$$

for any positive integer n.

Deduce that the solutions of the equation

$$\binom{2n+1}{1}x^n - \binom{2n+1}{3}x^{n-1} + \cdots + (-1)^n = 0$$

are

$$x = \cot^2\left(\frac{m\pi}{2n+1}\right)$$

where $m = 1, 2, \ldots, n$.

(ii) Hence show that

$$\sum_{m=1}^{n} \cot^2\left(\frac{m\pi}{2n+1}\right) = \frac{n(2n-1)}{3}.$$

(iii) Given that $0 < \sin\theta < \theta < \tan\theta$ for $0 < \theta < \frac{1}{2}\pi$, show that

$$\cot^2\theta < \frac{1}{\theta^2} < 1 + \cot^2\theta.$$

Hence show that

$$\sum_{m=1}^{\infty} \frac{1}{m^2} = \frac{\pi^2}{6}.$$

8 In this question, you should ignore issues of convergence.

(i) Let

$$I = \int_0^1 \frac{\mathrm{f}(x^{-1})}{1+x}\,\mathrm{d}x,$$

where $\mathrm{f}(x)$ is a function for which the integral exists.

Show that

$$I = \sum_{n=1}^{\infty} \int_n^{n+1} \frac{\mathrm{f}(y)}{y(1+y)}\,\mathrm{d}y$$

and deduce that, if $\mathrm{f}(x) = \mathrm{f}(x+1)$ for all x, then

$$I = \int_0^1 \frac{\mathrm{f}(x)}{1+x}\,\mathrm{d}x.$$

(ii) The *fractional part*, $\{x\}$, of a real number x is defined to be $x - \lfloor x \rfloor$ where $\lfloor x \rfloor$ is the largest integer less than or equal to x. For example $\{3.2\} = 0.2$ and $\{3\} = 0$.

Use the result of part (i) to evaluate

$$\int_0^1 \frac{\{x^{-1}\}}{1+x}\,\mathrm{d}x \quad \text{and} \quad \int_0^1 \frac{\{2x^{-1}\}}{1+x}\,\mathrm{d}x.$$

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[Turn over

Section B: Mechanics

9 A particle P of mass m is projected with speed u_0 along a smooth horizontal floor directly towards a wall. It collides with a particle Q of mass km which is moving directly away from the wall with speed v_0. In the subsequent motion, Q collides alternately with the wall and with P. The coefficient of restitution between Q and P is e, and the coefficient of restitution between Q and the wall is 1.

Let u_n and v_n be the velocities of P and Q, respectively, towards the wall after the nth collision between P and Q.

(i) Show that, for $n \geqslant 2$,

$$(1+k)u_n - (1-k)(1+e)u_{n-1} + e(1+k)u_{n-2} = 0\,. \qquad (*)$$

(ii) You are now given that $e = \frac{1}{2}$ and $k = \frac{1}{34}$, and that the solution of $(*)$ is of the form

$$u_n = A\left(\tfrac{7}{10}\right)^n + B\left(\tfrac{5}{7}\right)^n \qquad (n \geqslant 0)\,,$$

where A and B are independent of n. Find expressions for A and B in terms of u_0 and v_0.

Show that, if $0 < 6u_0 < v_0$, then u_n will be negative for large n.

10 A uniform disc with centre O and radius a is suspended from a point A on its circumference, so that it can swing freely about a horizontal axis L through A. The plane of the disc is perpendicular to L. A particle P is attached to a point on the circumference of the disc. The mass of the disc is M and the mass of the particle is m.

In equilibrium, the disc hangs with OP horizontal, and the angle between AO and the downward vertical through A is β. Find $\sin\beta$ in terms of M and m and show that

$$\frac{AP}{a} = \sqrt{\frac{2M}{M+m}}\,.$$

The disc is rotated about L and then released. At later time t, the angle between OP and the horizontal is θ; when P is higher than O, θ is positive and when P is lower than O, θ is negative. Show that

$$\tfrac{1}{2}I\dot{\theta}^2 + (1-\sin\beta)ma^2\dot{\theta}^2 + (m+M)ga\cos\beta\,(1-\cos\theta)$$

is constant during the motion, where I is the moment of inertia of the disc about L.

Given that $m = \frac{3}{2}M$ and that $I = \frac{3}{2}Ma^2$, show that the period of small oscillations is

$$3\pi\sqrt{\frac{3a}{5g}}\,.$$

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11 A particle is attached to one end of a light inextensible string of length b. The other end of the string is attached to a fixed point O. Initially the particle hangs vertically below O. The particle then receives a horizontal impulse.

The particle moves in a circular arc with the string taut until the acute angle between the string and the upward vertical is α, at which time it becomes slack. Express V, the speed of the particle when the string becomes slack, in terms of b, g and α.

Show that the string becomes taut again a time T later, where

$$gT = 4V \sin\alpha \,,$$

and that just before this time the trajectory of the particle makes an angle β with the horizontal where $\tan\beta = 3\tan\alpha$.

When the string becomes taut, the momentum of the particle in the direction of the string is destroyed. Show that the particle comes instantaneously to rest at this time if and only if

$$\sin^2\alpha = \frac{1+\sqrt{3}}{4} \,.$$

Section C: Probability and Statistics

12 A random process generates, independently, n numbers each of which is drawn from a uniform (rectangular) distribution on the interval 0 to 1. The random variable Y_k is defined to be the kth smallest number (so there are $k-1$ smaller numbers).

(i) Show that, for $0 \leqslant y \leqslant 1$,

$$\mathrm{P}(Y_k \leqslant y) = \sum_{m=k}^{n} \binom{n}{m} y^m (1-y)^{n-m} . \qquad (*)$$

(ii) Show that

$$m\binom{n}{m} = n\binom{n-1}{m-1}$$

and obtain a similar expression for $(n-m)\binom{n}{m}$.

Starting from $(*)$, show that the probability density function of Y_k is

$$n\binom{n-1}{k-1} y^{k-1} (1-y)^{n-k} .$$

Deduce an expression for $\displaystyle\int_0^1 y^{k-1}(1-y)^{n-k}\,\mathrm{d}y$.

(iii) Find $\mathrm{E}(Y_k)$ in terms of n and k.

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13 The random variable X takes only non-negative integer values and has probability generating function $\mathrm{G}(t)$. Show that

$$\mathrm{P}(X = 0 \text{ or } 2 \text{ or } 4 \text{ or } 6 \ \ldots) = \frac{1}{2}\big(\mathrm{G}\,(1) + \mathrm{G}\,(-1)\big).$$

You are now given that X has a Poisson distribution with mean λ. Show that

$$\mathrm{G}(t) = \mathrm{e}^{-\lambda(1-t)}\,.$$

(i) The random variable Y is defined by

$$\mathrm{P}(Y = r) = \begin{cases} k\mathrm{P}(X = r) & \text{if } r = 0,\ 2,\ 4,\ 6,\ \ldots\ , \\ 0 & \text{otherwise,} \end{cases}$$

where k is an appropriate constant.

Show that the probability generating function of Y is $\dfrac{\cosh \lambda t}{\cosh \lambda}$.

Deduce that $\mathrm{E}(Y) < \lambda$ for $\lambda > 0$.

(ii) The random variable Z is defined by

$$\mathrm{P}(Z = r) = \begin{cases} c\mathrm{P}(X = r) & \text{if } r = 0,\ 4,\ 8,\ 12,\ \ldots\ , \\ 0 & \text{otherwise,} \end{cases}$$

where c is an appropriate constant.

Is $\mathrm{E}(Z) < \lambda$ for all positive values of λ?

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Sixth Term Examination Papers **9475**
MATHEMATICS 3 Morning
Friday 21 June 2019 Time: 3 hours

3611115172

Additional Material: Answer Booklet

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ERRATUM

Question 5 part (ii)

The sentence

"Use the substitution $y = \frac{bx}{\sqrt{x^2+p}}$, where p is a suitably chosen constant, …"

Should read

"Use the substitution $y = \frac{cx}{\sqrt{x^2+p}}$, where p is a suitably chosen constant, …"

Section A: Pure Mathematics

1 The coordinates of a particle at time t are x and y. For $t \geqslant 0$, they satisfy the pair of coupled differential equations

$$\dot{x} = -x - ky$$
$$\dot{y} = x - y$$

where k is a constant. When $t = 0$, $x = 1$ and $y = 0$.

(i) Let $k = 1$. Find x and y in terms of t and sketch y as a function of t.

Sketch the path of the particle in the x-y plane, giving the coordinates of the point at which y is greatest and the coordinates of the point at which x is least.

(ii) Instead, let $k = 0$. Find x and y in terms of t and sketch the path of the particle in the x-y plane.

2 The definition of the derivative f' of a (differentiable) function f is

$$\mathrm{f}'(x) = \lim_{h \to 0} \frac{\mathrm{f}(x+h) - \mathrm{f}(x)}{h}\,. \qquad (*)$$

(i) The function f has derivative f' and satisfies

$$\mathrm{f}(x+y) = \mathrm{f}(x)\mathrm{f}(y)$$

for all x and y, and $\mathrm{f}'(0) = k$ where $k \neq 0$. Show that $\mathrm{f}(0) = 1$.

Using $(*)$, show that $\mathrm{f}'(x) = k\mathrm{f}(x)$ and find $\mathrm{f}(x)$ in terms of x and k.

(ii) The function g has derivative g' and satisfies

$$\mathrm{g}(x+y) = \frac{\mathrm{g}(x) + \mathrm{g}(y)}{1 + \mathrm{g}(x)\mathrm{g}(y)}$$

for all x and y, $|\mathrm{g}(x)| < 1$ for all x, and $\mathrm{g}'(0) = k$ where $k \neq 0$.

Find $\mathrm{g}'(x)$ in terms of $\mathrm{g}(x)$ and k, and hence find $\mathrm{g}(x)$ in terms of x and k.

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[Turn over

3 The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

(i) You are given that the transformation represented by $\mathbf{A}$ has a line L_1 of invariant points (so that each point on L_1 is transformed to itself). Let (x, y) be a point on L_1. Show that $\big((a-1)(d-1) - bc\big)xy = 0$.

Show further that $(a-1)(d-1) = bc$.

What can be said about $\mathbf{A}$ if L_1 does not pass through the origin?

(ii) By considering the cases $b \neq 0$ and $b = 0$ separately, show that if $(a-1)(d-1) = bc$ then the transformation represented by $\mathbf{A}$ has a line of invariant points. You should identify the line in the different cases that arise.

(iii) You are given instead that the transformation represented by $\mathbf{A}$ has an invariant line L_2 (so that each point on L_2 is transformed to a point on L_2) and that L_2 does not pass through the origin. If L_2 has the form $y = mx + k$, show that $(a-1)(d-1) = bc$.

4 The nth degree polynomial $\mathrm{P}(x)$ is said to be *reflexive* if:

(a) $\mathrm{P}(x)$ is of the form $x^n - a_1x^{n-1} + a_2x^{n-2} - \cdots + (-1)^n a_n$ where $n \geqslant 1$;
(b) $a_1, a_2, \ldots, a_n$ are real;
(c) the n (not necessarily distinct) roots of the equation $\mathrm{P}(x) = 0$ are $a_1, a_2, \ldots, a_n$.

(i) Find all reflexive polynomials of degree less than or equal to 3.

(ii) For a reflexive polynomial with $n > 3$, show that

$$2a_2 = -a_2^2 - a_3^2 - \cdots - a_n^2.$$

Deduce that, if all the coefficients of a reflexive polynomial of degree n are integers and $a_n \neq 0$, then $n \leqslant 3$.

(iii) Determine all reflexive polynomials with integer coefficients.

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5 **(i)** Let

$$\mathrm{f}(x) = \frac{x}{\sqrt{x^2 + p}},$$

where p is a non-zero constant. Sketch the curve $y = \mathrm{f}(x)$ for $x \geqslant 0$ in the case $p > 0$.

(ii) Let

$$I = \int \frac{1}{(b^2 - y^2)\sqrt{c^2 - y^2}}\,\mathrm{d}y,$$

where b and c are positive constants. Use the substitution $y = \dfrac{bx}{\sqrt{x^2 + p}}$, where p is a suitably chosen constant, to show that

$$I = \int \frac{1}{b^2 + (b^2 - c^2)x^2}\,\mathrm{d}x.$$

Evaluate

$$\int_1^{\sqrt{2}} \frac{1}{(3 - y^2)\sqrt{2 - y^2}}\,\mathrm{d}y.$$

[**Note:** $\displaystyle\int \frac{1}{a^2 + x^2}\,\mathrm{d}x = \frac{1}{a}\tan^{-1}\frac{x}{a} + \text{ constant.}$]

Hence evaluate

$$\int_{\frac{1}{\sqrt{2}}}^{1} \frac{y}{(3y^2 - 1)\sqrt{2y^2 - 1}}\,\mathrm{d}y.$$

(iii) By means of a suitable substitution, evaluate

$$\int_{\frac{1}{\sqrt{2}}}^{1} \frac{1}{(3y^2 - 1)\sqrt{2y^2 - 1}}\,\mathrm{d}y.$$

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[Turn over

6 The point P in the Argand diagram is represented by the the complex number z, which satisfies

$$zz^* - az^* - a^*z + aa^* - r^2 = 0.$$

Here, r is a positive real number and $r^2 \neq a^*a$. By writing $|z-a|^2$ as $(z-a)(z-a)^*$, show that the locus of P is a circle, C, the radius and the centre of which you should give.

(i) The point Q is represented by ω, and is related to P by $\omega = \dfrac{1}{z}$. Let C' be the locus of Q. Show that C' is also a circle, and give its radius and centre.

If C and C' are the same circle, show that

$$(|a|^2 - r^2)^2 = 1$$

and that either a is real or a is imaginary. Give sketches to indicate the position of C in these two cases.

(ii) Suppose instead that the point Q is represented by ω, where $\omega = \dfrac{1}{z^*}$. If the locus of Q is C, is it the case that either a is real or a is imaginary?

7 The *Devil's Curve* is given by

$$y^2(y^2 - b^2) = x^2(x^2 - a^2)\,,$$

where a and b are positive constants.

(i) In the case $a = b$, sketch the Devil's Curve.

(ii) Now consider the case $a = 2$ and $b = \sqrt{5}$, and $x \geqslant 0$, $y \geqslant 0$.

(a) Show by considering a quadratic equation in x^2 that either $0 \leqslant y \leqslant 1$ or $y \geqslant 2$.

(b) Describe the curve very close to and very far from the origin.

(c) Find the points at which the tangent to the curve is parallel to the x-axis and the point at which the tangent to the curve is parallel to the y-axis.

Sketch the Devil's Curve in this case.

(iii) Sketch the Devil's Curve in the case $a = 2$ and $b = \sqrt{5}$ again, but with $-\infty < x < \infty$ and $-\infty < y < \infty$.

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8 A pyramid has a horizontal rectangular base $ABCD$ and its vertex V is vertically above the centre of the base. The acute angle between the face AVB and the base is α, the acute angle between the face BVC and the base is β and the obtuse angle between the faces AVB and BVC is $\pi - \theta$.

(i) The edges AB and BC are parallel to the unit vectors $\mathbf{i}$ and $\mathbf{j}$, respectively, and the unit vector $\mathbf{k}$ is vertical. Find a unit vector that is perpendicular to the face AVB.

Show that

$$\cos\theta = \cos\alpha\cos\beta\,.$$

(ii) The edge BV makes an angle ϕ with the base. Show that

$$\cot^2\phi = \cot^2\alpha + \cot^2\beta.$$

Show also that

$$\cos^2\phi = \frac{\cos^2\alpha + \cos^2\beta - 2\cos^2\theta}{1 - \cos^2\theta} \geqslant \frac{2\cos\theta - 2\cos^2\theta}{1 - \cos^2\theta}$$

and deduce that $\phi < \theta$.

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[Turn over

Section B: Mechanics

9 In this question, **i** and **j** are perpendicular unit vectors and **j** is vertically upwards.

A smooth hemisphere of mass M and radius a rests on a smooth horizontal table with its plane face in contact with the table. The point A is at the top of the hemisphere and the point O is at the centre of its plane face.

Initially, a particle P of mass m rests at A. It is then given a small displacement in the positive **i** direction. At a later time t, when the particle is still in contact with the hemisphere, the hemisphere has been displaced by $-s\,\mathbf{i}$ and $\angle AOP = \theta$.

(i) Let **r** be the position vector of the particle at time t with respect to the initial position of O. Write down an expression for **r** in terms of a, θ and s and show that

$$\dot{\mathbf{r}} = (a\dot{\theta}\cos\theta - \dot{s})\,\mathbf{i} - a\dot{\theta}\sin\theta\,\mathbf{j}.$$

Show also that

$$\dot{s} = (1-k)\,a\dot{\theta}\cos\theta,$$

where $k = \dfrac{M}{m+M}$, and deduce that

$$\dot{\mathbf{r}} = a\dot{\theta}(k\cos\theta\,\mathbf{i} - \sin\theta\,\mathbf{j}).$$

(ii) Show that

$$a\dot{\theta}^2\left(k\cos^2\theta + \sin^2\theta\right) = 2g(1-\cos\theta).$$

(iii) At time T, when $\theta = \alpha$, the particle leaves the hemisphere. By considering the component of $\ddot{\mathbf{r}}$ parallel to the vector $\sin\theta\,\mathbf{i} + k\cos\theta\,\mathbf{j}$, or otherwise, show that at time T

$$a\dot{\theta}^2 = g\cos\alpha.$$

Find a cubic equation for $\cos\alpha$ and deduce that $\cos\alpha > \frac{2}{3}$.

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10 Two identical smooth spheres P and Q can move on a smooth horizontal table. Initially, P moves with speed u and Q is at rest. Then P collides with Q. The direction of travel of P before the collision makes an acute angle α with the line joining the centres of P and Q at the moment of the collision. The coefficient of restitution between P and Q is e where $e < 1$.

As a result of the collision, P has speed v and Q has speed w, and P is deflected through an angle θ.

(i) Show that

$$u \sin \alpha = v \sin(\alpha + \theta)$$

and find an expression for w in terms of v, θ and α.

(ii) Show further that

$$\sin \theta = \cos(\theta + \alpha) \sin \alpha + e \sin(\theta + \alpha) \cos \alpha$$

and find an expression for $\tan \theta$ in terms of $\tan \alpha$ and e.

Find, in terms of e, the maximum value of $\tan \theta$ as α varies.

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Section C: Probability and Statistics

11 The number of customers arriving at a builders' merchants each day follows a Poisson distribution with mean λ. Each customer is offered some free sand. The probability of any given customer taking the free sand is p.

(i) Show that the number of customers each day who take sand follows a Poisson distribution with mean $p\lambda$.

(ii) The merchant has a mass S of sand at the beginning of the day. Each customer who takes the free sand gets a proportion k of the remaining sand, where $0 \leqslant k < 1$. Show that by the end of the day the expected mass of sand taken is

$$\left(1 - \mathrm{e}^{-kp\lambda}\right)S.$$

(iii) At the beginning of the day, the merchant's bag of sand contains a large number of grains, exactly one of which is made from solid gold. At the end of the day, the merchant's assistant takes a proportion k of the remaining sand. Find the probability that the assistant takes the golden grain. Comment on the case $k = 0$ and on the limit $k \to 1$.

In the case $p\lambda > 1$ find the value of k which maximises the probability that the assistant takes the golden grain.

12 The set S is the set of all integers from 1 to n. The set T is the set of all distinct subsets of S, including the empty set $\varnothing$ and S itself. Show that T contains exactly 2^n sets.

The sets A_1, A_2, ..., A_m, which are not necessarily distinct, are chosen randomly and independently from T, and for each k $(1 \leqslant k \leqslant m)$, the set A_k is equally likely to be any of the sets in T.

(i) Write down the value of $\mathrm{P}(1 \in A_1)$.

(ii) By considering each integer separately, show that $\mathrm{P}(A_1 \cap A_2 = \varnothing) = \left(\frac{3}{4}\right)^n$.

Find $\mathrm{P}(A_1 \cap A_2 \cap A_3 = \varnothing)$ and $\mathrm{P}(A_1 \cap A_2 \cap \cdots \cap A_m = \varnothing)$.

(iii) Find $\mathrm{P}(A_1 \subseteq A_2)$, $\mathrm{P}(A_1 \subseteq A_2 \subseteq A_3)$ and $\mathrm{P}(A_1 \subseteq A_2 \subseteq \cdots \subseteq A_m)$.

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